

# Critical-Point Defects in Surreal Arithmetic

Large-cardinal embeddings, exact support thresholds,  
descent to the target model, omnific independence,  
surcomplex fields, and omnific integers

A merged research report from three manuscripts

Manuscripts 06, 07 and 09 of batch 30, each prepared for Vladimir Reshetnikov (07 and 09 with ChatGPT),  
23 September 2026; 07 is the base text

**Main result.** An elementary embedding  $j : V \rightarrow M$  with critical point  $\kappa$  induces two ordered-field embeddings  $J, H_j : \mathbf{No} \rightarrow \mathbf{No}$ . They agree on every real number, ordinal, and Conway monomial, but

$$J(x) = H_j(x) \iff |\text{supp}(x)| < \kappa.$$

Their images have an exactly determined intersection. A single coefficient of their discrepancy recovers the normal measure derived from  $j$ . [merge] For every individual normal form of length  $\lambda$ , the value  $H_j(x)$  lies in the target model exactly when  $j \restriction \lambda \in M$ ; for a normal-measure ultrapower this happens exactly when  $x$  has at most  $\kappa$  terms [06]. Without any large cardinal, almost-disjoint families give algebraically independent omnific integers over the whole short-support field [09].

**Scope and status.** All central assertions below have written proofs under explicit hypotheses. The general Hahn and large-cardinal ingredients are classical; the exact defect and intersection package is offered as a candidate original contribution. Publication priority has not been certified, and the article is neither refereed nor Lean-verified. It does not claim a solution to a named published conjecture or a new consistency proof for a large-cardinal axiom.

Repository context: VladimirReshetnikov/Surreal, snapshot 0865f043aec1.

Standalone source, compiled PDF, proof-status notes, and finite algebraic regression checks accompany this article. [merge] All three manuscripts were written against the same snapshot; the merged report was placed in commit 21375f8. Sources, notation and merge decisions: Section 1.5 and Appendix D; non-claims of every source: Appendix C.

## Abstract

Large-cardinal embeddings preserve every finitary arithmetic identity, but need not preserve the external strong sums that define surreal normal forms. We make this distinction exact. Let  $j : V \rightarrow M$  be an amenable elementary embedding into a transitive inner class, with critical point  $\kappa$ . Its restriction to surreal sign sequences gives an ordered-field embedding  $J$ . The same action on Conway monomials has a unique strongly  $\mathbb{R}$ -linear extension  $H_j$ . If  $x$  has normal form indexed by  $\theta$ , then  $J(x) - H_j(x)$  is precisely the subseries of the transformed normal form indexed by  $j(\theta) \setminus j \text{ “ } \theta$ . Consequently the equalizer of  $J$  and  $H_j$  is exactly the real-closed field of surreal numbers with fewer than  $\kappa$  outer normal-form terms. Moreover, their two image fields intersect exactly in the common image of that equalizer, and are its incomparable logarithmic and strong-sum closures. For a positive strongly summable family, preservation by  $J$  holds exactly when the index set has cardinality less than  $\kappa$ .

The defect has a first term at the first missing index  $\kappa$ . For  $h_A = \sum_{\alpha \in A} \omega^{-\alpha}$ ,  $A \subseteq \kappa$ , its coefficient at  $\omega^{-\kappa}$  is the indicator of membership in the normal measure derived from  $j$ . This gives a coefficient realization of classical ultrafilter evaluation and a precise measurable-cardinal criterion for the associated Hadamard mask algebra. The field results transfer to surcomplex numbers and to omnific integers; a purely infinite omnific defect encodes the same measure while its constant term vanishes. We also obtain a logarithmic defect homomorphism, an exact exponential compatibility locus, and a supercompact seed-coefficient construction. The article distinguishes these contributions from standard Hahn reindexing, classical derived measures, and the repository’s earlier work on birthday expansions and forcing. Twelve further research questions and a formalization plan conclude the account.

[merge] This report merges three manuscripts on the same pair of maps. The text above is the abstract of the base manuscript 07. Manuscript 06 adds descent: the companion value  $H_j(x)$  belongs to the target model exactly when the seed  $j \text{ “ } \lambda$  does, where  $\lambda$  is the normal-form length of  $x$ ; in a set ultrapower this is closure of  $M$  under  $\lambda$ -sequences, and for a normal-measure ultrapower it gives an exact three-way classification at support sizes  $< \kappa$ ,  $\kappa$ ,  $> \kappa$ , without GCH. It also adds bounded and omnific measure codes, and support-cover and exact-descent forms of strong compactness and supercompactness. Manuscript 09 adds a large-cardinal-free theory of the short-support field (real closed at every uncountable cardinal, closed and nowhere dense, with an omnific integer part), algebraically independent omnific masks over that whole field from almost-disjoint families ( $2^\kappa$  of them in one set-sized Hahn field when  $2^{<\kappa} = \kappa$ ), a compatible-field-embedding criterion for measurability, seed coefficients over arbitrary index sets, and an omnific prefix-tree form of weak compactness. The strong-compactness, supercompactness and weak-compactness statements are credited translations of classical ultrafilter and tree characterizations. As a consequence, if there is a measurable cardinal then there is a field embedding of **No** that preserves and reflects the omnific integers and their constant terms but is not strongly additive. There are 34 research questions in all, and a list of every source’s non-claims.

**Keywords.** Surreal number; Conway normal form; strong summation; measurable cardinal; elementary embedding; omnific integer; surcomplex number; Hahn field; ultrafilter; supercompactness.

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# 1 The questions answered and the novelty boundary

## 1.1 The distinction that drives the construction

For a set-theoretic elementary embedding, the image  $j(I)$  of an infinite set is generally larger than its pointwise image  $j \restriction I$ . For a surreal normal form, this is not merely a difference between two encodings of the same sum. The additional indices can carry additional nonzero monomials. The resulting discrepancy is visible in arithmetic, even when every original summand is individually fixed.

This article answers the following concrete question under a measurable-cardinal hypothesis: *how much of a surreal number is determined by the action of an embedding on all real coefficients and all monomials?* The answer is exactly its short-support part. There are two explicitly associated embeddings that agree on all those generators, and even on every ordinal, but disagree on every surreal whose outer normal form has at least  $\kappa$  terms. The distinction is a support-cardinality threshold, not a birthday cutoff.

Conway’s normal-form theorem and the generalized-series presentation of **No** are the classical background [1, 2, 3, 4]. Strongly additive maps and monomial reindexing also have a substantial literature. In particular, Kaplan–Krapp–Serra distinguish strong from non-strong automorphisms and study their relation to extra surreal structure [5]. The contribution proposed here is not the bare possibility of non-strong behavior. It is the exact critical-point formula, equalizer, intersection, and coefficient reconstruction for the specific large-cardinal construction.

## 1.2 Assumption and headline statements

Throughout the main construction, assume that

$$j : V \longrightarrow M \subseteq V \tag{1.1}$$

is an elementary class embedding,  $M$  is transitive and contains every ordinal, and  $\kappa = \text{crit}(j)$  exists. Class parameters are allowed in the Separation and Replacement instances used below. A sufficient, familiar instance is a normal ultrapower embedding from a measurable cardinal. One can carry out that instance in ZFC using the measure as a set parameter; the use of classes is notation for definable constructions. The general class-embedding version can be read in GBC with the indicated embedding as an amenable class. These hypotheses are discussed in Section 2.

Let

$$\mathcal{S}_{<\kappa} = \{x \in \mathbf{No} : |\text{supp}(x)| < \kappa\}, \tag{1.2}$$

where  $\text{supp}(x)$  means the set of *outer exponents* in its Conway normal form. Zero has empty support. Define  $J(x) = j(x)$  on surreal sign sequences, and define

$$H_j \left( \sum_{\alpha < \theta} r_\alpha \omega^{a_\alpha} \right) = \sum_{\alpha < \theta} r_\alpha \omega^{J(a_\alpha)}. \tag{1.3}$$

The central conclusions, proved in Sections 4–8, are

$$\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}, \tag{1.4}$$

$$J(\mathbf{No}) \cap H_j(\mathbf{No}) = J(\mathcal{S}_{<\kappa}) = H_j(\mathcal{S}_{<\kappa}), \tag{1.5}$$

$$J \left( \sum_{i \in I} x_i \right) = \sum_{i \in I} J(x_i) \iff |I| < \kappa \quad (x_i > 0 \text{ and strongly summable}). \tag{1.6}$$

The two image fields in (1.5) are incomparable, not merely different. Their common image is real closed. Coefficient projection canonically identifies the two images as immediate valued extensions of that common field (Theorem 6.4). An intrinsic version shows that every positive strong sum of at least  $\kappa$  image elements escapes the elementary image (Theorem 8.6).

For  $A \subseteq \kappa$ , put  $h_A = \sum_{\alpha \in A} \omega^{-\alpha}$ . One then obtains

$$[\omega^{-\kappa}] (J(h_A) - H_j(h_A)) = \begin{cases} 1, & \kappa \in j(A), \\ 0, & \kappa \notin j(A). \end{cases} \quad (1.7)$$

Thus the usual derived measure is recovered from a distinguished coefficient of the arithmetic defect. The derived measure itself is classical [6]; the coefficient realization is the new construction being proposed.

### 1.3 The headline statements of the other two sources [merge]

Manuscripts 06 and 09 prove the equalizer and the defect formula independently, under essentially the same hypotheses, and state them in their own headline theorems. We print both, with the numbering of this report in the proof pointers. The support cardinal  $|\text{supp}(x)|$ , the number of nonzero terms in the Conway normal form of  $x$ , is 06's  $\text{sc}(x)$ ; the short-support field  $\mathcal{S}_{<\kappa}$  is 06's  $K_{<\kappa}$  and 09's  $F_\kappa$  (Section 1.5).

**Theorem 1.1** (Main comparison and descent [06]). *Under the standing embedding hypotheses (2.3), the following assertions hold.*

- (i)  *$J$  and  $H_j$  are ordered-field embeddings fixing  $\mathbb{R}$  and agreeing on all monomials. The map  $H_j$  preserves every set-indexed strong sum;  $J$  preserves every such sum indexed by a set of cardinality  $< \kappa$ , but not every one indexed by  $\kappa$ .*
- (ii) *For every surreal  $x$ ,  $J(x) = H_j(x) \iff |\text{supp}(x)| < \kappa$ .*
- (iii) *For every normal form of ordinal length  $\lambda$ ,  $H_j(x) \in \mathbf{No}^M \iff j \text{ “ } \lambda \in M$ . When  $j$  is a set-ultrapower embedding, these conditions are equivalent to  ${}^\lambda M \cap V \subseteq M$ .*
- (iv) *If  $j$  is the ultrapower by a normal measure on  $\kappa$ , then  $H_j(x) \in \mathbf{No}^M \iff |\text{supp}(x)| \leq \kappa$ . Consequently equality, unequal descent, and failure of descent occur exactly at support sizes  $< \kappa$ ,  $\kappa$ , and  $> \kappa$ .*

The proof is distributed among Theorem 4.3, Theorem 5.1, Theorem 8.1, Theorem 9.1, Lemma 9.3 and Theorem 9.7. No cardinal-arithmetic hypothesis such as GCH is needed for the last clause [06].

Two further manifestations of the same phenomenon are in 06's introduction. For  $x = p(x) + r(x) + \varepsilon(x)$ , with  $p(x)$  purely infinite,  $r(x) \in \mathbb{R}$  and  $\varepsilon(x)$  infinitesimal,

$$\frac{H_j(\exp x)}{\exp(H_j x)} = \exp(J(p(x)) - H_j(p(x))),$$

so the Hahn lift commutes with exponentiation at  $x$  exactly when the positive-exponent part has fewer than  $\kappa$  terms (Theorem 15.3, proved by 06 and 07). Exact descent of  $H_j(h_\lambda)$  characterizes a supercompactness witness, whereas an internally short binary-support *cover* characterizes a strong compactness witness; multiplication by a monomial gives the same tests inside the omnific integers (Section 17) [06].

**Theorem 1.2** (Main package [09]). *Assume that  $\kappa$  is measurable and let  $j : V \rightarrow M$  be the transitive ultrapower embedding by a normal measure on  $\kappa$ . There are ordered field embeddings  $J, H_j : \mathbf{No} \rightarrow \mathbf{No}$  with the following properties.*

- (i) Both fix  $\mathbb{R}$  and agree on every Conway monomial. The map  $J$  commutes with Conway's omega-map, and  $H_j$  preserves every set-indexed strongly summable family.
- (ii) Their exact equalizer is  $\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}$ . For a normal form of length  $\theta$ , the difference is the subseries indexed by  $j(\theta) \setminus j$  “ $\theta$  in the normal form of  $J(x)$ ”.
- (iii) The map  $J$  preserves every strongly summable family of cardinality less than  $\kappa$ , but fails to preserve a specified family of size  $\kappa$  consisting of purely infinite omnific monomials.
- (iv) The equalizers on omnific integers and on the specified complexification are  $\mathbf{Oz} \cap \mathcal{S}_{<\kappa}$  and  $\mathcal{S}_{<\kappa}(i)$ , respectively.
- (v) There are  $2^\kappa$  explicitly described omnific integers, algebraically independent over  $\mathcal{S}_{<\kappa}$ , on each of which  $J$  and  $H_j$  disagree.
- (vi) The normal measure can be recovered from one designated normal-form coefficient of  $J(Y_A)$ , where  $A \subseteq \kappa$  and  $Y_A$  is an explicit omnific code of  $A$ .

Proof pointers: (i) Proposition 2.10 and Theorem 4.3; (ii) Theorem 5.1; (iii) Theorems 8.1 and 8.4; (iv) Theorems 11.1 and 12.1; (v) Theorem 13.5 and Corollary 13.6; (vi) Theorem 10.3. According to 09, part (ii), its full difference formula, and the coefficient-independent algebraic construction underlying part (v) are its main proposed contributions. The hypotheses and the exact formulas matter. Merely constructing a Hahn lift, observing that elementary embeddings fail to preserve some long sequences, or translating an ultrafilter into a numerical code would not by itself constitute the same result [09]. Part (ii) is proved independently by all three sources and printed once, as Theorem 5.1.

## 1.4 What is not being asserted

We do not establish that arbitrary non-strong surreal embeddings require a measurable cardinal. We do not prove a new consistency result about measurability, supercompactness, or elementary self-embeddings of the set universe. We do not identify the two embeddings as automorphisms. Nor do we claim that their equality on monomials implies equality on a completed field: the whole point is that the relevant external sums need not be preserved by  $J$ .

The proofs are mathematical arguments, not output from a theorem prover. The accompanying finite checks test finite identities used in the arguments; they cannot test an elementary embedding of the universe or a nonprincipal complete ultrafilter. A targeted literature and repository comparison appears in Section 20; it is not an exhaustive priority search.

[09] The results do not establish the existence or consistency of any large cardinal. They do not produce a nontrivial elementary embedding  $(V, \epsilon) \rightarrow (V, \epsilon)$ , and do not settle the outstanding structure questions for all surreal exponential automorphisms. The research questions at the end include genuine gaps in the present theory; they are not presented as an exhaustive list of previously published open problems.

[06] Manuscript 06 is an AI-assisted research draft with written proofs, not a refereed or machine-checked paper. Its proposed contributions are the exact comparison, defect and descent statements, not the underlying Hahn-lifting or ultrapower-seed machinery. They were not located in the sources it inspected; this is not a priority certification or a claim to resolve a named published conjecture. All its large-cardinal implications retain their hypotheses; no consistency strength beyond the corresponding classical ultrafilter principle is claimed. Appendix C lists every non-claim of every source.

## 1.5 Sources, conventions and notation [merge]

**Three manuscripts.** The report is built from manuscripts 06, 07 and 09 of batch 30. All three were written on 23 September 2026 against the repository snapshot 0865f04, and all three study the same pair of maps  $(J, H_j)$  induced by an elementary embedding with critical point  $\kappa$ . They prove the same spine independently: the companion lift, the first missing index, the exact defect formula, the short-support equalizer, the small-family summation threshold and the measure coefficient.

| Source | Manuscript                                                                                                                                                | Contributes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07     | <i>Critical-Point Defects in Surreal Arithmetic: Large-cardinal embeddings, exact support thresholds, surcomplex fields, and omnific integers</i>         | The base text: every section not marked otherwise, with its exact image intersection and projection isomorphism, positive-family threshold and escape theorem, mask algebra and mask criterion, omnific and surcomplex transfers, logarithmic defect, exponential locus and two closures, supercompact seed, twelve questions, formalization plan and finite regression.                                                                                                                               |
| 06     | <i>Critical Supports and Large Cardinals in Surreal Arithmetic: Ultrapower transport, Hahn lifts, and omnific descent</i>                                 | A proof of the critical-point facts (Lemma 2.3); descent of $H_j(x)$ to $M$ , the landing spectrum, the seed–closure lemma, the cover obstruction and the normal-measure trichotomy (Section 9); bounded and omnific measure codes with a standard-part reading; formal evaluation of power series and the exponential scale corollary; support covers and exact descent for compactness principles (Section 17); the fixed class of $H_j$ ; nine questions and an audit checklist.                    |
| 09     | <i>Large Cardinals and the Support Geometry of Surreal Embeddings: Exact equalizers, additional normal-form terms, and omnific algebraic independence</i> | A second proof of the absoluteness lemma with its limit step exposed; the short-support field for every uncountable $\kappa$ (Section 3); omnific monomial witnesses; the omega-map incompatibility; order continuity; almost-disjoint omnific independence and the $2^\kappa$ family (Section 13); the compatible-field-embedding criterion; seed coefficients over arbitrary index sets; weak compactness as an omnific prefix principle (Section 18); the visibility hierarchy; thirteen questions. |

**How the merge is marked.** A statement proved by several sources is printed once, with its sources in brackets, for example [06, 07, 09]. Material from 06 or 09 carries its source tag, and its labels carry the sub-prefixes `lce:cs:` (06, “critical supports”) and `lce:sg:` (09, “support geometry”); labels of the base text carry `lce:` alone. Where two sources give genuinely different proofs, the second is printed as a marked second route. Text written for the merge, rather than taken from a source, is marked [merge]. Base-text passages without a tag are 07’s. Every proof imported from 06 or 09 was re-read against the hypotheses of this report; the places where a source’s hypotheses differ are noted at the statement.

**Hypotheses of the three sources.** 07 assumes an elementary class embedding  $j : V \rightarrow M \subseteq V$  into a transitive class containing every ordinal, with a critical point, and class-parameter Separation and Replacement (an amenable class in GBC, or a normal ultrapower in ZFC). 06 assumes the same, stated as (2.3) with  $M \models \text{ZFC}$  added. 09 states its main package for the ultrapower by a normal measure, and says that its Sections on the equalizer, the threshold, the omnific and surcomplex consequences and the measure apply to any such

class embedding with uncountable critical point. On the shared theorems 06 and 07 thus have the weakest stated hypotheses, and 09's arguments need no more; 07 is the base because it is the broader of 06 and 07 (placement commit 21375f8). 09's independence theorems assume no embedding at all, and 06's descent results add the hypotheses stated with them (a set-ultrapower presentation, or an index set of size  $\kappa$ ).

**Notation.** The base text's symbols are kept. Symbols of 06 and 09 were renamed where they collide with the base text or with each other; no normalization changed. The table lists every renaming.

| Source symbol                                                                            | Here                                                             | Reason and meaning                                                                                                             |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 09's $T, D$                                                                              | $H_j, D_j$                                                       | the strong companion and the defect $J - H_j$ , as in 06 and 07                                                                |
| 06's $K_{<\kappa}$ , 09's $F_\kappa$                                                     | $\mathcal{S}_{<\kappa}$                                          | surreals with fewer than $\kappa$ normal-form terms; 06's $K_{\leq\kappa}$ is $\mathcal{S}_{\leq\kappa}$                       |
| 09's $I_\kappa$                                                                          | $\mathcal{I}_{<\kappa} = \mathbf{Oz} \cap \mathcal{S}_{<\kappa}$ | short-support omnific integers                                                                                                 |
| 06's $\text{sc}(x)$                                                                      | $ \text{supp}(x) $                                               | the support cardinal                                                                                                           |
| 06's $\ell_{\text{nf}}(x)$ , 09's $\text{len}_{\text{NF}}(x)$                            | $\text{len}(x)$                                                  | normal-form length; 06 writes $\lambda$ and 09 writes $\mu$ for it, kept as $\lambda$ in Section 9                             |
| 06's $u_\lambda$ , 09's $Q_\theta$                                                       | $h_\lambda, h_\theta$                                            | $\sum_{\alpha < \lambda} \omega^{-\alpha}$ , 07's (7.1)                                                                        |
| 06's $c_A$                                                                               | $\tilde{h}_A = h_\kappa + h_A$                                   | the bounded code $\sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha))\omega^{-\alpha}$ ; the letter $c$ is kept for coefficients |
| 06's $z_A = \omega^\kappa c_A$                                                           | $\tilde{z}_A = z_\kappa + z_A$                                   | 07's $z_A$ is $\omega^\kappa h_A$                                                                                              |
| 06's $v_\lambda = \omega^\lambda u_\lambda$                                              | $\zeta_\lambda = \omega^\lambda h_\lambda$                       | $v$ is 07's valuation; $\zeta_\kappa = z_\kappa$                                                                               |
| 06's $x^\dagger$                                                                         | $p(x)$                                                           | the purely infinite part, as in 07                                                                                             |
| 06's LM, LC                                                                              | LT                                                               | leading monomial and coefficient enter only through the leading term                                                           |
| 06's binary series $Q, Q^+$                                                              | $\widehat{B}, \widehat{B}^+$                                     | 07's Question 21.1 uses $P, Q$ for embeddings                                                                                  |
| 06's $W_s$                                                                               | $U_s$                                                            | $\{X : s \in j(X)\}$ , as in 07 and 09; 06's letter $W$ for a fine ultrafilter is kept locally                                 |
| 09's $e_\alpha$ and $q_\alpha$                                                           | $q_\alpha = \omega^{-\alpha}$                                    | 09 had two names for one number; $e$ is 07's enumeration of an index set                                                       |
| 09's $u_\alpha = \omega^{e_\alpha}$                                                      | $\tau_\alpha = \omega^{q_\alpha}$                                | $u$ is 06's detector and 07's infinitesimal                                                                                    |
| 09's $Z_\theta, Z_A, W_f, Z_{A,h}$                                                       | $Y_\theta, Y_A, Y_f, Y_{A,e}$                                    | omnific masks $\sum \tau_\alpha$ ; $z_A$ is 07's shifted mask                                                                  |
| 09's $c_\gamma(x)$                                                                       | $[\omega^{q_\gamma}]x$                                           | the coefficient of $\tau_\gamma$ ; 07's coefficient notation                                                                   |
| 09's bijection $h : \theta \rightarrow I$                                                | $e$                                                              | 07's name; $h$ is 07's mask                                                                                                    |
| 09's $E = \bigoplus \mathbb{Q}e_\alpha, H$                                               | $G_\kappa, \Xi$                                                  | $E$ is 09's embedding in Theorem 10.10; $H$ collides with $H_j$                                                                |
| 09's polynomial $P$ ;<br>exponents $h_0, h$ ; coefficient $a_0$                          | $\Phi; d_0, d'; \varrho$                                         | $P_\theta$ is 07's missing-index set; $a$ is an exponent                                                                       |
| 09's index $j$ in $\bigcup_{j \neq i} A_j$                                               | $k$                                                              | $j$ is the embedding                                                                                                           |
| 09's $\chi(A)$                                                                           | $\chi_E(A)$                                                      | 07's $\chi$ acts on the mask algebra                                                                                           |
| 09's strings $s \in 2^\alpha$ , tree $S$ ,<br>injections $q_\alpha$ , lengths $D_\alpha$ | $\sigma, \mathcal{R}, \iota_\alpha, \Delta_\alpha$               | $s$ is a seed, $\mathcal{S}_{<\kappa}$ the short field, $q_\alpha$ an exponent, $D_j$ the defect                               |
| 09's $P_\kappa \lambda, \text{Ord}$                                                      | $P_\kappa(\lambda), \mathbf{On}$                                 | base-text spelling                                                                                                             |

Some tempting readings are false and are excluded throughout. The equalizer  $\mathcal{S}_{<\kappa}$  is not a fixed field:  $J(\kappa) = H_j(\kappa) = j(\kappa) \neq \kappa$  (Warning 5.4). Short support is not a birthday bound: every monomial lies in  $\mathcal{S}_{<\kappa}$  (Section 3). The Hadamard product  $\odot$  is not surreal multiplication.  $\omega^a$  is always Conway's omega-map, never

$\exp(a \log \omega)$ . Two different letters denote measure codes:  $h_A$  has coefficients 0, 1 and  $\tilde{h}_A$  has coefficients 1, 2; the omnific masks  $z_A$  (exponents  $\kappa - \alpha$ ) and  $Y_A$  (exponents  $\omega^{-\alpha}$ ) are different numbers. The class  $\mathcal{L}(j)$  of Section 9 is a landing spectrum, not the logarithmic defect  $L_j$ .

## 2 Foundations, normal forms, and the meaning of a sum

### 2.1 Surreal conventions

We use the sign-sequence realization of **No**: a surreal is a function from an ordinal to  $\{-, +\}$ . Its birthday  $b(x)$  is the domain of that function. The numerical order is the first-disagreement order, with  $- < \text{undefined} < +$ . Simplicity is proper prefix inclusion. Ordinals are identified with all-plus sequences. The distinction between an ordinal and its all-plus sign code will normally be suppressed, but all applications of  $j$  commute with that identification.

The Conway monomial map is denoted by

$$\Omega(a) = \omega^a.$$

Here  $\omega^a$  is the Conway omega-map, not an abbreviation for  $\exp(a \log \omega)$  with an arbitrarily chosen exponential structure. It satisfies  $\omega^{a+b} = \omega^a \omega^b$ , and is strictly increasing. Every nonzero surreal has a unique normal form

$$x = \sum_{\alpha < \theta} r_\alpha \omega^{a_\alpha}, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad a_\alpha > a_\beta \quad (\alpha < \beta < \theta), \quad (2.1)$$

where  $\theta$  is an ordinal. All supports in this article are sets, even though the full coefficient and exponent universes are classes. Write  $\text{len}(x) = \theta$  for the length of (2.1) and  $\text{supp}(x) = \{a_\alpha : \alpha < \theta\}$ . For  $x \neq 0$ , define

$$\text{LT}(x) = r_0 \omega^{a_0}, \quad v(x) = -a_0.$$

This is the additive natural valuation with values in  $(\mathbf{No}, +)$ ;  $v(0) = \infty$ . The infinitesimal ideal of the finite-surreal valuation ring is

$$\mathfrak{m} = \{0\} \cup \{x \in \mathbf{No} : v(x) > 0\}.$$

These conventions use decreasing omega-exponents; sources that use  $t^g$  with increasing support have the opposite sign convention for exponents.

**Definition 2.1** (Strong summability). A set-indexed family  $(x_i)_{i \in I}$  is *strongly summable* if  $\bigcup_{i \in I} \text{supp}(x_i)$  is reverse well ordered and each exponent occurs in the support of only finitely many members. Its strong sum is the unique normal form obtained by adding those finitely many coefficients at each exponent and discarding zero coefficients. A map is *strongly  $\mathbb{R}$ -linear* if it is  $\mathbb{R}$ -linear and preserves all such external set-indexed sums, including their summability.

The field laws for generalized series, in particular the finiteness of contributions in a product, are part of the Hahn–Neumann support theorem [2, 3, 5]. We use them for sets of exponents. There is no summation over a proper-class support.

*Remark 2.2* (Not order-topological convergence). A displayed strong sum is not defined as the supremum of its partial sums, and is not asserted to be an order-topological limit in all of **No**. For example, the sum  $h_\kappa$  constructed below exists by Definition 2.1. This makes no assertion that its partial sums converge against every positive surreal neighborhood. Confusing these two notions would invalidate the threshold argument.

## 2.2 Elementary embeddings and external sets

The standard large-cardinal facts used here are that the critical point of (1.1) is an uncountable inaccessible cardinal, that  $j$  fixes  $V_\kappa$ , and that a measurable cardinal supplies such an embedding by a normal ultrapower [6]. In particular,

$$j(\mathbb{R}) = \mathbb{R}, \quad j(r) = r \ (r \in \mathbb{R}), \quad \mathbb{R}^M = \mathbb{R}^V. \quad (2.2)$$

We take these large-cardinal facts as background, not as results of this article. When a normal measure  $U$  is chosen, the ultrapower is first formed on set functions with domain  $\kappa$  and then collapsed to its transitive class model. Countable completeness gives well-foundedness. All individual images and all set-indexed images used below are sets.

We distinguish  $j(I)$ , an element of  $M$ , from the external pointwise image  $j \text{ `` } I = \{j(i) : i \in I\}$ , formed in  $V$ . The latter need not belong to  $M$ . Every cardinality without a superscript is computed in  $V$ . In particular,  $|\text{supp}(x)| < \kappa$  always concerns the original external support. No cardinal arithmetic in  $M$  is silently substituted for it.

**06's class conventions and standing hypotheses [06].** The concrete large-cardinal statements concern a countably complete ultrafilter on a set and its well-founded ultrapower. They can be read in ZFC using definable classes. For statements about an arbitrary class embedding, work in GBC with the embedding as a class parameter, or assume the corresponding class-parameter Separation and Replacement instances. In particular, restrictions of  $j$  to sets and their ranges are sets. No object consisting of all class embeddings is formed. 06's standing hypotheses are

$$j : V \longrightarrow M \subseteq V, \quad M \text{ transitive}, \quad M \models \text{ZFC}, \quad \mathbf{On}^M = \mathbf{On}^V, \quad \text{crit}(j) = \kappa, \quad (2.3)$$

with  $j$  elementary for set theory; the canonical sign-sequence presentation identifies  $\mathbf{No}^M$  with a subclass of  $\mathbf{No}^V$ , and when  $j$  comes from an ultrapower its values are understood after the Mostowski collapse. For a function  $a : \lambda \rightarrow V$ ,

$$j(a)(j(\alpha)) = j(a(\alpha)) \quad (\alpha < \lambda), \quad (2.4)$$

but  $j(a)$  has values at every index below  $j(\lambda)$ , not just at indices in  $j \text{ `` } \lambda$ . Hypotheses (2.3) are those of (1.1), with the model-theoretic reading made explicit.

Where 07 takes the critical-point facts as background, 06 proves them; the proof is short and is printed here, since Section 9 uses the power-set clause.

**Lemma 2.3** (Critical point facts [06]). *Under (2.3), the cardinal  $\kappa$  is uncountable, regular, and strong limit. The embedding fixes every real number, and  $M$  and  $V$  have the same reals. Moreover  $\mathcal{P}(\kappa)^M = \mathcal{P}(\kappa)^V$ .*

*Proof.* Define the derived ultrafilter  $U_j = \{A \subseteq \kappa : \kappa \in j(A)\}$ . It is a set under the class conventions. Preservation of complements makes it an ultrafilter. It is nonprincipal since  $j(\alpha) = \alpha$  for  $\alpha < \kappa$ . For a family of fewer than  $\kappa$  members,  $j$  fixes the indexing ordinal, so  $\kappa$  belongs to the image of their intersection. Thus  $U_j$  is  $\kappa$ -complete. No set of size  $< \kappa$  belongs to it, by intersecting the complements of its singletons. The critical point cannot be finite or  $\omega$ , since these ordinals are fixed by their set-theoretic definitions.

A cofinal sequence in  $\kappa$  of length  $< \kappa$  would partition  $\kappa$  into fewer than  $\kappa$  bounded, hence null, sets, contradicting completeness. For the strong-limit property, suppose  $\mu < \kappa$  and choose distinct subsets  $A_\xi \subseteq \mu$  for  $\xi < \kappa$ . For each coordinate  $\eta < \mu$ , choose the  $U_j$ -large side of the partition of  $\kappa$  according to whether  $\eta \in A_\xi$ . Intersecting these  $\mu$  large sets yields a large set on which all  $A_\xi$  agree, contradicting their distinctness. Consequently  $2^\mu < \kappa$ . The same standard argument shows that  $\kappa$  is a cardinal; alternatively, a bijection from a smaller ordinal onto  $\kappa$  contradicts the definition of the critical point after applying  $j$ .

By induction on rank below  $\kappa$ , using inaccessibility and the fact that sets of size  $< \kappa$  have pointwise images,  $j$  fixes  $V_\kappa$  pointwise. In particular it fixes all real numbers. Every ambient real therefore belongs to  $M$ , and transitivity gives the other inclusion. Finally, for  $A \subseteq \kappa$ , the identity  $A = j(A) \cap \kappa$  puts  $A$  in  $M$ . Since every member of the internal power set is an actual subset, the two power sets agree.  $\square$

*Remark 2.4* (Why the target has all real coefficients [09]). A second route to the reals clause: a real can be coded by a subset of  $\omega$ , and  $j$  fixes that subset coordinate by coordinate since  $j(\omega) = \omega$ . Thus every real of  $V$  belongs to  $M$  and  $\mathbb{R}^M = \mathbb{R}^V$ , so Lemma 2.6 applies to the target. It does not assert that  $M$  contains every set of reals or every set of old surreals.

**Lemma 2.5** (The first missing index [06, 07, 09]). *For any set  $I$ ,*

$$j(I) = j \text{ `` } I \iff |I| < \kappa.$$

*For any ordinal  $\theta$ , define  $P_\theta = j(\theta) \setminus j \text{ `` } \theta$ . Then*

$$P_\theta = \emptyset \quad (\theta < \kappa), \quad \min P_\theta = \kappa \quad (\theta \geq \kappa).$$

*Proof.* Choose a bijection  $e : \mu \rightarrow I$ , where  $\mu$  is the cardinality of  $I$ . If  $\mu < \kappa$ , then  $j(\mu) = \mu$ , and every member of  $j(I)$  has the form  $j(e)(\alpha) = j(e(\alpha))$  for some  $\alpha < \mu$ . Conversely, suppose  $\mu \geq \kappa$ . The ordinal  $\kappa$  belongs to  $j(\mu)$ , but is not  $j(\alpha)$  for any ordinal  $\alpha$ : below  $\kappa$  all ordinals are fixed, while for  $\alpha \geq \kappa$  we have  $j(\alpha) \geq j(\kappa) > \kappa$ . Thus  $j(e)(\kappa)$  is in  $j(I)$  but not in  $j \text{ `` } I$ , by injectivity of  $j(e)$ .

For ordinal  $\theta < \kappa$ , every index is fixed. If  $\theta \geq \kappa$ , then  $j(\theta) > \kappa$ , all smaller indices are in  $j \text{ `` } \theta$ , and  $\kappa$  is not. This proves both assertions.  $\square$

06 (its Lemma “The first omitted index”) and 09 state and prove the ordinal clause with the same argument; the first clause, for an arbitrary set  $I$ , is 07’s.

### 2.3 The absoluteness needed to compare the two fields

There are two universe changes in play. Applying  $j$  moves an input into  $M$ . Then the identity inclusion regards the resulting old surreal sign sequence as a surreal of  $V$ . We need the standard constructions to agree under this second passage.

**Lemma 2.6** (Absolute surreal constructions [06, 07, 09]). *Let  $M \subseteq V$  be transitive models or inner classes of ZFC with the same ordinals and the same reals. On inputs that belong to  $M$ , the identity inclusion  $\mathbf{No}^M \subseteq \mathbf{No}^V$  preserves order, the field operations, the Conway omega-map, and canonical normal forms. An  $M$ -set strongly summable family in  $M$  is externally strongly summable in  $V$ , with the same sum.*

*Proof.* Order and prefix comparison are comparisons of ordinal-length sign functions, so they are absolute. The simplest separator of a given set cut is constructed by following the forced signs until the first permissible termination; this construction depends only on its input sign sequences. Thus the cut construction is absolute. Well-founded recursion on canonical sign options gives the same negation, addition, multiplication, and omega-map in both universes. Inverses then agree by uniqueness in the ordered field.

For normal forms we use the constructive sign-sequence version of the Conway–Gonshor normal-form theorem [2, Chapter 5], not just the existence of an unspecified isomorphism with a Hahn field. The conversion from a reverse well-ordered list of real-coefficient monomials to its surreal sign sequence is given by the normal-form sign expansion: ordinal sign concatenations with the prescribed deletions determined by the input exponent signs and preceding terms. The ordinal operations, restrictions, and sign comparisons in

that construction are absolute on a given set of inputs. Hence an  $M$ -normal form represents the same sign sequence in  $V$ . Uniqueness of the normal form in  $V$  shows that it acquires no extra terms on passing to  $V$ .

Finally, a reverse well ordering of an  $M$ -set has an ordinal order-type witness in  $M$ , which remains a witness in  $V$ ; finiteness of an old fiber is absolute. The old family is therefore still strongly summable. Its coefficientwise finite sums agree, as do the normal forms just considered.  $\square$

All three sources prove this lemma. 06's proof follows the same constructive route as the one above, adds that the sign-recursion algorithm and its ordinal stopping stage are absolute because the rank bound on the option sign lengths bounds the construction in both universes, and includes Gonshor's exponential in the statement: it is treated by its canonical option recursion with finite Taylor polynomials, the real constants and natural indices agreeing (this is Lemma 15.1). 06 states the lemma under (2.3), and 09 for a transitive inner class of ZFC containing all ordinals and all reals of  $V$ . 09's proof is genuinely different at the limit step and does not invoke the sign-expansion algorithm; it is printed as a second route.

*Second route* [09]. Sign comparison and sign restriction are absolute. Every sign prefix of an old sign sequence is old: it is obtained by restricting that sequence to an ordinal in  $M$ .

For the simplest-number operation, let a nonempty convex class be given by an absolute condition on old parameters and suppose that  $M$  has a member of that class. Its externally simplest member is an initial sign segment of every member, in particular of the internally simplest member. It therefore belongs to  $M$ . Membership is absolute, so internal minimality forces the two simplest members to coincide. Applied recursively to the canonical option sets, this proves absoluteness of addition, negation, multiplication and the omega-map. Inversion then follows from uniqueness of a multiplicative inverse. The relevant recursions use sign prefixes, finite arithmetic, and sets of options; their domains for any specified input are sets. For the omega-map one can use its defining cut with positive integers and their reciprocals, so there is no quantification over newly available coefficients.

For normal forms, let  $(r_\alpha, a_\alpha)_{\alpha < \theta} \in M$  be a normal-form sequence, and argue by induction on  $\theta$ . The zero and successor steps use finite field operations. Suppose that  $\theta$  is a nonzero limit and all proper partial sums  $p_\eta$  have already been identified in the two universes. Consider the convex class defined by

$$|z - p_{\alpha+1}| < \frac{\omega^{a_\alpha}}{n} \quad (\alpha < \theta, 1 \leq n < \omega). \quad (2.5)$$

Both universes have their own proposed full sum in this class. The condition is absolute on old  $z$  because its quantifiers range over the old ordinal  $\theta$  and the ordinary natural numbers, and every boundary in (2.5) is already absolute.

In either universe, a member of this class has the prescribed sequence as an initial segment of its normal form. To see this, compare its normal form with the prescribed sequence at the first exponent or coefficient at which they differ; the corresponding inequality in (2.5) fails. A discrepancy after all prescribed terms is allowed, but then the prescribed full sum is a normal-form truncation of that member. The classical truncation-simplicity theorem says that a proper normal-form truncation is simpler than the original surreal; see [14, Theorem 19.2(4)]. It follows that the full sum is the simplest member of (2.5) in each universe. The simplest-member argument of the first paragraph now identifies the two full sums. This completes the induction. This reasoning is also consistent with the usual recursive construction of normal-form sums, whose limit stage chooses a simplest approximant [14, Theorem 10.2].

For  $x \in M$ , take its normal-form sequence computed in  $M$ ; it sums externally to the same  $x$ , and uniqueness of normal forms gives the claim about extraction of coefficients and exponents. Finally, for a strongly summable family in  $M$ , its union of supports has in  $M$  an ordinal enumeration witnessing reverse well-ordering. This witness is still valid in  $V$ . The finitely many occurrences at each exponent and their

coefficient sums are absolute. The resulting normal-form sequence belongs to  $M$ , so its sum is absolute by the preceding argument.  $\square$

09 stresses the point of this route: it is not enough to say that normal forms are definable and therefore absolute, since general definable objects need not be absolute [09]. 06 adds that the lemma is a foundational interface, not a claim that the full Hahn fields of  $M$  and  $V$  coincide.

*Remark 2.7* (Old monomials, new sums [06]). An external sequence of old monomials may fail to be an element of  $M$ , and its sum may be a new surreal. That failure is precisely what the descent theorem of Section 9 detects.

*Remark 2.8* (Where a classical theorem is used). The normal-form sign-expansion theorem is a classical input to Lemma 2.6; it is not reproved from the axioms of surreal games here. This is the one place where an arbitrary Hahn-field identification would be insufficient: the identification must be the canonical Conway normal form. A formal verification should first prove this absoluteness lemma from that canonical construction, rather than assuming that every isomorphism chosen in  $M$  is absolute.

*Remark 2.9* (An instance for the definable-surreals report [merge]). Lemma 2.6 supplies the absoluteness that the definable-surreals report leaves unproved in its question on intermediate support fields (`dsn:q:support` in `docs/foundations-and-computation/definable-surreals-and-omnific-integers`). There, a proper subfield  $K \subsetneq \mathbf{No}$  with  $\omega \in K$ ,  $\omega^{-(\alpha+1)} \in K$  for every ordinal  $\alpha$ , and  $\text{supp}(a) \subseteq K$  for  $0 \neq a \in K$  (`dsn:eq:Khyp`) carries an **On**-indexed algebraically independent omnific family (`dsn:thm:amplify`), and the report records that  $\mathbf{No}^M$  for a proper same-ordinal inner model  $M$  “would satisfy” these hypotheses if the omega map and normal forms were absolute, while the universes report proves only the arithmetic and cuts (`univ:prop:absolute`). Let  $M$  be a transitive inner class of ZFC with  $\mathbf{On}^M = \mathbf{On}^V$  and  $\mathbb{R}^M = \mathbb{R}^V$ , and  $M \neq V$ . Then  $K = \mathbf{No}^M$  satisfies `dsn:eq:Khyp`: it is a subfield containing the ordinal  $\omega$  by the arithmetic part of the lemma; each  $\omega^{-(\alpha+1)}$  computed in  $M$  is the ambient one by the omega-map part; and the normal form of  $a \in \mathbf{No}^M$  computed in  $M$  is its ambient normal form, so its exponents are elements of  $\mathbf{No}^M$ . It is proper: a sign sequence of length  $\alpha$  codes a subset of  $\alpha$ , so  $\mathbf{No}^M = \mathbf{No}^V$  would put every set of ordinals of  $V$  into  $M$ , and a transitive inner model of ZFC containing every set of ordinals is all of  $V$ . Hence the support-subfield amplification applies to  $\mathbf{No}^M$ . The targets  $M$  of the embeddings (1.1) are instances (Lemma 2.3); for the ultrapower by a normal measure  $U$ ,  $M \neq V$  since  $U \notin M$ . Same-ordinal inner models without all reals, such as  $L$  when  $\mathbb{R} \not\subseteq L$ , and HOD when it lacks a real, are not covered: all three sources state the lemma for models with the same reals (06 through (2.3) and Lemma 2.3), and this report does not claim more.

**Proposition 2.10** (The elementary-induced surreal embedding [06, 07, 09]). *The map  $J : \mathbf{No} \rightarrow \mathbf{No}$ ,  $J(x) = j(x)$  on sign codes, is an ordered-field embedding that fixes  $\mathbb{R}$ , and*

$$J(\omega^a) = \omega^{J(a)}, \quad \mathbf{b}(J(x)) = j(\mathbf{b}(x)).$$

*It preserves and reflects prefix simplicity. It is elementary in the ordered-field language.*

*Proof.* Elementary transfer into  $M$  preserves the defining surreal constructions; Lemma 2.6 then identifies their values with the values in  $V$ . The all-plus coding of ordinals is definable, so it commutes with  $j$ . The domain of  $j(x)$  is  $j(\text{dom } x)$ , proving the birthday formula. Prefix simplicity is a definable statement about sign functions and is absolute after transfer. Fixing the reals follows from (2.2). An embedding between real-closed ordered fields is elementary by quantifier elimination for RCF. Applied formula by formula, the same argument works for these class fields: it makes no demand for a class-sized satisfaction set.  $\square$

06 (its Theorem “The two lift constructions”) and 09 (its Proposition “The set-induced action”) prove the same statement, and add two clauses: the range of  $J$  is contained in  $\mathbf{No}^M$ , and  $J$  maps **Oz** into **Oz**

(Theorem 11.1). 09 also records here the external normal form (5.1) of  $J(x)$ , whose domain is  $j(\theta)$ , not  $j \text{ “ } \theta$ , and notes that injectivity follows either from elementarity of equality or from strict order preservation [06, 09].

*Remark 2.11* (The first moved birthday [09]). Every surreal of birthday less than  $\kappa$  is fixed by  $J$ : its sign sequence has a fixed domain of length less than  $\kappa$  and fixed values. The ordinal  $\kappa$  itself is moved to  $j(\kappa)$ . Thus  $\kappa$  is also the least birthday of a surreal moved by  $J$ . This is a *fixed-point* statement about  $J$ , distinct from the *equalizer* statement about  $J$  and  $H_j$  below.

### 3 The field of short normal forms [09, 06]

This section needs no elementary embedding:  $\kappa$  is an arbitrary uncountable cardinal, and  $\mathcal{S}_{<\kappa}$  is defined by (1.2). 07 proves that  $\mathcal{S}_{<\kappa}$  is real closed through the two embeddings (Theorem 6.1); 09 proves it directly for every uncountable  $\kappa$ , regular or singular. Because  $\kappa$  is a cardinal,

$$\text{len}(x) < \kappa \iff |\text{supp}(x)| < \kappa, \quad (3.1)$$

since an ordinal of cardinality below  $\kappa$  is itself below  $\kappa$ . We use the “large monomial” convention: for a set-sized divisible ordered subgroup  $G \subseteq (\mathbf{No}, +)$ , the normal forms supported in  $G$  form the Hahn field  $\mathbb{R}((\omega^G))$  with reverse well-ordered supports; this is the increasing-support convention with  $t^g = \omega^{-g}$ . The standard Hahn-field theorem says that  $\mathbb{R}((\omega^G))$  is real closed when  $G$  is divisible [1, 2, 14].

**Theorem 3.1** (Short-support real closed field [09]). *For every uncountable cardinal  $\kappa$ , regular or singular,  $\mathcal{S}_{<\kappa}$  is a real closed ordered subfield of  $\mathbf{No}$ . It contains  $\mathbb{R}$ , every Conway monomial and every ordinal. It is closed under normal-form truncation, but is not a birthday cutoff  $\{x : \text{b}(x) < \lambda\}$  for any ordinal  $\lambda$ .*

*Proof.* Let  $S$  be the union of the supports of finitely many elements of  $\mathcal{S}_{<\kappa}$ . Then  $|S| < \kappa$ . The rational vector space  $G = \text{span}_{\mathbb{Q}}(S) \subseteq \mathbf{No}$  is a set-sized divisible ordered group, with  $|G| \leq \max(|S|, \aleph_0) < \kappa$ . Every Hahn series supported in  $G$  consequently belongs to  $\mathcal{S}_{<\kappa}$ . The selected elements lie in the real closed field  $\mathbb{R}((\omega^G))$ .

The fields so obtained form a directed family: replace two exponent groups by their rational span to include both. Their union is exactly  $\mathcal{S}_{<\kappa}$ . This proves closure under the field operations. Positive square roots and roots of odd-degree polynomials can be found in one such real closed subfield containing the finitely many coefficients. Thus the union is real closed. No union of  $\text{cf}(\kappa)$  unrelated supports was taken, so no regularity assumption has been used.

Real numbers and monomials have support of size at most one. The Cantor normal form of an ordinal is finite and agrees with its surreal normal form, so ordinals also belong to  $\mathcal{S}_{<\kappa}$ . Truncation only deletes support. Finally,  $\mathcal{S}_{<\kappa}$  contains ordinals of arbitrarily large birthday but omits a normal form of length  $\kappa$ , for instance  $h_\kappa$  of (7.1) (09 uses the omnific number  $Y_\kappa$  of Theorem 8.4). No ordinal birthday cutoff contains ordinals of arbitrarily large birthday, while the full class  $\mathbf{No}$  also contains the omitted number.  $\square$

**Corollary 3.2** (Truncation-closed short fields [06]). *Let  $\kappa$  be the critical point of (2.3). The class  $\mathcal{S}_{<\kappa}$  is exactly the equalizer of  $J$  and  $H_j$ , and it is a truncation-closed subfield containing every monomial. The class  $\mathcal{S}_{\leq\kappa} = \{x \in \mathbf{No} : |\text{supp}(x)| \leq \kappa\}$  is also a truncation-closed subfield.*

*Proof.* The equalizer assertion is Theorem 5.1, and an equalizer of field embeddings is a subfield. Direct support counting also proves both field statements. Unions and pairwise sums of fewer than  $\kappa$  support exponents still have size  $< \kappa$ . For an inverse write  $x = r\omega^a(1+\varepsilon)$  and use  $(1+\varepsilon)^{-1} = \sum_{n \geq 0} (-\varepsilon)^n$ . Its support lies in the additive monoid generated by  $\text{supp}(\varepsilon)$ , of cardinality at most  $\max(\aleph_0, |\text{supp}(\varepsilon)|)$ . Regularity and

uncountability give the strict bound. The same estimate gives the nonstrict bound  $\kappa$  for  $\mathcal{S}_{\leq \kappa}$ . Truncation only removes terms. Every monomial has one term.  $\square$

[merge] The counting proof uses only that  $\kappa$  is uncountable: the product of two cardinals below an infinite  $\kappa$  is below  $\kappa$ , and the monoid bound needs no regularity. So both field statements hold for every uncountable cardinal, in agreement with Theorem 3.1; 06 states them at the critical point, where they are needed.

For example,  $\omega^a$  belongs to the equalizer no matter how large the birthday or support of  $a$  is. The map need not fix this monomial: it sends it to  $\omega^{J(a)}$ . Equality of the two lifts is different from being a fixed point of either lift [06].

*Remark 3.3* (The countable boundary [09]). The finite-support collection  $\mathcal{S}_{< \omega}$  is not a field. For example,  $1 - \omega^{-1}$  has finite support, but its inverse is the strongly summable geometric series  $\sum_{n < \omega} \omega^{-n}$ . This explains the uncountability hypothesis in Theorem 3.1.

**Corollary 3.4** (Individual transcendence [09]). *If  $x \in \mathbf{No} \setminus \mathcal{S}_{< \kappa}$ , then  $x$  is transcendental over  $\mathcal{S}_{< \kappa}$ .*

*Proof.* A real closed field has no proper ordered algebraic extension. A surreal algebraic over  $\mathcal{S}_{< \kappa}$  would generate such an extension and must already belong to  $\mathcal{S}_{< \kappa}$ .  $\square$

The corollary is much weaker than algebraic independence of many specified long-support numbers. Two transcendental numbers can satisfy a polynomial relation. Section 13 proves the stronger assertion by an explicit coefficient-separation argument [09].

**Proposition 3.5** (Order-closed but not an order-dense approximation [09]). *For uncountable  $\kappa$ , the class  $\mathcal{S}_{< \kappa}$  is closed and nowhere dense in the order topology of  $\mathbf{No}$ . It is not definable, even with finitely many surreal parameters, in the pure ordered-field language.*

*Proof.* Take  $x \notin \mathcal{S}_{< \kappa}$  and its first  $\kappa$  normal-form terms  $r_\alpha \omega^{a_\alpha}$ . Choose an exponent  $b < a_\alpha$  for all  $\alpha < \kappa$ , using a cut below this set of exponents. If  $|y - x| < \omega^b$ , then every exponent in the support of  $y - x$  is at most  $b$ , and hence is smaller than every  $a_\alpha$ . Thus all the first  $\kappa$  terms of  $x$  remain unchanged in  $y$ , and  $y \notin \mathcal{S}_{< \kappa}$ . The complement is open.

To see that  $\mathcal{S}_{< \kappa}$  has no interior, fix a nonzero  $z \notin \mathcal{S}_{< \kappa}$ . If an interval about  $y \in \mathcal{S}_{< \kappa}$  of radius  $\varepsilon > 0$  were contained in  $\mathcal{S}_{< \kappa}$ , choose a positive ordinal  $\beta > |z|/\varepsilon$ . Then  $\beta \in \mathcal{S}_{< \kappa}$  and  $0 < |z/\beta| < \varepsilon$ , but  $y + z/\beta \notin \mathcal{S}_{< \kappa}$ : otherwise field closure would imply  $z \in \mathcal{S}_{< \kappa}$ . This is a contradiction. Closedness and empty interior give nowhere density.

Quantifier elimination for real closed fields makes every definable one-variable set a finite union of intervals and points [20]. The class  $\mathcal{S}_{< \kappa}$  is infinite and has no interior, so it cannot be such a set.  $\square$

The proposition explains why agreement on monomials plus order continuity does not force the two embeddings to agree everywhere (Proposition 8.5). Strong summability is a different extension principle from approximation in the order topology [09].

**Proposition 3.6** (Short-support integer part [09]). *For uncountable  $\kappa$ , the ring  $\mathcal{I}_{< \kappa} = \mathbf{Oz} \cap \mathcal{S}_{< \kappa}$  is an integer part of  $\mathcal{S}_{< \kappa}$ : it is discretely ordered and, for every  $x \in \mathcal{S}_{< \kappa}$ , there is  $z \in \mathcal{I}_{< \kappa}$  with  $z \leq x < z + 1$ .*

*Proof.* Write  $x = p + r + \varepsilon$ , where  $p$  is its positive-exponent part,  $r \in \mathbb{R}$  is its constant coefficient, and  $\varepsilon$  is its negative-exponent part. The last term is zero or infinitesimal. Put  $n = \lfloor r \rfloor$ , except that if  $r \in \mathbb{Z}$  and  $\varepsilon < 0$ , put  $n = r - 1$ . Then  $z = p + n$  has the required inequalities. Its support is a subset of  $\text{supp}(x)$  together with possibly zero, so  $z \in \mathcal{I}_{< \kappa}$ . The normal-form description (11.1) of  $\mathbf{Oz}$  shows closure under the ring operations.

A nonzero omnific integer is either an ordinary nonzero integer or has infinite magnitude, so 1 is the least positive one.  $\square$

*Warning 3.7* (Three unrelated bounds [09]). The conditions  $b(x) < \kappa$ ,  $|\text{supp}(x)| < \kappa$ , and “the support of  $x$  is bounded in an exponent group” are not interchangeable. The short-support field has no birthday bound and includes arbitrarily large exponents. In the main independence example (Theorem 13.3), all selected exponent supports lie in the order interval  $(0, 1]$ , while their cardinality is  $\kappa$ .

## 4 The strong companion

### 4.1 General Hahn reindexing

We record the standard mechanism in the form needed here. It does not use large cardinals.

**Lemma 4.1** (Reindexing exponents). *Let  $f : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$  be an additive order embedding. Then*

$$T_f \left( \sum_a r_a \omega^a \right) = \sum_a r_a \omega^{f(a)}$$

*defines a strongly  $\mathbb{R}$ -linear ordered-field embedding of  $\mathbf{No}$  into itself. Its image consists exactly of the surreal numbers whose exponent supports are contained in  $f(\mathbf{No})$ . It preserves and reflects the condition of being an omnific integer.*

*Proof.* An order embedding carries each reverse well-ordered set to a reverse well-ordered set of the same order type. Thus the displayed normal form exists and is unique. Coefficientwise addition and scalar multiplication commute with reindexing. For multiplication,  $f(a + b) = f(a) + f(b)$ , and injectivity shows that exactly the same finitely many contributing pairs occur at corresponding exponents. Hence  $T_f(xy) = T_f(x)T_f(y)$ . Also  $T_f(1) = 1$ , so inverses are preserved by the field equation  $T_f(x)T_f(x^{-1}) = 1$  when  $x \neq 0$ . A nonzero leading coefficient is unchanged; signs, strict order, and injectivity follow.

For a strongly summable family, the union of its supports and every finite coefficient fiber are simply reindexed by  $f$ . This proves strong  $\mathbb{R}$ -linearity for arbitrary external set-indexed families.

One inclusion in the image description is immediate. For the other, take a set support  $S \subseteq f(\mathbf{No})$ . Injectivity and class-parameter Replacement form the set of its unique inverse images. That set is reverse well ordered, and reindexing its series gives the desired preimage. No assertion is made that the inverse-image class of an arbitrary class is a set.

An omnific integer has only nonnegative exponents and has integer coefficient at exponent zero. An additive order embedding fixes zero, preserves and reflects nonnegativity, and does not change coefficients. This proves the last assertion.  $\square$

The preceding proof is the usual Hahn reindexing argument; compare [3, 5] and the exponent-embedding results described in the repository [8]. Its role is to produce a canonical comparison map, not to supply the critical-point obstruction.

**Definition 4.2** (Strong companion and defect). For the embedding in (1.1), let  $H_j = T_j$  and set

$$D_j(x) = J(x) - H_j(x), \quad \Gamma_j = J(\mathbf{No}).$$

The subgroup  $\Gamma_j$  is being used as an *exponent group* when it occurs inside a support condition. Write

$$\mathcal{H}(\Gamma_j) = \{y \in \mathbf{No} : \text{supp}(y) \subseteq \Gamma_j\}.$$

**Theorem 4.3** (Existence and uniqueness of the companion [06, 07, 09]). *The map  $H_j$  is a strongly  $\mathbb{R}$ -linear ordered-field embedding and*

$$H_j(\mathbf{No}) = \mathcal{H}(\Gamma_j). \quad (4.1)$$

*It is the unique strongly  $\mathbb{R}$ -linear map that agrees with  $J$  on every Conway monomial. The maps  $J$  and  $H_j$  agree on every real number and on every ordinal, as well as on every monomial.*

*Proof.* Apply Lemma 4.1 to the additive order embedding  $J$ . For any strongly  $\mathbb{R}$ -linear map with the stated monomial values, applying strong linearity to the normal form (2.1) forces (1.3), proving uniqueness.

Agreement on monomials follows from Proposition 2.10. Real coefficients are fixed. Every ordinal has a finite Cantor normal form  $\omega^{\beta_0}n_0 + \cdots + \omega^{\beta_{m-1}}n_{m-1}$ , which is also its Conway normal form. Finite additivity therefore gives agreement on every ordinal, irrespective of its size.  $\square$

06 (Theorem “The two lift constructions”) and 09 (Proposition “Canonical strong lift”) prove the same theorem with the same reindexing argument; 09 notes that Replacement for  $j$  makes the image of a support a set, and that the proof is the usual Hahn lifting argument, included to identify exactly which hypotheses are used. Its novelty is not claimed: the proposed contribution is its comparison with  $J$  [06, 09]. Both also note that  $H_j$  maps  $\mathbf{Oz}$  into  $\mathbf{Oz}$  (Lemma 4.1).

*Remark 4.4* (A crucial noncommutation). The formula for the companion is

$$H_j(\omega^a) = \omega^{J(a)},$$

not  $\omega^{H_j(a)}$  in general. In the former expression,  $J$  acts on an *exponent*; in  $H_j(x)$ , it reindexes the outer exponents of  $x$ . Failing to distinguish these two levels would incorrectly make the two embeddings identical.

## 5 The exact normal-form defect

### 5.1 The transformed index set

Fix a nonzero  $x$  with normal form (2.1). Regard the coefficient list  $r$  and the exponent list  $a$  as set functions with domain  $\theta$ . Put

$$r^* = j(r), \quad a^* = j(a), \quad P_\theta = j(\theta) \setminus j \text{ “ } \theta.$$

The transformed lists have domain  $j(\theta)$ . By elementary transfer and Lemma 2.6, the full *external* normal form of  $J(x)$  is

$$J(x) = \sum_{\beta < j(\theta)} r_\beta^* \omega^{a_\beta^*}. \quad (5.1)$$

For every  $\alpha < \theta$ ,

$$r_{j(\alpha)}^* = r_\alpha, \quad a_{j(\alpha)}^* = J(a_\alpha). \quad (5.2)$$

The extra indices in  $P_\theta$  will be called *missing-image indices*. They are ordinary set indices in  $V$ , not informal nonstandard positions.

**Theorem 5.1** (Exact defect and first term [06, 07, 09]). *For  $x$  as above,*

$$D_j(x) = \sum_{\beta \in P_\theta} r_\beta^* \omega^{a_\beta^*}. \quad (5.3)$$

*Every exponent occurring on the right lies outside  $\Gamma_j = J(\mathbf{No})$ . Consequently*

$$D_j(x) = 0 \iff \theta < \kappa \iff |\text{supp}(x)| < \kappa. \quad (5.4)$$

If  $\theta \geq \kappa$ , then

$$\text{LT}(D_j(x)) = r_\kappa^* \omega^{a_\kappa^*}, \quad v(D_j(x)) = -a_\kappa^*. \quad (5.5)$$

The same statements include  $x = 0$  by giving it empty support.

*Proof.* The normal form (5.1) is a strongly summable family. Split it into the subfamilies on  $j \prec \theta$  and  $P_\theta$ . Subfamilies of a strongly summable family remain strongly summable. By (5.2), the first subfamily sums to  $H_j(x)$ . Subtracting it leaves exactly (5.3).

Suppose, toward a contradiction, that  $a_\beta^* = J(b)$  for some  $\beta \in P_\theta$  and  $b \in \mathbf{No}$ . Then in  $M$ ,  $j(b) \in \text{ran}(j(a))$ . Elementarity gives  $b \in \text{ran}(a)$  in  $V$ , so  $b = a_\alpha$  for some  $\alpha < \theta$ . But then  $a_\beta^* = J(a_\alpha) = a_{j(\alpha)}^*$ . Since  $a^*$  is strictly decreasing, it is injective, and  $\beta = j(\alpha)$ , a contradiction. Thus all the missing-image exponents lie outside  $\Gamma_j$ .

All  $r_\beta^*$  are nonzero, and all  $a_\beta^*$  are distinct. There is therefore no cancellation between the terms of (5.3). The defect vanishes precisely when  $P_\theta$  is empty, which by Lemma 2.5 is precisely when  $\theta < \kappa$ . Since  $\kappa$  is an initial ordinal, the latter condition is equivalent to  $|\theta| < \kappa$ . The bijection  $\alpha \mapsto a_\alpha$  converts it into the last condition of (5.4).

For  $\theta \geq \kappa$ , the least missing-image index is  $\kappa$ . The exponent list is strictly decreasing, so its term at that index is the leading term of the defect, proving (5.5).  $\square$

This is 06's ‘‘missing-index formula’’ and ‘‘exact equalizer’’ and 09's ‘‘exact normal-form discrepancy’’, proved by all three sources by splitting the transformed normal form into its pointwise-image subseries and the complementary subseries. 09 adds that the complement of a subset of a well-ordered ordinal is still well ordered, so both subseries are legitimate Hahn sums read in the inherited increasing order of the indices. The exclusion of image exponents is 07's. All three sources stress that every coefficient of the transformed normal form is nonzero; otherwise a missing index could disappear and the equalizer conclusion would fail.

**Corollary 5.2** (Same leading scale, different tails [06, 07, 09]). *For every  $x \neq 0$ ,*

$$\text{LT}(J(x)) = \text{LT}(H_j(x)) = r_0 \omega^{J(a_0)}, \quad v(J(x)) = v(H_j(x)) = J(v(x)).$$

*If  $D_j(x) \neq 0$ , its valuation is strictly larger than this common value.*

*Proof.* Index zero is fixed and is never a missing-image index. The first missing index, when present, is later in a strictly decreasing exponent list.  $\square$

06 states the same as  $D_j(x) \prec H_j(x)$  and  $J(x)/H_j(x) \in 1 + o(1)$ , where  $y \prec z$  means  $|y| < c|z|$  for every positive real  $c$ ; 09 as (5.6) below. 09 adds a sharper form.

**Corollary 5.3** (Agreement to all earlier normal-form positions [09]). *If  $\text{len}(x) \geq \kappa$ , then  $J(x)$  and  $H_j(x)$  have the same first  $\kappa$  terms, but are not equal. The discrepancy is strictly dominated by every monomial occurring at one of those earlier positions:*

$$|D_j(x)| \prec \omega^{J(a_\alpha)} \quad (\alpha < \kappa).$$

*For all  $x \neq 0$ , the two images have the same leading term and*

$$\frac{J(x)}{H_j(x)} = 1 + \varepsilon_x, \quad \varepsilon_x = 0 \text{ or } |\varepsilon_x| < 1/n \text{ for every } n \geq 1. \quad (5.6)$$

*Proof.* Every index below  $\kappa$  is fixed and lies in the pointwise-image subseries. The first missing exponent lies strictly below all their exponents. Division by the common leading monomial establishes the stated Archimedean comparisons. If  $x$  has fewer than  $\kappa$  terms the images are equal; otherwise the same argument at index zero proves (5.6).  $\square$

This separates leading-scale information from exact Hahn information. Even the relative error is always infinitesimal, while the absolute discrepancy can itself be an infinite omnific integer (Theorem 8.4) [09]. The word “small” needs care here: the defect is infinitesimal *relative to the original value*; it need not itself be an infinitesimal. Multiplying a bounded witness by a large monomial can make its defect purely infinite without changing its leading coefficient [06].

**Warning 5.4** (Equalizer is not fixed field [09]). An element of  $\mathcal{S}_{<\kappa}$  need not be fixed by either map. For example,  $\kappa$  has finite Cantor normal form, so it belongs to  $\mathcal{S}_{<\kappa}$  and  $H_j(\kappa) = J(\kappa) = j(\kappa) \neq \kappa$ . The equalizer consists of inputs on which the two procedures agree, not inputs on which they are the identity.

**Proposition 5.5** (The fixed class of the Hahn lift [06]). *For a real or complex normal form  $x = \sum_{\alpha < \lambda} r_\alpha \omega^{a_\alpha}$ ,*

$$H_j(x) = x \iff J(a_\alpha) = a_\alpha \text{ for every } \alpha < \lambda.$$

*Thus the fixed class of the Hahn lift consists exactly of the set-supported series over the fixed exponent group  $\text{Fix}(J)$ .*

*Proof.* The two normal forms have the same length and the same coefficient array. Equality of their values forces equality of their unique ordered exponent arrays, giving necessity. Sufficiency is termwise. The fixed exponent group is an ordered subgroup because  $J$  preserves addition and negation.  $\square$

The measure codes  $h_A$  and  $\tilde{h}_A$  of Section 10 exemplify this observation: their exponents are fixed, even though transporting the entire code can detect the measure. There is no conflict with the equalizer theorem. “Fixed by  $H_j$ ” and “sent to the same value by  $J$  and  $H_j$ ” are different conditions [06]. (06 places this proposition in its surcomplex section; the complex case uses Section 12.)

**Corollary 5.6** (Exact omega-map compatibility locus [06, 07, 09]). *For every  $x \in \mathbf{No}$ ,*

$$H_j(\omega^x) = \omega^{H_j(x)} \iff x \in \mathcal{S}_{<\kappa}.$$

*Thus  $J$  preserves the omega-map globally but is not strongly additive, whereas  $H_j$  is strongly additive but does not preserve the omega-map globally.*

*Proof.* By Proposition 2.10 and the companion formula,  $H_j(\omega^x) = \omega^{J(x)}$ . The omega-map is injective, so the desired equality is equivalent to  $J(x) = H_j(x)$ . Apply Theorem 5.1. The existence of a long-support witness is established explicitly in Section 7.  $\square$

06 (its Corollary on compatibility with the monomial map) and 09 (the first part of its Theorem “Exact omega-equivariance locus of the lift”) prove the same criterion. 09 adds an incompatibility for the prescribed monomial action.

**Theorem 5.7** (Omega-map versus strong additivity [09]). *There is no ordered field embedding  $E : \mathbf{No} \rightarrow \mathbf{No}$  agreeing with  $J$  on all monomials that both commutes with the omega-map everywhere and preserves every strongly summable family of size  $\kappa$ .*

*Proof.* If  $E$  had the two stated properties, then for every  $x$ ,

$$\omega^{E(x)} = E(\omega^x) = J(\omega^x) = \omega^{J(x)},$$

so  $E = J$  by injectivity of the omega-map. The monomial family  $(\tau_\alpha)_{\alpha < \kappa}$  of Theorem 8.4 (or the family  $(\omega^{-\alpha})_{\alpha < \kappa}$  of Theorem 7.1) contradicts the asserted summation property.  $\square$

This is an incompatibility for a *specified* monomial action. It is not a theorem that all omega-preserving strongly additive embeddings are impossible; the identity is an immediate counterexample to that broader statement [09].

## 5.2 Why small support is not small birthday

Every monomial belongs to  $\mathcal{S}_{<\kappa}$ , however large its exponent or birthday. For example,  $\omega^b \in \mathcal{S}_{<\kappa}$  for every surreal  $b$ . This excludes any interpretation of (5.4) as agreement only on a set-sized initial segment of  $\mathbf{No}$ . Conversely, the simplest witness below has  $\kappa$  outer terms and falls outside the equalizer even though all its individual monomials are fixed by  $J$ . The cardinal bound applies to the number of terms, not to the size of a single exponent.

The distinction from the classical bounded birthday fields  $\mathbf{No}(\lambda)$  is substantial. Van den Dries–Ehrlich analyze those fields and their restricted analytic and exponential closures [3, Proposition 4.7 and Corollary 5.5]. We are not renaming those set-sized fields;  $\mathcal{S}_{<\kappa}$  is a proper class because it contains every monomial. Section 3 develops  $\mathcal{S}_{<\kappa}$  without the embedding, and Warning 3.7 separates it from birthday bounds and from order-bounded supports [09].

## 6 The equalizer field and the exact image intersection

**Theorem 6.1** (Equalizer and common image). *Let  $C_j = J(\mathcal{S}_{<\kappa}) = H_j(\mathcal{S}_{<\kappa})$ . Then*

$$\text{Eq}(J, H_j) = \mathcal{S}_{<\kappa}, \quad (6.1)$$

$$J(\mathbf{No}) \cap H_j(\mathbf{No}) = C_j, \quad (6.2)$$

$$C_j = \{y \in \mathbf{No} : |\text{supp}(y)| < \kappa \text{ and } \text{supp}(y) \subseteq \Gamma_j\}. \quad (6.3)$$

*Both  $\mathcal{S}_{<\kappa}$  and  $C_j$  are real-closed subfields of  $\mathbf{No}$ .*

*Proof.* The equalizer assertion is Theorem 5.1. Suppose  $y = J(x) \in H_j(\mathbf{No})$ . The image description (4.1) says that all exponents in the normal form of  $y$  lie in  $\Gamma_j$ . If  $x$  had at least  $\kappa$  normal-form terms, Theorem 5.1 would exhibit a nonzero term of  $J(x)$  with exponent outside  $\Gamma_j$ . Hence  $x \in \mathcal{S}_{<\kappa}$ , and  $y = J(x) = H_j(x) \in C_j$ . The opposite inclusion is immediate. Reindexing preserves the cardinality of support, and the inverse-support construction in Lemma 4.1 proves (6.3).

The equalizer of two field embeddings is a subfield, so  $\mathcal{S}_{<\kappa}$  is a subfield. To prove real closedness without relying on a separate bounded-support theorem, let  $0 \neq p \in \mathcal{S}_{<\kappa}[X]$ . If  $r_1 < \dots < r_m$  are all its distinct real roots in  $\mathbf{No}$ , then  $J(r_1) < \dots < J(r_m)$  and  $H_j(r_1) < \dots < H_j(r_m)$  are both the full ordered list of real roots of the same transformed polynomial. Indeed, the images of  $\mathbf{No}$  under these embeddings are real closed, and an ordered extension of a real-closed field cannot add a new real algebraic element. Each ordered position therefore agrees:  $J(r_i) = H_j(r_i)$ . Every such root lies in  $\mathcal{S}_{<\kappa}$ .

In particular, each positive member of  $\mathcal{S}_{<\kappa}$  has its positive square root in  $\mathcal{S}_{<\kappa}$ , and each odd-degree polynomial over  $\mathcal{S}_{<\kappa}$  has a root there. This characterizes real closedness. The isomorphic common image  $C_j$  is real closed as well.  $\square$

The real closedness of  $\mathcal{S}_{<\kappa}$  has a second, embedding-free proof, valid at every uncountable cardinal: Theorem 3.1 [09].

**Remark 6.2** (Domain equalizer versus image intersection). The equalizer is  $\mathcal{S}_{<\kappa}$ , whereas the intersection is its *image*  $C_j$ . The two are not being identified. An arbitrary short-support series can have exponents outside  $J(\mathbf{No})$ , and then it is not in  $C_j$ . In (6.3), both the support bound and the exponent-image condition are essential.

**Corollary 6.3** (Transcendental separation). *Every element of  $J(\mathbf{No}) \setminus C_j$  is transcendental over  $H_j(\mathbf{No})$ , and every element of  $H_j(\mathbf{No}) \setminus C_j$  is transcendental over  $J(\mathbf{No})$ . In particular, each is transcendental over  $C_j$ .*

*Proof.* Both image fields are real closed. An element of one outside their intersection is not in the other. A real-closed field has no proper algebraic ordered extension, so that element cannot be algebraic over the other field.  $\square$

The same argument over the domain field is 09's Corollary 3.4: every surreal outside  $\mathcal{S}_{<\kappa}$  is transcendental over  $\mathcal{S}_{<\kappa}$ .

## 6.1 A canonical isomorphism by erasing nonimage exponents

Define a coefficient projection on all surreals by

$$\Pi_j(y) = \sum_{a \in \text{supp}(y) \cap \Gamma_j} ([\omega^a] y) \omega^a. \quad (6.4)$$

This retains a subseries; it is not generally an initial normal-form truncation. Its support is a set, formed by Separation with the embedding parameter.

**Theorem 6.4** (Isomorphic immediate image fields). *The projection  $\Pi_j$  is strongly  $\mathbb{R}$ -linear and idempotent, with image  $H_j(\mathbf{No})$ , and*

$$\Pi_j \circ J = H_j. \quad (6.5)$$

*Its restriction is an ordered-field isomorphism*

$$\Pi_j \upharpoonright J(\mathbf{No}) : J(\mathbf{No}) \xrightarrow{\cong} H_j(\mathbf{No}) \quad (6.6)$$

*that fixes  $C_j$  pointwise and fixes every Conway monomial in either image. Both image fields are immediate valued extensions of  $C_j$ : they have the same value group  $\Gamma_j$  and residue field  $\mathbb{R}$ . The projection is not a field homomorphism on all of  $\mathbf{No}$ .*

*Proof.* Taking a subseries is compatible with coefficientwise finite sums, and commutes with an arbitrary strong sum: one either retains or deletes all contributions at each exponent. This proves strong  $\mathbb{R}$ -linearity. The image and idempotence follow from (4.1). The defect theorem states that  $J(x)$  consists of  $H_j(x)$  supported on  $\Gamma_j$ , plus a series supported outside  $\Gamma_j$ . Therefore  $\Pi_j(J(x)) = H_j(x)$ .

The maps  $J$  and  $H_j$  are ordered-field isomorphisms onto their respective images. Hence (6.5) identifies the restriction in (6.6) with  $H_j \circ J^{-1}$ , which is an ordered-field isomorphism. It fixes every element of the intersection, since those supports are contained in  $\Gamma_j$ . The monomials in either image are exactly  $\{\omega^{J(a)} : a \in \mathbf{No}\}$ , and they already lie in  $C_j$ . For the assertion about the monomials of  $J(\mathbf{No})$ , a monomial image must have normal-form length one and leading coefficient one, so its preimage also does.

Every nonzero member of either image has valuation in  $\Gamma_j$ , and the monomials just listed realize every value in  $\Gamma_j$ . They belong to  $C_j$ , so its value group is the same. All three fields contain the constant copy of  $\mathbb{R}$  and have residue field  $\mathbb{R}$ . Equality of these value groups and residue fields is precisely immediacy; the value groups here are allowed to be classes.

Finally choose  $a \notin \Gamma_j$ , for instance  $a = \kappa$ . Then also  $-a \notin \Gamma_j$ , and

$$\Pi_j(\omega^a) = \Pi_j(\omega^{-a}) = 0, \quad \Pi_j(\omega^a \omega^{-a}) = \Pi_j(1) = 1.$$

Thus global multiplicativity fails. The fact that multiplicativity holds on  $J(\mathbf{No})$  is a genuine restriction on the tails of that image field.  $\square$

This gives transcendence, not algebraic independence between the two image fields. Exact intersection alone does not establish linear disjointness; that stronger question is left open in Section 21.

## 7 Explicit witnesses and incomparability

For every ordinal  $\eta$ , define the genuine strong sum

$$h_\eta = \sum_{\alpha < \eta} \omega^{-\alpha}. \quad (7.1)$$

The set  $\{-\alpha : \alpha < \eta\}$  is strictly decreasing in its displayed order, and each exponent occurs once, so this is a legitimate normal form. We always use surreal negation and subtraction in exponents; they are not ordinal subtraction.

**Theorem 7.1** (The basic witness). *For  $h_\kappa$  one has*

$$H_j(h_\kappa) = h_\kappa, \quad (7.2)$$

$$J(h_\kappa) = h_{j(\kappa)}, \quad (7.3)$$

$$D_j(h_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{-\beta} = \omega^{-\kappa}(1 + \varepsilon), \quad 0 < \varepsilon \in \mathfrak{m}. \quad (7.4)$$

Moreover,

$$h_\kappa \in H_j(\mathbf{No}) \setminus J(\mathbf{No}), \quad h_{j(\kappa)} \in J(\mathbf{No}) \setminus H_j(\mathbf{No}).$$

Thus  $J(\mathbf{No})$  and  $H_j(\mathbf{No})$  are incomparable real-closed subfields of  $\mathbf{No}$ , although the embeddings agree on all reals, ordinals, and monomials.

*Proof.* Each  $\alpha < \kappa$  is fixed, so  $J(-\alpha) = -\alpha$ , and (7.2) follows. The definition of the whole list in (7.1) transfers to the list of length  $j(\kappa)$ , giving (7.3). Their difference is (7.4). Factoring out its first monomial leaves 1 plus a nonempty series with strictly negative exponents and positive coefficients. That remainder is a positive infinitesimal. It is nonempty because  $j(\kappa)$  is a limit cardinal strictly above  $\kappa$ .

If  $h_\kappa = J(x)$ , then the length of its normal form is  $j(\text{len}(x)) = \kappa$ , contradicting that  $\kappa$  is not in the ordinal range of  $j$ . Hence  $h_\kappa$  is not in  $J(\mathbf{No})$ . On the other hand, the normal form of  $h_{j(\kappa)}$  contains the exponent  $-\kappa$ . This exponent is not  $J(b)$  for any surreal  $b$ : if it were,  $J(-b) = \kappa$ , and elementarity would make  $-b$  an ordinal, again contradicting the missing-index fact. The image description for  $H_j$  therefore excludes  $h_{j(\kappa)}$  from its image.  $\square$

**Example 7.2** (A short-support element outside the common image). By Proposition 2.10,  $J(\kappa) = j(\kappa) > \kappa$ . Consequently  $\omega^{-\kappa}$  is not in  $C_j$ : its exponent is outside  $\Gamma_j$ . This is a one-term surreal, so it nevertheless belongs to  $\mathcal{S}_{<\kappa}$ . It concretely illustrates Remark 6.2. The two assertions “ $x$  has short support” and “ $x$  lies in the common image” really are different assertions.

*Remark 7.3* (Not a geometric series [06]). The expression  $h_\lambda$  is a transfinite Hahn sum. It must not be simplified as though every ordinal exponent were a natural multiple of 1. In particular the ordinary geometric-series identity for  $\sum_{n < \omega} \omega^{-n}$  does not identify  $h_\kappa$  with that countable sum. Surreal addition of ordinal exponents is not ordinary ordinal addition.

**Proposition 7.4** (Both maps are proper [06]). *Both  $J$  and  $H_j$  are proper embeddings of  $\mathbf{No}^V$  into itself.*

*Proof.* The normal-form length of  $J(x)$  is  $j(\text{len}(x))$ . No ordinal has image  $\kappa$ , by Lemma 2.5; hence  $h_\kappa$  is not in the range of  $J$ . Every exponent in the support of an  $H_j$ -image belongs to  $J(\mathbf{No})$ . The monomial  $\omega^{h_\kappa}$  therefore cannot be in the range of  $H_j$ . More explicitly, injectivity on exponent supports would force a preimage to have one term, with exponent  $a$  satisfying  $J(a) = h_\kappa$ , which is impossible.  $\square$

The first half is contained in Theorem 7.1, whose second half uses the witness  $h_{j(\kappa)}$  instead of  $\omega^{h_\kappa}$ . Thus  $J$ , although preserving the monomial map and exponentiation, is not an automorphism. These results do not answer the automorphism classification in Question 5.6 of [5]. Nor does an elementary embedding of sets into a proper inner model contradict a theorem excluding a nontrivial elementary embedding of the universe into itself [06].

## 8 The exact strong-summation threshold

The previous results compare the images of a *single normal form*. There is a separate, equally sharp statement about an arbitrary family whose individual members may themselves have very long supports.

**Theorem 8.1** (Positive-family threshold [06, 07, 09]). *Let  $F : I \rightarrow \mathbf{No}$  be an external set-indexed strongly summable family. Then  $(J(F(i)))_{i \in I}$  is strongly summable. If  $|I| < \kappa$ , then*

$$J\left(\sum_{i \in I} F(i)\right) = \sum_{i \in I} J(F(i)). \quad (8.1)$$

*If in addition  $F(i) > 0$  for every  $i \in I$ , then (8.1) holds if and only if  $|I| < \kappa$ . For  $|I| \geq \kappa$  in this positive case,*

$$J\left(\sum_{i \in I} F(i)\right) - \sum_{i \in I} J(F(i)) > 0. \quad (8.2)$$

*Proof.* The family  $j(F)$  is strongly summable over  $j(I)$  in  $M$ , and hence externally in  $V$  by Lemma 2.6. Its sum is  $J(\sum_{i \in I} F(i))$ . Restrict it to the old-index part  $j \restriction I$ : its values there are  $j(F)(j(i)) = J(F(i))$ . This proves external summability of the termwise image family. Splitting the full family yields

$$J\left(\sum_{i \in I} F(i)\right) - \sum_{i \in I} J(F(i)) = \sum_{u \in j(I) \setminus j \restriction I} j(F)(u). \quad (8.3)$$

If  $|I| < \kappa$ , the right side is empty by Lemma 2.5.

Suppose now that every  $F(i) > 0$ . Transfer gives  $j(F)(u) > 0$  for all  $u \in j(I)$ . A nonempty strongly summable family of positive surreals has positive sum: take the largest exponent in the union of its supports. Every term contributing at that exponent has it as a leading exponent, so its coefficient is positive; only finitely many contribute, and their sum is positive. When  $|I| \geq \kappa$ , the missing-index part is nonempty by Lemma 2.5. Applying this observation to (8.3) proves (8.2).  $\square$

**Remark 8.2** (Why the positivity hypothesis is necessary). Let  $I = \kappa \times \{0, 1\}$  and put  $F(\alpha, 0) = \omega^{-\alpha}$ ,  $F(\alpha, 1) = -\omega^{-\alpha}$ . This is strongly summable and has sum zero. The termwise image also sums to zero, so preservation holds even though  $|I| = \kappa$ . One cannot replace the positive-family conclusion by an unconditional cardinality criterion for signed families. The exact criterion in general is that the missing-index sum in (8.3) vanish.

**Corollary 8.3** (First failure cardinal [06, 07, 09]). *The least cardinality of a positive strongly summable family on which  $J$  fails to commute with strong summation is exactly  $\kappa$ .*

*Proof.* All smaller families are preserved, and the monomial family defining  $h_\kappa$  is a positive witness of size  $\kappa$ .  $\square$

06 (its strong-additivity threshold) and 09 (the first part of its “Exact threshold”) prove the small-family half by enumerating the family by an ordinal  $\mu < \kappa$ : the transported family has domain  $j(\mu) = \mu$  and term  $J(x_\alpha)$  at each old index, and Lemma 2.6 turns the internal identity into the external one. There is no restriction on the support of any individual member. Both give the witness  $h_\kappa$  (06’s  $u_\kappa$ , 09’s  $Q_\kappa$ ): the coefficient at exponent  $-\kappa$  is one in  $J(h_\kappa)$  and zero in  $\sum_{\alpha < \kappa} J(\omega^{-\alpha})$ .

There is no conflict between this theorem and the defect of an individual long-support  $x$ . A singleton family  $\{x\}$  is always preserved by  $J$ . Its normal-form decomposition may have  $\kappa$  or more terms, and that is the different family on which  $J$  and  $H_j$  disagree.

### 8.1 An omnific monomial witness [09]

For every ordinal  $\alpha$  put

$$q_\alpha = \omega^{-\alpha}, \quad \tau_\alpha = \omega^{q_\alpha} = \omega^{\omega^{-\alpha}}. \quad (8.4)$$

Both  $(q_\alpha)$  and  $(\tau_\alpha)$  are strictly decreasing monomial families on every ordinal domain. Each  $q_\alpha$  is positive, so every  $\tau_\alpha$  is larger than all positive real numbers. In particular,  $\tau_\alpha \in \mathbf{Oz}$ . For an ordinal  $\theta$ , define

$$Y_\theta = \sum_{\alpha < \theta} \tau_\alpha. \quad (8.5)$$

It is a strong sum of length  $\theta$  and is purely infinite whenever  $\theta > 0$ . Here “purely infinite” means that its normal form has only positive exponents, not that its sign must be positive for every possible choice of coefficients;  $Y_\theta$  is positive because all its coefficients equal one.

**Theorem 8.4** (Explicit threshold witnesses [09]). *The map  $J$  does not preserve all strongly summable families of cardinality  $\kappa$ . More explicitly,*

$$J(h_\kappa) = h_{j(\kappa)}, \quad H_j(h_\kappa) = h_\kappa, \quad D_j(h_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{-\beta}, \quad (8.6)$$

$$J(Y_\kappa) = Y_{j(\kappa)}, \quad H_j(Y_\kappa) = Y_\kappa, \quad D_j(Y_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \tau_\beta. \quad (8.7)$$

*The leading discrepancies are respectively  $\omega^{-\kappa}$  and  $\tau_\kappa = \omega^{\omega^{-\kappa}}$ . Every individual summand in the original  $\kappa$ -families is fixed by  $J$ .*

*Proof.* The constructions (8.4) and (8.5) are uniform definable ordinal-indexed constructions. Hence  $J(h_\kappa) = h_{j(\kappa)}$  and  $J(Y_\kappa) = Y_{j(\kappa)}$ . For each  $\alpha < \kappa$  the ordinal  $\alpha$  is fixed, and the omega identity gives  $J(q_\alpha) = q_\alpha$  and  $J(\tau_\alpha) = \tau_\alpha$ . The strong lift therefore fixes both original sums. Subtraction leaves exactly the indicated tails, whose first indices are  $\kappa$ .  $\square$

The row (8.6) is 07’s Theorem 7.1. Its discrepancy is infinitesimal even in absolute magnitude. The discrepancy in (8.7) is instead an infinite omnific integer. It is still infinitesimal *relative* to the common leading term  $\tau_0 = \omega$ . This supplies two different geometric manifestations of the same missing-index mechanism.

For a cardinal  $\lambda$ , let “ $< \lambda$ -strong” mean preservation of every strongly summable family indexed by a set of size less than  $\lambda$ . Theorem 8.1 and Theorem 8.4 state that  $J$  is  $< \kappa$ -strong and is not  $< \lambda$ -strong for any cardinal  $\lambda > \kappa$ . They do not claim that every individual family of size  $\kappa$  fails; many do not [09] (compare the signed family of the remark after Theorem 8.1).

**Proposition 8.5** (Order continuity does not enforce strong additivity [09]). *Both  $J$  and  $H_j$  are continuous for the order topology on  $\mathbf{No}$ . Nevertheless,  $J$  is not strongly additive.*

*Proof.* The failure of strong additivity is Theorem 8.4. For continuity at zero, take any  $\delta > 0$  and choose a positive ordinal  $\beta > 1/\delta$ . Elementary embeddings are nondecreasing on ordinal magnitudes:  $j(\beta) \geq \beta$ , as follows by induction from order preservation. The ordinal  $\beta$  has finite normal form, so  $H_j(\beta) = J(\beta) = j(\beta)$ . For either map  $E \in \{J, H_j\}$  and  $|x| < 1/\beta$ , therefore,

$$|E(x)| < 1/E(\beta) \leq 1/\beta < \delta.$$

This proves continuity at zero; translation gives continuity everywhere. This is an interval-based statement about a class order, not an assertion that its topology is a set of open sets.  $\square$

## 8.2 An intrinsic closure theorem for the image

The threshold can be stated without first specifying a family in the domain of  $J$ . This strengthens the noncommutation result to an actual failure of membership in the image.

**Theorem 8.6** (Strong-sum closure and positive escape). *Let  $(y_i)_{i \in I}$  be an external strongly summable family in  $J(\mathbf{No})$ , and let  $x_i$  be the unique preimage with  $J(x_i) = y_i$ . Then  $(x_i)$  is strongly summable and*

$$\sum_{i \in I} y_i \in J(\mathbf{No}) \iff \sum_{i \in I} y_i = J\left(\sum_{i \in I} x_i\right). \quad (8.8)$$

*In particular,  $J(\mathbf{No})$  is closed under all external strongly summable families of cardinality less than  $\kappa$ . If every  $y_i > 0$ , then*

$$\sum_{i \in I} y_i \in J(\mathbf{No}) \iff |I| < \kappa. \quad (8.9)$$

*In contrast,  $H_j(\mathbf{No})$  is closed under every external set-indexed strong sum of its members.*

*Proof.* Replacement with the embedding parameter gives the set-indexed preimage family. For every  $a$ , the normal-form defect theorem implies

$$[\omega^{J(a)}] y_i = [\omega^a] x_i.$$

Thus  $J \left( \bigcup_i \text{supp}(x_i) \right)$  is a subset of the reverse well-ordered union  $\bigcup_i \text{supp}(y_i)$ . Pulling this suborder back by the order embedding  $J$  proves reverse well ordering of  $\bigcup_i \text{supp}(x_i)$ . An exponent  $a$  occurs in  $x_i$  exactly when  $J(a)$  occurs in  $y_i$ , so the coefficient fibers are finite as well. Hence the preimage family is strongly summable.

Let  $x = \sum_i x_i$  and  $y = \sum_i y_i$ . At every exponent  $J(a)$  the finite coefficient sums give

$$[\omega^{J(a)}] y = \sum_i [\omega^a] x_i = [\omega^a] x = [\omega^{J(a)}] J(x).$$

Therefore  $J(x) - y$  has support disjoint from  $\Gamma_j$ . But every nonzero element of  $J(\mathbf{No})$  has its leading exponent in  $\Gamma_j$ , by Corollary 5.2. If  $y \in J(\mathbf{No})$ , the difference  $J(x) - y$  is in that field and must consequently be zero. This proves the forward implication in (8.8); the reverse implication is immediate.

For  $|I| < \kappa$ , Theorem 8.1 gives equality, without requiring positive terms. If all  $y_i > 0$ , then all  $x_i > 0$ , and that theorem makes equality equivalent to  $|I| < \kappa$ . Finally, a strong sum of series supported on  $\Gamma_j$  is again supported on  $\Gamma_j$ ; (4.1) proves the closure assertion for  $H_j(\mathbf{No})$ .  $\square$

## 9 Descent of the companion to the target model [06]

The value  $H_j(x)$  always exists in the ambient field. The question of this section, which is 06's, is whether it exists in the target model. In a set-ultrapower the answer measures considerably more than the size of an arithmetic error: it detects closure under external sequences. In this section  $\lambda$  denotes the normal-form length of  $x$ , and by (5.1) and (5.2),

$$H_j(x) = \sum_{\alpha < \lambda} r_\alpha \omega^{J(a_\alpha)} = \sum_{\beta \in j^{\text{“}} \lambda} j(r)(\beta) \omega^{j(a)(\beta)}, \quad (9.1)$$

the latter sum read in increasing order of the index  $\beta$ . It exists externally because  $j$  preserves the exponent ordering. Only  $J$  is *a priori* valued in  $\mathbf{No}^M$ .

### 9.1 Landing is equivalent to the seed

**Theorem 9.1** (Universal landing criterion [06]). *Let  $x$  have normal-form length  $\lambda$ . Then*

$$H_j(x) \in \mathbf{No}^M \iff j^{\text{“}} \lambda \in M. \quad (9.2)$$

*In particular the answer depends only on the cardinality of the support, not on the actual coefficients or exponents. At a fixed support cardinality there are no exceptional normal forms that land when others do not.*

*Proof.* Suppose first that  $s = j^{\text{“}} \lambda$  belongs to  $M$ . The transported arrays  $j(r)$  and  $j(a)$  are in  $M$ . Restrict them to  $s$  inside  $M$ , using its increasing enumeration to obtain an ordinal-indexed normal form. The value of this normal form belongs to  $\mathbf{No}^M$  and, by Lemma 2.6, equals the external expression (9.1). It is  $H_j(x)$ .

Conversely suppose  $y = H_j(x)$  belongs to  $\mathbf{No}^M$ . Its normal form computed in  $M$  is its external normal form. In particular its exponent support  $S$  belongs to  $M$ . The array  $j(a)$  belongs to  $M$  and is injective on  $j(\lambda)$ . By Separation in  $M$ , the set

$$t = \{\beta < j(\lambda) : j(a)(\beta) \in S\}$$

belongs to  $M$ . Externally, the support formula (9.1) and injectivity of  $j(a)$  identify  $t$  exactly with  $j^{\text{“}} \lambda$ . This proves the converse, including the empty support case.

Finally, if  $f : \mu \rightarrow \lambda$  is a bijection, then  $j(f)$  maps  $j^{\text{“}} \mu$  bijectively onto  $j^{\text{“}} \lambda$ . Each seed is obtained from the other using  $j(f)$  or its inverse, both belonging to  $M$ . Thus seed membership, and hence landing, is unchanged when the support length is replaced by its cardinality.  $\square$

The forward implication is stronger than a statement about canonical 0–1 series alone. Even an elaborate normal form with arbitrary nonzero real coefficients cannot conceal a missing seed. Its exponents, compared with the transported original exponent array, recover that seed exactly [06].

**Corollary 9.2** (Landing spectrum [06]). *Define the class of cardinals*

$$\mathcal{L}(j) = \{\mu : j^{\text{“}} \mu \in M\}.$$

*It is downward closed and contains every cardinal  $\mu \leq \kappa$ . Moreover*

$$H_j^{-1}(\mathbf{No}^M) = \{x : |\text{supp}(x)| \in \mathcal{L}(j)\}$$

*is a truncation-closed subfield of  $\mathbf{No}^V$ .*

*Proof.* For ordinals  $\mu \leq \lambda$ ,  $j^{\text{“}} \mu = j^{\text{“}} \lambda \cap j(\mu)$ , giving downward closure. Below  $\kappa$  the seed equals the ordinal, and  $j^{\text{“}} \kappa = \kappa$ , so all these seeds belong to  $M$ . The displayed equality follows from Theorem 9.1. The inverse image of a subfield under a field embedding is a subfield. Truncation decreases the support cardinality.  $\square$

## 9.2 Why the seed measures sequence closure in an ultrapower

The next lemma is a standard ultrapower fact, included with its proof because it explains precisely what is being imported from set theory. It is stated as Fact 7 in [12].

**Lemma 9.3** (Seed–closure equivalence [06]). *Suppose  $j : V \rightarrow M$  is the well-founded ultrapower by a countably complete ultrafilter on a set  $I$ . For any ordinal  $\lambda$ ,*

$$j \text{ “ } \lambda \in M \iff {}^\lambda M \cap V \subseteq M. \quad (9.3)$$

*Here the right side means that every ambient set function with domain  $\lambda$  and values in  $M$  belongs to  $M$ .*

*Proof.* If  $M$  is closed under such sequences, the sequence  $\langle j(\alpha) : \alpha < \lambda \rangle$  belongs to  $M$ , so its range  $j \text{ “ } \lambda$  does too. This direction does not require the ultrapower hypothesis.

For the other direction, let  $s = j \text{ “ } \lambda \in M$  and take an ambient sequence  $\langle x_\alpha : \alpha < \lambda \rangle$  of elements of  $M$ . Choose representatives  $f_\alpha : I \rightarrow V$  with  $x_\alpha = [f_\alpha]_U$ , where brackets include the transitive collapse. This choice is legitimate for a set of indices: choose bounded-rank representatives first and then apply ordinary Choice. Pack them into a single set function  $F$  with  $F(\alpha, i) = f_\alpha(i)$ . Let  $d = [\text{id}_I]_U \in M$  be the usual ultrapower seed for  $I$ . The evaluation identity for ultrapowers gives

$$j(F)(j(\alpha), d) = x_\alpha.$$

The increasing enumeration  $e : \lambda \rightarrow s$  belongs to  $M$ ; its order type is the actual ordinal  $\lambda$ , by absoluteness of well-orders. Inside  $M$  form

$$\alpha \mapsto j(F)(e(\alpha), d).$$

This is exactly the original ambient sequence. Hence it belongs to  $M$ .  $\square$

**Corollary 9.4** (One number tests all sequences [06]). *For a set-ultrapower embedding and a surreal  $x$  with support cardinal  $\mu$ ,*

$$H_j(x) \in \mathbf{No}^M \iff {}^\mu M \cap V \subseteq M.$$

*Thus descent of one such normal form implies descent of every surreal whose support has at most  $\mu$  terms.*

*Proof.* Combine Theorem 9.1, Lemma 9.3 and Corollary 9.2.  $\square$

The word “one” refers to a test relative to the specified embedding and its target. It does not mean that a single pure-field formula with no access to  $j$ ,  $M$ , or normal-form data defines supercompactness [06].

## 9.3 Normal measures: the exact three-way classification

**Lemma 9.5** (Cover obstruction [06]). *Let  $j : V \rightarrow M$  be a set-ultrapower on an index set  $I$  with critical point  $\kappa$ . Let  $\lambda \geq \kappa$  be an ambient cardinal. Suppose that  $s \in M$  satisfies*

$$j \text{ “ } \lambda \subseteq s \subseteq j(\lambda), \quad M \models |s| < j(\kappa).$$

*Then*

$$\lambda \leq |I| \cdot \kappa. \quad (9.4)$$

*Proof.* Represent  $s$  by a function  $f : I \rightarrow V$ . By the ultrapower theorem, the assertion  $s \in j(P_\kappa(\lambda))$  implies that  $f(i) \in P_\kappa(\lambda)$  on a measure-one set. Change  $f$  to the empty set outside that set. For every  $\alpha < \lambda$ , membership  $j(\alpha) \in s$  says

$$\{i \in I : \alpha \in f(i)\} \in U.$$

In particular this set is nonempty. Consequently  $\lambda = \bigcup_{i \in I} f(i)$ . Each  $f(i)$  has cardinality  $< \kappa$ , so the cardinality of the union is at most  $|I| \cdot \kappa$ .  $\square$

This elementary counting argument is useful because it does not confuse an internal support bound with an external one. An internally short cover can be externally large, but not too large for its representing index set [06].

**Lemma 9.6** (The image of the critical point is large [06]). *Under (2.3),  $j(\kappa) > (2^\kappa)^V$ .*

*Proof.* The ordinal  $j(\kappa)$  is a strong-limit cardinal in  $M$  by Lemma 2.3 and elementarity. Let  $\delta = (2^\kappa)^M$ . The set  $\mathcal{P}(\kappa)$  is the same in both models by Lemma 2.3, so a bijection in  $M$  from  $\delta$  onto that power set is an actual bijection in  $V$ . Thus  $\delta \geq (2^\kappa)^V$  as ordinals. Since  $M$  regards  $\kappa < j(\kappa)$  and  $j(\kappa)$  as strong limit, it has  $\delta < j(\kappa)$ . Combine these inequalities.  $\square$

**Theorem 9.7** (Normal-measure trichotomy [06]). *Suppose  $U$  is a normal nonprincipal  $\kappa$ -complete ultrafilter on  $\kappa$  and  $j : V \rightarrow M = \text{Ult}(V, U)$  is its transitive ultrapower. For every surreal  $x$ ,*

$$H_j(x) \in \mathbf{No}^M \iff |\text{supp}(x)| \leq \kappa. \quad (9.5)$$

*More precisely:*

| Support cardinal | Comparison         | Descent of $H_j(x)$                |
|------------------|--------------------|------------------------------------|
| $< \kappa$       | $J(x) = H_j(x)$    | belongs to $\mathbf{No}^M$         |
| $= \kappa$       | $J(x) \neq H_j(x)$ | belongs to $\mathbf{No}^M$         |
| $> \kappa$       | $J(x) \neq H_j(x)$ | does not belong to $\mathbf{No}^M$ |

*In all three rows  $J(x)$  belongs to  $\mathbf{No}^M$ .*

*Proof.* By Corollary 9.2, every cardinal at most  $\kappa$  belongs to the landing spectrum. We prove that the ambient successor  $\kappa^+$  does not. Suppose to the contrary that  $s = j \restriction \kappa^+ \in M$ . As an ordered set of ordinals,  $s$  has order type  $\kappa^+$ , and this order type is absolute to  $M$ . By Lemma 9.6,

$$\kappa^+ \leq (2^\kappa)^V < j(\kappa).$$

The increasing enumeration therefore witnesses in  $M$  that  $s$  has cardinality  $< j(\kappa)$ . Apply Lemma 9.5 with  $I = \kappa$  and  $\lambda = \kappa^+$ . It gives  $\kappa^+ \leq \kappa \cdot \kappa = \kappa$ , a contradiction.

Downward closure now excludes every larger cardinal. The universal landing criterion proves (9.5), and the exact equalizer theorem gives the three rows.  $\square$

**Remark 9.8** (Slightly stronger scope [06]). Normality is not needed in the counting proof of Theorem 9.7. The same conclusion holds for a well-founded ultrapower on an index set of cardinality  $\kappa$  with critical point  $\kappa$ . The normal-measure formulation is retained because it also identifies the recovered measure in Section 10 with the original ultrafilter.

**Corollary 9.9** (Landing domain versus equalizer [06]). *For a normal-measure ultrapower on  $\kappa$ , the maximal domain on which the Hahn lift lands in the target is exactly  $\mathcal{S}_{\leq \kappa}$ , whereas the equalizer of the two lifts is exactly  $\mathcal{S}_{< \kappa}$ . The target is closed under ambient  $\kappa$ -sequences but not under all ambient  $\kappa^+$ -sequences.*

*Proof.* Use Theorem 9.7, Theorem 5.1 and Lemma 9.3.  $\square$

No assertion that  $j(\kappa)$  is inaccessible in  $V$  was used. It is inaccessible in  $M$ , and that is exactly the property needed for the internal support bound. These two assertions should not be interchanged [06].

## 10 Recovering a normal measure from one coefficient

### 10.1 Boolean masks at ordinal exponents

For each  $A \subseteq \kappa$ , define

$$h_A = \sum_{\alpha \in A} \omega^{-\alpha}. \quad (10.1)$$

The notation is unambiguous: the index set is ordered by the ordinal order, and zeros are omitted from the full  $\kappa$ -list. In particular,  $h_\emptyset = 0$  and  $h_\kappa$  agrees with (7.1).

**Theorem 10.1** (The measure-coefficient formula). *For every  $A \subseteq \kappa$ ,*

$$H_j(h_A) = h_A, \quad (10.2)$$

$$J(h_A) = \sum_{\beta \in j(A)} \omega^{-\beta}, \quad (10.3)$$

$$[\omega^{-\kappa}] D_j(h_A) = \mathbf{1}_{\{\kappa \in j(A)\}}. \quad (10.4)$$

*Consequently*

$$U_j = \{A \subseteq \kappa : [\omega^{-\kappa}] D_j(h_A) = 1\} \quad (10.5)$$

*is the nonprincipal normal  $\kappa$ -complete ultrafilter on  $\kappa$  derived from  $j$ .*

*Proof.* The original exponents  $-\alpha$ ,  $\alpha < \kappa$ , are fixed by  $J$ , so (10.2) follows from the companion formula. Transferring the whole mask construction gives (10.3). Since  $\kappa$  is not an original index,  $h_A$  has no coefficient at  $\omega^{-\kappa}$ . The coefficient in  $J(h_A)$  is 1 precisely when  $\kappa \in j(A)$ . This proves (10.4) and identifies (10.5) with the classical derived measure.

For completeness, its properties can be read directly from this last identification. Since  $\kappa \in j(\kappa)$ , the whole space is in  $U_j$  and the empty set is not. Elementary transfer of complements and finite intersections makes it an ultrafilter. For  $\alpha < \kappa$ ,  $j(\{\alpha\}) = \{\alpha\}$  does not contain  $\kappa$ , so it is nonprincipal. If  $\mu < \kappa$  and  $A_\xi \in U_j$  for  $\xi < \mu$ , then  $j(\mu) = \mu$  and

$$\kappa \in \bigcap_{\xi < \mu} j(A_\xi) = j\left(\bigcap_{\xi < \mu} A_\xi\right),$$

which proves  $\kappa$ -completeness.

Finally, suppose  $A \in U_j$  and  $f : A \rightarrow \kappa$  is regressive, meaning  $f(\alpha) < \alpha$  for all nonzero  $\alpha \in A$ . Then  $j(f)(\kappa) < \kappa$ , so  $j(f)(\kappa) = \eta = j(\eta)$  for some  $\eta < \kappa$ . Therefore

$$\kappa \in j(\{\alpha \in A : f(\alpha) = \eta\}).$$

The indicated fiber is in  $U_j$ , proving normality.  $\square$

The ultrafilter properties just proved are standard properties of derived measures [6]. The new information is their simultaneous realization by (10.4) within one explicitly given surreal coefficient space. No real-valued probability measure on all surreal sets is being introduced.

All three sources recover the derived normal measure from one coefficient, with the same proof of its ultrafilter, completeness and normality properties (06 in Lemma 2.3 and its recovery theorem, 09 in its “Normal-measure coefficient formula”). They use three different codes. 07’s  $h_A$  has coefficients 0, 1 and is bounded. 06’s code has coefficients 1, 2, so that its support has exactly  $\kappa$  terms for every  $A$ ; 09’s code is omnific from the start. We print the two further codes with the computations that differ.

**Theorem 10.2** (A bounded code of constant length [06]). *For  $A \subseteq \kappa$  put*

$$\tilde{h}_A = h_\kappa + h_A = \sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha)) \omega^{-\alpha}. \quad (10.6)$$

*Every coefficient is one or two, so the support has exactly  $\kappa$  terms, independently of  $A$ , and  $1 < \tilde{h}_A < 3$ . For any embedding satisfying (2.3),  $U_j$  is a normal measure on  $\kappa$ ,  $H_j(\tilde{h}_A) = \tilde{h}_A$ , and*

$$[\omega^{-\kappa}] D_j(\tilde{h}_A) = 1 + \mathbf{1}_{U_j}(A), \quad (10.7)$$

$$\text{st}(\omega^\kappa D_j(\tilde{h}_A)) = 1 + \mathbf{1}_{U_j}(A), \quad (10.8)$$

*where st is the standard part of a finite surreal. If  $j$  is the ultrapower by a normal measure  $U$  on  $\kappa$ , then  $U_j = U$ .*

*Proof.* The bounds hold because the tail after the constant term is a nonzero positive infinitesimal and the constant term is at most two. The measure properties are those proved in Theorem 10.1 (06 proves them in Lemma 2.3 and by the same regressive-function argument). All exponents  $-\alpha$  with  $\alpha < \kappa$  and all coefficients in (10.6) are fixed by  $J$ , so its Hahn lift is  $\tilde{h}_A$  itself. The transported whole code has normal form

$$J(\tilde{h}_A) = \sum_{\beta < j(\kappa)} (1 + \mathbf{1}_{j(A)}(\beta)) \omega^{-\beta}.$$

The first  $\kappa$  terms are exactly  $\tilde{h}_A$ . Its first remaining coefficient, at  $-\kappa$ , is two if  $\kappa \in j(A)$  and one otherwise. This proves (10.7). Multiplication by  $\omega^\kappa$  shifts that exponent to zero; all later exponents become negative. The standard part therefore extracts the same coefficient, proving (10.8).

For a normal ultrapower on  $\kappa$ , the class of the identity function is the ordinal  $\kappa$ . Indeed every smaller ultrapower ordinal has a regressive representative on a large set and hence is constant modulo  $U$ , whereas every constant below  $\kappa$  is smaller than  $[\text{id}]_U$ . Thus  $\kappa \in j(A)$  if and only if  $A \in U$ , which gives  $U_j = U$ .  $\square$

The same seed calculation is the step “the measure derived at the seed  $\kappa$  is the original measure” in the proof of Corollary 10.4; 09 cites it from [17].

**Theorem 10.3** (An omnific code [09]). *For  $A \subseteq \kappa$  define the omnific mask*

$$Y_A = \sum_{\alpha \in A} \omega^{q_\alpha} = \sum_{\alpha \in A} \tau_\alpha, \quad (10.9)$$

*with  $q_\alpha$  and  $\tau_\alpha$  as in (8.4). Its support is  $\{q_\alpha : \alpha \in A\}$  and its coefficients are all one; for  $A = \kappa$  it is  $Y_\kappa$  of (8.5). Then*

$$[\omega^{q_\kappa}] J(Y_A) = [\omega^{q_\kappa}] D_j(Y_A) = \begin{cases} 1, & A \in U_j, \\ 0, & A \notin U_j, \end{cases} \quad (10.10)$$

*where  $[\omega^{q_\kappa}] x$  is the coefficient of  $\tau_\kappa = \omega^{q_\kappa}$  in  $x$ .*

*Proof.* Uniform definability of the mask construction and Lemma 2.6 give

$$J(Y_A) = \sum_{\beta \in j(A)} \tau_\beta.$$

For  $\alpha < \kappa$  all the exponents  $q_\alpha$  are fixed by  $J$ , so  $H_j(Y_A) = Y_A$ . Since  $Y_A$  has no term  $\tau_\kappa$ , subtracting it does not change the coefficient at  $\tau_\kappa$ . That coefficient is one exactly when  $\kappa \in j(A)$ .  $\square$

This gives a faithful numerical presentation of a normal measure once the set-induced embedding is available. It is not a proof that an arbitrary field embedding produces a normal measure. The set-theoretic origin and its action on every subset code are indispensable hypotheses [09].

**Corollary 10.4** (Distinguishing normal measures [06, 07]). *Suppose  $U$  and  $W$  are distinct normal measures on the same  $\kappa$ , with normal ultrapower embeddings  $j_U$  and  $j_W$ . Their coefficient functionals*

$$A \mapsto [\omega^{-\kappa}] D_{j_U}(h_A), \quad A \mapsto [\omega^{-\kappa}] D_{j_W}(h_A)$$

*are distinct. Thus the first failure cardinal does not determine the coefficient-level defect data.*

*Proof.* Choose  $A$  in exactly one of  $U, W$ . In the normal ultrapower construction, the measure derived at the seed  $\kappa$  is the original measure. Apply Theorem 10.1.  $\square$

06 proves the same with its bounded codes, and adds that the Hahn lifts fix those codes pointwise.

**Corollary 10.5** (Faithful bounded coding [06]). *Distinct normal measures on the same  $\kappa$  induce different transport maps on the set  $\{\tilde{h}_A : A \subseteq \kappa\} \subset (1, 3)$ . Every associated Hahn lift fixes that entire set pointwise.*

*Proof.* Choose  $A$  which belongs to one measure and not to the other. The coefficient in (10.7), and hence the coefficient of the transported code at  $-\kappa$ , differs. The fixed-point assertion was proved in Theorem 10.2.  $\square$

This recovers the ultrafilter from a coefficient observable rather than by recovering all of set theory from birthday data. The coefficient extraction and the monomial scale are specified additional structure; neither is claimed definable in the pure field language [06].

*Remark 10.6* (A zero probe does not mean a zero defect [07, 09]). If  $A \subseteq \kappa$  has cardinality  $\kappa$  but does not belong to  $U_j$ , then  $D_j(h_A) \neq 0$  by the exact support theorem, although its coefficient at  $\omega^{-\kappa}$  is zero. Such sets exist: partition  $\kappa$  into two sets of size  $\kappa$ , at most one of which is in the ultrafilter. The first missing *normal-form index* is always  $\kappa$  for a long support, but its associated *exponent* depends on the exponent list. The fixed mask probe need not be the leading nonzero term of every mask defect. 09 makes the same warning for its masks: if  $A$  has size  $\kappa$  and  $A \notin U_j$ , then  $[\omega^{\kappa}] D_j(Y_A) = 0$  but  $D_j(Y_A) \neq 0$ ; other additional exponents remain, and the distinguished coefficient records measure membership, not the entirety of the support discrepancy.

## 10.2 Real-valued masks and the Hadamard product

For  $f : \kappa \rightarrow \mathbb{R}$ , set

$$h_f = \sum_{\alpha < \kappa} f(\alpha) \omega^{-\alpha}. \quad (10.11)$$

This includes zero coefficients with the convention that they are suppressed in the canonical normal form. Let  $\mathcal{B}_\kappa$  denote the set of all these masks. Give it its coefficientwise, or Hadamard, product

$$h_f \odot h_g = h_{fg}, \quad (fg)(\alpha) = f(\alpha)g(\alpha). \quad (10.12)$$

Together with ordinary addition and real scalar multiplication, this makes  $\mathcal{B}_\kappa$  an  $\mathbb{R}$ -algebra isomorphic to  $\mathbb{R}^\kappa$ . Its multiplicative unit is  $h_\kappa = h_1$ , not the surreal number 1. The map from scalars into this algebra is  $r \mapsto rh_\kappa$ .

**Theorem 10.7** (Ultrafilter evaluation as a defect coefficient [07, 09]). *The functional*

$$\chi_j(h_f) = [\omega^{-\kappa}] (J(h_f) - H_j(h_f)) \quad (10.13)$$

*satisfies*

$$\chi_j(h_f) = j(f)(\kappa). \quad (10.14)$$

*It is a unital  $\mathbb{R}$ -algebra homomorphism  $(\mathcal{B}_\kappa, +, \odot) \rightarrow \mathbb{R}$ . Its value is the unique  $r \in \mathbb{R}$  for which  $f^{-1}(\{r\}) \in U_j$ .*

*Proof.* All original exponents are fixed, so  $H_j(h_f) = h_f$ . The whole transformed mask is

$$J(h_f) = \sum_{\beta < j(\kappa)} j(f)(\beta) \omega^{-\beta}.$$

Reading its coefficient at  $-\kappa$  proves (10.14). The value lies in  $\mathbb{R}$ , since  $j(f)$  takes values in  $j(\mathbb{R}) = \mathbb{R}$ . If  $r = j(f)(\kappa)$ , then  $j(r) = r$  and

$$\kappa \in j(f^{-1}(\{r\})),$$

which gives the asserted fiber. Two distinct fibers cannot both belong to an ultrafilter. Transferring pointwise addition and multiplication gives

$$j(f + g)(\kappa) = j(f)(\kappa) + j(g)(\kappa), \quad j(fg)(\kappa) = j(f)(\kappa)j(g)(\kappa).$$

Constant functions take their constant real value. These are exactly the claimed algebra-homomorphism laws under (10.12).  $\square$

09 proves the same evaluation for weighted omnific masks, with a different proof that the fiber exists.

**Proposition 10.8** (Weighted omnific evaluation [09]). *For  $f : \kappa \rightarrow \mathbb{R}$  put*

$$Y_f = \sum_{\alpha < \kappa} f(\alpha) \tau_\alpha, \quad (10.15)$$

*with zero coefficients omitted. The map  $f \mapsto Y_f$  is an injective  $\mathbb{R}$ -linear map into the purely infinite part of  $\mathbf{Oz}$ , and*

$$[\omega^{q\kappa}] J(Y_f) = j(f)(\kappa). \quad (10.16)$$

*There is a unique  $r \in \mathbb{R}$  such that  $f^{-1}(\{r\}) \in U_j$ , and the right side of (10.16) equals that  $r$ . Hence this coefficient is the character of  $\mathbb{R}^\kappa$  associated with  $U_j$ , when  $\mathbb{R}^\kappa$  has pointwise addition and multiplication.*

*Proof.* The normal-form coefficient at  $\tau_\kappa$  after applying  $j$  to (10.15) is exactly  $j(f)(\kappa)$ , including the possibility that it is zero. Since  $\kappa$  is measurable (by Lemma 2.3, or by Theorem 10.1),  $|\mathbb{R}| < \kappa$ . The partition of  $\kappa$  into the nonempty fibers of  $f$  has fewer than  $\kappa$  parts. If no fiber belonged to  $U_j$ , the intersection of their complements would belong to  $U_j$  by completeness and would be empty. At most one fiber can belong to an ultrafilter, proving existence and uniqueness. Applying  $j$  to that fiber identity gives  $j(f)(\kappa) = j(r) = r$ . Values of sums and pointwise products are therefore the sums and products of these selected real values, either directly by  $j$  or by intersecting the corresponding large fibers.  $\square$

The image of  $\mathbb{R}^\kappa$  in (10.15) is naturally an algebra under coefficientwise, or Hadamard, multiplication  $Y_f \odot Y_g = Y_{fg}$ . It is *not* a subalgebra under ordinary surreal multiplication. For masks,  $Y_A \odot Y_B = Y_{A \cap B}$ , but in general  $Y_A Y_B \neq Y_{A \cap B}$ : the latter product adds exponents and creates terms  $\omega^{q\alpha + q\beta}$ . Thus the ultrafilter interpretation uses explicitly specified coefficient operations; it cannot be smuggled into the ordinary ring language by treating masks as idempotents. A field has no nontrivial ordinary idempotents [09].

**Product warning.** The character theorem concerns the Hadamard product  $\odot$ . It does not say that coefficient extraction at  $\omega^{-\kappa}$  is multiplicative for ordinary surreal multiplication. Ordinary multiplication convolves exponents; Hadamard multiplication multiplies coefficients at matching exponents.

Infinite direct-product homomorphisms and their ultrafilter factorizations are a classical subject; see, for example, Bergman [7]. The next result records an exact large-cardinal criterion in this particular mask presentation. Its underlying ultrafilter characterization is not claimed to be new.

**Theorem 10.9** (A mask criterion for measurability). *Let  $\kappa$  be an uncountable cardinal, without initially assuming an embedding  $j$ . The following are equivalent.*

- (i)  $\kappa$  is measurable.
- (ii) *There is a unital  $\mathbb{R}$ -algebra homomorphism  $\chi : (\mathcal{B}_\kappa, +, \odot) \rightarrow \mathbb{R}$  such that  $\chi(h_{\{\alpha\}}) = 0$  for every  $\alpha < \kappa$ , and for every pairwise disjoint family  $(A_\xi)_{\xi < \mu}$  with  $\mu < \kappa$ ,*

$$\chi\left(h_{\bigcup_{\xi < \mu} A_\xi}\right) = \sum_{\xi < \mu} \chi(h_{A_\xi}). \quad (10.17)$$

*The scalar sum in (10.17) has at most one nonzero term, so no infinite real summation convention is needed. Under measurability,  $\chi$  can be chosen to be the defect functional (10.13) of a normal ultrapower embedding.*

*Proof.* Assume (i) and choose a normal measure and its ultrapower. The functional from Theorem 10.7 vanishes on singleton masks. Its value on each indicator mask is 0 or 1 according to membership in the measure. For fewer than  $\kappa$  disjoint sets, their union belongs to a  $\kappa$ -complete ultrafilter exactly when one of them belongs to it: otherwise all their complements have intersection in the ultrafilter. This proves (10.17).

Conversely, assume (ii). Each  $h_A$  is a Hadamard idempotent, so  $\chi(h_A)^2 = \chi(h_A)$  and hence  $\chi(h_A) \in \{0, 1\}$ . Define  $U = \{A \subseteq \kappa : \chi(h_A) = 1\}$ . The identities

$$h_A \odot h_B = h_{A \cap B}, \quad h_{\kappa \setminus A} = h_\kappa - h_A$$

show that  $U$  is an ultrafilter. For example, upward closure follows from  $h_A \odot h_B = h_A$  when  $A \subseteq B$ , and exactly one of a set and its complement has value 1. The singleton hypothesis makes it nonprincipal.

To verify  $\kappa$ -completeness, let  $A_\xi \in U$  for  $\xi < \mu < \kappa$ . Set  $B_\xi = \kappa \setminus A_\xi$ , and disjointify the family by

$$C_\xi = B_\xi \setminus \bigcup_{\eta < \xi} B_\eta.$$

Each  $C_\xi \subseteq B_\xi$  has value zero: multiplying its idempotent by  $h_{B_\xi}$  leaves it unchanged. By (10.17),  $\bigcup_{\xi < \mu} C_\xi = \bigcup_{\xi < \mu} B_\xi$  also has value zero. Its complement  $\bigcap_{\xi < \mu} A_\xi$  is therefore in  $U$ . Thus  $U$  is a nonprincipal  $\kappa$ -complete ultrafilter on  $\kappa$ , which is the usual measure characterization of measurability.  $\square$

This criterion detects measurability only because the mask algebra and its small-union condition explicitly retain a  $\kappa$ -indexed Boolean algebra. It is not a characterization of measurability in the pure ordered-field language of **No**.

### 10.3 A compatible-field-embedding criterion [09]

09 gives a second criterion, of a different form: its witness is an ordered field embedding of all of **No**, not a functional on a mask algebra, and its extra operation is a Hadamard product on all surreals. For arbitrary surreals define

$$x \odot y = \sum_{\alpha \in \text{supp}(x) \cap \text{supp}(y)} [\omega^\alpha] x [\omega^\alpha] y \omega^\alpha.$$

The intersection is reverse well-ordered, so this operation is always defined. It restricts to the operation on the masks  $h_f$  and  $Y_f$  introduced above.

**Theorem 10.10** (Compatible-field-embedding criterion for measurability [09]). *Fix an uncountable cardinal  $\kappa$ . In a class framework with Separation and Replacement for the maps in question, the following are equivalent.*

- (i) *The cardinal  $\kappa$  is measurable.*
- (ii) *There is an ordered field embedding  $E : \mathbf{No} \rightarrow \mathbf{No}$  fixing  $\mathbb{R}$  and commuting with Conway's omega-map such that:*
  - (a)  $E(\tau_\alpha) = \tau_\alpha$  for every  $\alpha < \kappa$ ;
  - (b)  $E$  preserves every strongly summable family of cardinality less than  $\kappa$ ;
  - (c) for all  $A, B \subseteq \kappa$ ,  $E(Y_{A \cap B}) = E(Y_A) \odot E(Y_B)$ ;
  - (d)  $[\omega^{\kappa}] E(Y_\kappa) = 1$ .

*In the forward direction the witness  $E$  can be chosen definable from a normal measure. In the reverse direction no elementary embedding of set universes is assumed.*

*Proof.* Assume measurability, choose a normal measure and take  $E = J$ . Theorem 8.1 and Theorem 8.4 give (a) and (b), and (8.7) gives (d). Hadamard multiplication is definable from normal forms and real coefficient multiplication. Its value for old inputs is absolute by Lemma 2.6; thus  $J$  preserves it. Since  $Y_A \odot Y_B = Y_{A \cap B}$ , condition (c) follows.

Conversely, let  $E$  satisfy (ii) and put

$$\chi_E(A) = [\omega^{\kappa}] E(Y_A), \quad U_E = \{A \subseteq \kappa : \chi_E(A) = 1\}.$$

Separation for the graph of  $E$  makes  $U_E$  a set. Taking  $A = B$  in (c) shows  $\chi_E(A)^2 = \chi_E(A)$ . As a real idempotent,  $\chi_E(A)$  is zero or one. Condition (d) gives  $\chi_E(\kappa) = 1$ , and additivity of the field embedding, applied to  $Y_A + Y_{\kappa \setminus A} = Y_\kappa$ , gives  $\chi_E(A) + \chi_E(\kappa \setminus A) = 1$ . Condition (c) further gives  $\chi_E(A \cap B) = \chi_E(A) \chi_E(B)$ . These identities show that  $U_E$  is an ultrafilter. Its singleton values are zero by (a), since  $\tau_\alpha$  has no coefficient at  $\tau_\kappa$  for  $\alpha < \kappa$ . Thus  $U_E$  is nonprincipal.

To prove completeness, take  $\nu < \kappa$  and sets  $A_\xi \in U_E$  for  $\xi < \nu$ . Put  $B_\xi = \kappa \setminus A_\xi$  and disjointify them:

$$C_\xi = B_\xi \setminus \bigcup_{\eta < \xi} B_\eta.$$

Each  $C_\xi \subseteq B_\xi$  has  $\chi_E(C_\xi) = 0$  by the ultrafilter identities. The family  $(Y_{C_\xi})_{\xi < \nu}$  is strongly summable: its support union is a subset of the fixed decreasing scale  $(q_\alpha)_{\alpha < \kappa}$  and each exponent occurs at most once. Its sum is  $Y_B$ , where  $B = \bigcup_{\xi < \nu} B_\xi$ . By (b) its image family is strongly summable and sums to  $E(Y_B)$ . Taking the coefficient at  $\tau_\kappa$  gives zero, since every summand has zero there and coefficient extraction of a strong sum is a finite coefficient calculation. Consequently  $\chi_E(B) = 0$ , so  $\bigcap_{\xi < \nu} A_\xi \in U_E$ .

We have produced a nonprincipal  $\kappa$ -complete ultrafilter on  $\kappa$ , which is exactly measurability. The extra ordered-field and omega-map compatibility in (ii) was not needed for this converse, but the forward construction shows that it can be required simultaneously.  $\square$

The implication from the coefficient identities to a measure is the classical ultrafilter criterion expressed in normal-form coordinates. The simultaneous compatibility with ordinary surreal field arithmetic, the omega-map, and all small strong sums is supplied by the large-cardinal construction. The theorem is an infinitary criterion with a specified coefficient operation and all subset masks, not a first-order criterion in the bare field or omnific-ring language. Determining which of these extra requirements can be removed is Question 21.22 [09]. [merge] Theorem 10.9 and Theorem 10.10 share the ultrafilter argument (idempotents, disjointification, small unions); they differ in what carries it, a character of an abstract mask algebra (07) or an ordered field embedding of **No** compatible with  $\odot$  on masks (09). Neither removes the coefficient algebra.

## 11 Omnific integers and a purely infinite measure code

### 11.1 The ring and its short-support part

The omnific integers form the discretely ordered subring

$$\mathbf{Oz} = \left\{ \sum_a r_a \omega^a : \text{supp}(x) \subseteq \mathbf{No}^{\geq 0}, \quad r_0 \in \mathbb{Z} \right\}. \quad (11.1)$$

A missing zero exponent has coefficient zero. Nonzero coefficients at positive exponents are arbitrary reals, not necessarily ordinary integers. This is Conway's normal-form description [1, 2]. We use it rather than an informal criterion such as “having no fractional part,” which can hide a convention at exponent zero. Put

$$\mathcal{J}_{<\kappa} = \mathbf{Oz} \cap \mathcal{S}_{<\kappa}.$$

**Theorem 11.1** (Omnific equalizer, intersection, and fractions [06, 07, 09]). *The maps  $J$  and  $H_j$  preserve and reflect membership in  $\mathbf{Oz}$ , and*

$$\{z \in \mathbf{Oz} : J(z) = H_j(z)\} = \mathcal{J}_{<\kappa}, \quad (11.2)$$

$$J(\mathbf{Oz}) \cap H_j(\mathbf{Oz}) = J(\mathcal{J}_{<\kappa}) = H_j(\mathcal{J}_{<\kappa}), \quad (11.3)$$

$$\text{Frac}(\mathcal{J}_{<\kappa}) = \mathcal{S}_{<\kappa}, \quad (11.4)$$

$$\text{Frac}(J(\mathbf{Oz}) \cap H_j(\mathbf{Oz})) = C_j. \quad (11.5)$$

Also  $\text{ct}(J(z)) = \text{ct}(H_j(z)) = \text{ct}(z)$  for every  $z \in \mathbf{Oz}$ .

*Proof.* For  $H_j$  this preservation and reflection is part of Lemma 4.1. For  $J$ , transfer the canonical normal form and the condition in (11.1). Nonnegativity of all exponents and the integer condition on the constant coefficient are absolute; equivalently, apply elementarity to the normal-form description and then Lemma 2.6. A coefficient at exponent zero is fixed, and a new missing-image exponent cannot equal zero because  $0 \in \Gamma_j$ . The same reasoning gives the constant-term identity for both maps.

The equalizer formula is Theorem 5.1 restricted to  $\mathbf{Oz}$ . If  $y \in J(\mathbf{Oz}) \cap H_j(\mathbf{Oz})$ , write  $y = J(z)$  with  $z \in \mathbf{Oz}$ . The field intersection theorem forces  $z \in \mathcal{S}_{<\kappa}$ , so  $z \in \mathcal{J}_{<\kappa}$ . Conversely, both maps have the same value on any  $z \in \mathcal{J}_{<\kappa}$ . This proves (11.3).

To prove (11.4), take nonzero  $x \in \mathcal{S}_{<\kappa}$ . Its support is a set, so choose  $d > 0$  such that  $d > -a$  for every  $a \in \text{supp}(x)$ . The monomial  $q = \omega^d$  and the shifted series  $p = \omega^d x$  have only strictly positive exponents.

Their constant coefficients vanish, and both are in  $\mathbf{Oz}$ . Their supports have cardinality less than  $\kappa$ , so  $p, q \in \mathcal{J}_{<\kappa}$  and  $x = p/q$ . The reverse inclusion follows because  $\mathcal{S}_{<\kappa}$  is a field containing  $\mathcal{J}_{<\kappa}$ . Applying the field embedding  $J$  gives (11.5).  $\square$

Preservation of  $\mathbf{Oz}$  and the equalizer (11.2) are also proved by 06 (Theorem 11.4) and 09 (its “Restricted equalizers”). The constant-term identity is 09’s Proposition “Constant coefficients survive”, stated for all  $x \in \mathbf{No}$ :  $[\omega^0] J(x) = [\omega^0] H_j(x) = [\omega^0] x$ , so  $D_j(z)$  has constant coefficient zero for every  $z \in \mathbf{Oz}$ . For  $J$ , coefficient extraction is absolute by Lemma 2.6 and definable, so its value is the image of the fixed real  $[\omega^0] x$ ; for  $H_j$ , injectivity of the exponent map and  $J(0) = 0$  show that only the old exponent zero can contribute [09]. It is also the case  $a = 0$  of Proposition 14.1. For arbitrary surreals the constant-coefficient operation is not a field homomorphism: positive and negative exponents can multiply to zero. On  $\mathbf{Oz}$ , whose exponents are nonnegative, it is a ring homomorphism to  $\mathbb{Z}$ . The large discrepancies here therefore lie in its purely infinite kernel. This does not conflict with claims about maps to set-sized rings in the repository: the embeddings in this article have proper-class codomains [09].

## 11.2 A defect with no finite residue

For  $A \subseteq \kappa$ , define

$$z_A = \omega^\kappa h_A = \sum_{\alpha \in A} \omega^{\kappa - \alpha}. \quad (11.6)$$

Every exponent is strictly positive, since  $\alpha < \kappa$ . Thus  $z_A$  is a purely infinite omnific integer when  $A$  is nonempty, and is zero otherwise. Here and below  $\kappa - \alpha$  is the difference in the surreal ordered field.

**Theorem 11.2** (Omnific recovery of the measure). *For  $A \subseteq \kappa$ ,*

$$D_j(z_A) = \omega^{j(\kappa)} D_j(h_A), \quad (11.7)$$

$$[\omega^{j(\kappa) - \kappa}] D_j(z_A) = \mathbf{1}_{\{A \in U_j\}}, \quad (11.8)$$

$$\text{ct}(D_j(z_A)) = 0. \quad (11.9)$$

In particular, for  $z_\kappa = \omega^\kappa h_\kappa$ ,

$$D_j(z_\kappa) = \sum_{\kappa \leq \beta < j(\kappa)} \omega^{j(\kappa) - \beta} > 0, \quad \text{LT}(D_j(z_\kappa)) = \omega^{j(\kappa) - \kappa}.$$

Every nonzero term of every  $D_j(z_A)$  is purely infinite.

*Proof.* The two embeddings agree on  $\omega^\kappa$ , with common value  $\omega^{j(\kappa)}$ , and both are multiplicative. This proves (11.7). Multiplication by that monomial shifts exponents by  $j(\kappa)$ , so (11.8) follows from (10.4). Moreover  $j(A) \cap \kappa = A$ , so the defect of  $h_A$  has its possible indices only in  $j(A) \setminus A \subseteq [\kappa, j(\kappa))$ . After shifting, all exponents  $j(\kappa) - \beta$  are strictly positive, which proves (11.9) and the last assertion. Taking  $A = \kappa$  gives the explicit nonempty series and its leading term.  $\square$

The constant-term map therefore loses this large-cardinal information, even though one specified infinite-scale coefficient recovers it. This is compatible with the repository’s emphasis on constant-term factorizations in omnific arithmetic [11]; it does not supply a new factorization theorem for set-sized quotient rings.

06 shifts its bounded code the same way. 09’s code  $Y_A$  of Theorem 10.3 is omnific from the start, with exponents  $q_\alpha \in (0, 1]$  instead of  $\kappa - \alpha$ .

**Corollary 11.3** (Omnific measure codes [06]). *The elements*

$$\tilde{z}_A = \omega^\kappa \tilde{h}_A = z_\kappa + z_A = \sum_{\alpha < \kappa} (1 + \mathbf{1}_A(\alpha)) \omega^{\kappa - \alpha} \quad (11.10)$$

are purely infinite omnific integers. Their transport defects satisfy

$$[\omega^{j(\kappa) - \kappa}] D_j(\tilde{z}_A) = 1 + \mathbf{1}_{U_j}(A). \quad (11.11)$$

Thus the same measure is recoverable from the two induced maps on  $\mathbf{Oz}$ .

*Proof.* For  $\alpha < \kappa$ , the surreal difference  $\kappa - \alpha$  is positive. This proves membership in  $\mathbf{Oz}$ . Both embeddings send  $\omega^\kappa$  to  $\omega^{j(\kappa)}$ . Multiplicativity gives  $D_j(\tilde{z}_A) = \omega^{j(\kappa)} D_j(\tilde{h}_A)$ , and (11.11) follows by shifting the coefficient in (10.7). Preservation of the ring by both embeddings is Theorem 11.1.  $\square$

**Theorem 11.4** (Omnific restrictions [06]). *Both  $J$  and  $H_j$  preserve the ring  $\mathbf{Oz}$  of omnific integers and restrict to injective unital ring maps. Their surcomplex extensions preserve  $\mathbf{Oz}[i]$ . The equalizer and landing criteria remain exact on each of these rings, with support computed in its real or complex normal form. For a normal-measure ultrapower, all three support regimes of Theorem 9.7 occur already in  $\mathbf{Oz}$ .*

*Proof.* In the normal-form description (11.1),  $J$  carries the transported array of positive exponents to an array of positive exponents by elementarity and absoluteness. The unique constant coefficient, when present, is an ordinary integer and is fixed. Thus  $J(x)$  is omnific. The same argument for  $H_j$  is termwise: positivity of exponents and the integer constant term are preserved. Ring preservation and injectivity follow from the field-embedding theorem.

For  $\mathbf{Oz}[i]$  apply the real result to the two coordinates. The exact criteria follow by restriction of the real and complex theorems (Theorem 5.1, Theorem 9.1 and Theorem 12.3). For the three regimes, the omnific normal forms  $\zeta_\mu = \omega^\mu h_\mu = \sum_{\alpha < \mu} \omega^{\mu - \alpha}$  have support cardinality  $|\mu|$ . Take respectively a nonzero finite cardinal,  $\kappa$ , and  $\kappa^+$ .  $\square$

*Remark 11.5* (Preserving is not fixing [06]). Preserving  $\mathbf{Oz}$  does not mean preserving its elements pointwise. Nor is this a theorem that all endomorphisms of the omnific integers come from large cardinals. It constructs and classifies the behavior of a specified pair arising from an elementary universe embedding. The distinction is important in comparison with the repository's independent work on omnific automorphisms and set-sized quotients.

*Remark 11.6* (An  $\mathbf{Oz}$ -reflecting embedding that is not strong [merge]). The omnific-preserving-automorphisms report asks whether every field embedding  $f : \mathbf{No} \hookrightarrow \mathbf{No}$  with  $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$  must be strongly additive, and what changes if also  $\text{ct} \circ f = \text{ct}$  (`opa:as:q:embeddingstrong` in `docs/surreal/omnific-preserving-automorphisms`, where  $J$  is written  $\hat{j}$ ). If there is a measurable cardinal, or more generally an embedding as in (1.1) available as a class, then  $J$  answers both clauses negatively. It is an ordered field embedding fixing  $\mathbb{R}$  (Proposition 2.10); it preserves and reflects  $\mathbf{Oz}$ , and preserves constant terms (Theorem 11.1); it carries every strongly summable family to a strongly summable family, but the family  $(\omega^{-\alpha})_{\alpha < \kappa}$  has image sum  $h_\kappa \neq J(h_\kappa) = h_{j(\kappa)}$  (Theorem 8.1, Theorem 7.1; the omnific family  $(\tau_\alpha)_{\alpha < \kappa}$  of Theorem 8.4 does the same inside  $\mathbf{Oz}$ ). So  $J$  is not strongly additive. It is not onto (Proposition 7.4), so there is no conflict with that report's theorem that every ring automorphism of  $\mathbf{Oz}$  extends uniquely to a strongly  $\mathbb{R}$ -linear automorphism of  $\mathbf{No}$  (`opa:as:thm:main`). A positive answer to the first clause is therefore not provable in the base theory unless that theory with a measurable cardinal is inconsistent. Whether a counterexample exists without large cardinals is not addressed here, and nothing here bears on that question's clauses about dense images or images that are neither closed nor discrete. The pair  $(J, H_j)$  also shows that preservation and reflection of  $\mathbf{Oz}$  together with agreement on all monomials does not determine an embedding:  $H_j$  has the same properties and is strong.

## 12 Surcomplex numbers and the Gaussian omnific lattice

Let  $\mathbf{SC} = \mathbf{No}(i)$  with a fixed named element  $i$  satisfying  $i^2 = -1$ , and write conjugation as  $\overline{x + iy} = x - iy$ . A surcomplex normal form has complex coefficients and a reverse well-ordered set of surreal exponents. Its support is the union of the supports of its real and imaginary coordinates: a complex coefficient vanishes exactly when both real coordinates vanish. Define

$$J_{\mathbb{C}}(x + iy) = J(x) + iJ(y), \quad H_{j,\mathbb{C}}(x + iy) = H_j(x) + iH_j(y).$$

These definitions fix the chosen orientation  $i$ ; they are not a statement about arbitrary automorphisms of the pure surcomplex field.

**Theorem 12.1** (Surcomplex transfer [06, 07, 09]). *The maps  $J_{\mathbb{C}}, H_{j,\mathbb{C}} : \mathbf{SC} \rightarrow \mathbf{SC}$  are field embeddings fixing  $\mathbb{C}$  and commuting with conjugation. The second is strongly  $\mathbb{C}$ -linear. They are elementary in the pure field language, and*

$$\text{Eq}(J_{\mathbb{C}}, H_{j,\mathbb{C}}) = \mathcal{S}_{<\kappa}(i), \quad (12.1)$$

$$J_{\mathbb{C}}(\mathbf{SC}) \cap H_{j,\mathbb{C}}(\mathbf{SC}) = C_j(i). \quad (12.2)$$

*Both fields on the right are algebraically closed. The image fields on the left of (12.2) are incomparable. Their intersection with the Gaussian omnific lattice satisfies*

$$J_{\mathbb{C}}(\mathbf{Oz}[i]) \cap H_{j,\mathbb{C}}(\mathbf{Oz}[i]) = J(\mathcal{J}_{<\kappa})[i]. \quad (12.3)$$

*Proof.* The quadratic-extension multiplication law proves the field-homomorphism claims coordinatewise. Uniqueness of real and imaginary parts gives injectivity and commutation with conjugation. The strong summation claim is the real statement on both coordinates. Since  $\mathbf{No}$  is real closed,  $\mathbf{SC}$  and both image fields are algebraically closed of characteristic zero. Quantifier elimination for  $\text{ACF}_0$  gives elementarity, formula by formula, as before.

Equality of  $J_{\mathbb{C}}(x + iy)$  and  $H_{j,\mathbb{C}}(x + iy)$  is equivalent to both  $J(x) = H_j(x)$  and  $J(y) = H_j(y)$ . This is precisely  $x, y \in \mathcal{S}_{<\kappa}$ . Equivalently, the union of the two supports has size less than  $\kappa$ : a finite union of sets of size less than an infinite cardinal still has size less than that cardinal. This proves (12.1). The same coordinate comparison and Theorem 6.1 prove (12.2). The fields  $\mathcal{S}_{<\kappa}$  and  $C_j$  are real closed, so adjoining  $i$  makes them algebraically closed.

The real witnesses  $h_\kappa$  and  $h_{j(\kappa)}$  still witness incomparability. If a real element is the image of  $x + iy$ , its imaginary coordinate is zero, forcing  $y = 0$ ; allowing a complex preimage cannot repair the failure of a real preimage. Finally apply Theorem 11.1 separately to the two integer coordinates to obtain (12.3).  $\square$

The exact defect formula also holds directly with complex coefficients: for  $z = \sum_{\alpha < \theta} c_\alpha \omega^{a_\alpha}$ , set  $c^* = j(c)$  after identifying  $\mathbb{C}$  with  $\mathbb{R}^2$ . Then

$$J_{\mathbb{C}}(z) - H_{j,\mathbb{C}}(z) = \sum_{\beta \in P_\theta} c_\beta^* \omega^{a_\beta^*}.$$

Nonzero complex coefficients at distinct exponents cannot cancel. Thus long complex support, not a sign or order on complex numbers, is what forces disagreement. The Boolean measure masks from Section 10 already lie on the real axis and require no complex choice beyond the named inclusion.

**Corollary 12.2** (Conjugation and modulus). *If  $|z| = (z\bar{z})^{1/2} \in \mathbf{No}^{\geq 0}$ , then*

$$|J_{\mathbb{C}}(z)| = J(|z|), \quad |H_{j,\mathbb{C}}(z)| = H_j(|z|).$$

*Every surcomplex number in either image outside  $C_j(i)$  is transcendental over  $C_j(i)$ .*

*Proof.* Both maps preserve conjugation and the unique nonnegative square root. The transcendence assertion follows because  $C_j(i)$  is algebraically closed.  $\square$

06 and 09 prove the transfer and the equalizer  $\mathcal{S}_{<\kappa}(i)$  as well; 09 notes that the same  $\kappa$ -sized real omnific family  $(\tau_\alpha)$  witnesses the failure of full strong additivity for  $J_{\mathbb{C}}$ , and 06 that properness is witnessed by the real elements of Proposition 7.4: a nonreal preimage of a real value is impossible, since the injective real-coordinate map would have to send a nonzero imaginary part to zero. 06 adds descent.

**Theorem 12.3** (Surcomplex descent [06]). *For every  $z \in \mathbf{SC}$  of complex normal-form length  $\lambda$  (the support being the union of the supports of the two coordinates),*

$$H_{j,\mathbb{C}}(z) \in \mathbf{SC}^M \iff j \text{ “ } \lambda \in M.$$

*The missing-index formula and the normal-measure trichotomy (Theorem 9.7) hold with complex coefficients. In a set-ultrapower, descent is again equivalent to closure under sequences of the support cardinality.*

*Proof.* Apply  $j$  to the packed complex coefficient and real exponent arrays. The transformed coefficient array remains nonzero at every index. The termwise image retains exactly the indices  $j \text{ “ } \lambda$ , so normal-form uniqueness proves the equalizer and defect formulas. For descent, if the complex number belongs to  $\mathbf{SC}^M$ , its two coordinates and their normal forms belong to  $M$ , hence their combined complex support does as well. The preimage under  $j(a)$  used in Theorem 9.1 recovers  $j \text{ “ } \lambda$ . Conversely an internal seed permits evaluation of both coordinates inside  $M$ . The sequence and normal-measure conclusions follow unchanged, by Lemma 9.3 and the proof of Theorem 9.7.  $\square$

The local analytic conclusion of Proposition 15.2 therefore applies to complex formal power series at surcomplex infinitesimals. No choice of a global surcomplex logarithm or exponential at arbitrary infinite arguments is needed. In particular no assertion about such a global function is hidden in the notation  $\mathbf{SC}$  [06]. For surcomplex numbers the field-theoretic transfer is automatic once the real result is known, but descent still depends on the real exponent support and on the same seed [06].

## 13 Algebraic independence over the short-support field [09]

This section does not assume a measurable cardinal or an elementary embedding. Fix any uncountable cardinal  $\kappa$ . The exponent scale  $q_\alpha = \omega^{-\alpha}$  of (8.4) has two useful features: it is decreasing, and its members are linearly independent over  $\mathbb{Q}$ . The latter follows immediately from uniqueness of finite Conway normal forms, since the  $q_\alpha$  are distinct monomials. The omnific masks  $Y_A$  are those of (10.9).

**Definition 13.1** (Almost disjoint families [09]). A family  $\mathcal{A} \subseteq [\kappa]^\kappa$  is  $\kappa$ -almost disjoint if  $|A \cap B| < \kappa$  whenever  $A, B \in \mathcal{A}$  are distinct.

### 13.1 A coordinate separation lemma

**Lemma 13.2** (Avoiding all coefficient directions [09]). *Let  $G_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}q_\alpha$  and let  $\Xi \subseteq (\mathbf{No}, +)$  be a rational vector subspace of cardinality less than  $\kappa$ . There is a set  $B \subseteq \kappa$  of size less than  $\kappa$  such that, for every  $\alpha \notin B$ , the coordinate map*

$$G_\kappa \longrightarrow \mathbb{Q}, \quad \sum_{\xi} c_\xi q_\xi \longmapsto c_\alpha$$

*extends to a linear functional  $\ell_\alpha : \Xi + G_\kappa \rightarrow \mathbb{Q}$  that vanishes on  $\Xi$ .*

*Proof.* Choose a rational basis of  $\Xi \cap G_\kappa$ . Every vector in this basis has finite support in the displayed basis of  $G_\kappa$ . Let  $B$  be the union of these finite supports. The number of basis vectors is at most  $|\Xi|$ , so  $|B| < \kappa$ , with the finite cases interpreted in the evident way. This estimate does not require  $\kappa$  to be regular.

For  $\alpha \notin B$  the  $\alpha$ -coordinate vanishes on  $\Xi \cap G_\kappa$ . Define  $\ell_\alpha(\xi + v)$  to be that coordinate of  $v$  for  $\xi \in \Xi$  and  $v \in G_\kappa$ . If  $\xi + v = \xi' + v'$ , then  $v - v' \in \Xi \cap G_\kappa$ , so the coordinate is independent of the representation. The resulting map is linear and kills  $\Xi$ .  $\square$

The lemma is what allows arbitrary short-support surreal coefficients, not just coefficients whose exponents already lie in  $G_\kappa$  [09].

## 13.2 The independence theorem

**Theorem 13.3** (Almost-disjoint omnific independence [09]). *For every uncountable cardinal  $\kappa$  and every  $\kappa$ -almost-disjoint family  $\mathcal{A} \subseteq [\kappa]^\kappa$ , the family*

$$\{Y_A : A \in \mathcal{A}\} \quad (13.1)$$

*is algebraically independent over the entire field  $\mathcal{S}_{<\kappa}$  inside **No**. Its members are positive purely infinite omnific integers of normal-form length  $\kappa$ .*

*Proof.* Choose distinct  $A_1, \dots, A_n \in \mathcal{A}$  and a nonzero polynomial

$$\Phi(X_1, \dots, X_n) = \sum_{\mathbf{m}} c_{\mathbf{m}} X_1^{m_1} \cdots X_n^{m_n} \in \mathcal{S}_{<\kappa}[X_1, \dots, X_n]. \quad (13.2)$$

Only finitely many coefficients are nonzero. Let  $S$  be the union of their normal-form supports and set  $\Xi = \text{span}_{\mathbb{Q}}(S)$ . Then  $|\Xi| < \kappa$ , exactly as in the proof of Theorem 3.1. Obtain  $B$  and the coordinate functionals from Lemma 13.2.

For each  $i$  choose

$$\alpha_i \in A_i \setminus \left( B \cup \bigcup_{k \neq i} A_k \right). \quad (13.3)$$

Such a choice is possible because the removed part of  $A_i$  has cardinality less than  $\kappa$ : it is a finite union of small intersections and the small set  $B$ . The indices  $\alpha_i$  are distinct. More importantly,  $q_{\alpha_i}$  occurs in the support of  $Y_{A_i}$  and in none of the other selected masks.

Choose a multi-index  $\mathbf{m} = (m_1, \dots, m_n)$  with nonzero coefficient whose total degree  $d = m_1 + \dots + m_n$  is maximum among the nonzero terms of  $\Phi$ . Choose one exponent  $d_0 \in \text{supp}(c_{\mathbf{m}})$ , and let  $\varrho \neq 0$  be its real coefficient. We claim that the exponent

$$\gamma = d_0 + \sum_{i=1}^n m_i q_{\alpha_i} \quad (13.4)$$

has coefficient exactly  $\varrho$  in  $\Phi(Y_{A_1}, \dots, Y_{A_n})$ .

A contribution to this coefficient from a term with multi-index  $\mathbf{n}$  consists of an exponent  $d' \in \text{supp}(c_{\mathbf{n}})$  and, for each of its  $n_i$  factors of  $Y_{A_i}$ , a choice of one exponent  $q_\beta$  with  $\beta \in A_i$ . The total exponent is  $d'$  plus all these chosen exponents. It belongs to the set-sized group  $\Xi + G_\kappa$ . Apply the  $n$  functionals  $\ell_{\alpha_i}$  to a putative equality of this total exponent with (13.4). Each functional kills  $d'$  and  $d_0$ . It counts exactly the number of selected factors whose exponent is  $q_{\alpha_i}$ . That count must be  $m_i$ .

The counts sum to  $d$ , so at least  $d$  factors have already been used. On the other hand the maximality of  $d$  gives  $\sum_i n_i \leq d$ . Hence exactly  $d$  factors occur, all of them at the specified private indices. Because  $\alpha_i$  belongs only to  $A_i$  among the selected sets, the equality forces  $n_i = m_i$  for every  $i$ . There are no further factors, and the exponent equation now forces  $d' = d_0$ .

Within the powers  $Y_{A_i}^{m_i}$  all  $m_i$  factors must select the same term  $\tau_{\alpha_i}$ . The coefficient of that choice is one; there is no further multinomial multiplicity. Thus the only contribution to the coefficient at  $\gamma$  is  $\varrho$ , as claimed. All convolutions used here are legitimate finite coefficient convolutions by Hahn summability. The argument also covers  $d = 0$ , in which case the polynomial is a nonzero constant. We have proved that every nonzero polynomial (13.2) evaluates nontrivially.

Each mask has positive exponents and coefficients one, so is a positive purely infinite omnific integer. A subset of  $\kappa$  of cardinality  $\kappa$  has inherited order type  $\kappa$ : its order type is at most  $\kappa$  and cannot be a smaller ordinal of the same cardinality. Therefore its normal-form length is  $\kappa$ .  $\square$

*Remark 13.4* (Why cancellation cannot be hidden in the coefficients [09]). A coefficient in  $\mathcal{S}_{<\kappa}$  can contain exponents unrelated to any  $q_\alpha$ . Such exponents can create complicated additive relations. The proof does not assume they are absent. It puts their entire rational span into  $\Xi$  and chooses private coordinates annihilating  $\Xi$ . The maximal *total* degree then exhausts every factor, leaving exactly one nonzero coefficient. Using only distinct leading exponents of the masks would not establish this stronger coefficient-field assertion.

### 13.3 Maximal cardinality inside a fixed Hahn field

**Theorem 13.5** (A  $2^\kappa$  family [09]). *Suppose  $\kappa$  is uncountable and  $2^{<\kappa} = \kappa$ . Let*

$$G_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}q_\alpha, \quad K_\kappa = \mathbb{R}((\omega^{G_\kappa})).$$

*Then  $K_\kappa$  contains  $2^\kappa$  positive purely infinite omnific integers, algebraically independent over  $\mathcal{S}_{<\kappa}$ . Moreover*

$$|K_\kappa| = 2^\kappa, \quad \text{trdeg}(\mathcal{S}_{<\kappa}K_\kappa/\mathcal{S}_{<\kappa}) = 2^\kappa. \quad (13.5)$$

*The compositum in (13.5) is the subfield of  $\mathbf{No}$  generated by  $\mathcal{S}_{<\kappa}$  and  $K_\kappa$ .*

*Proof.* The binary tree  $2^{<\kappa} = \bigcup_{\alpha < \kappa} 2^\alpha$  has cardinality  $\kappa$ . Fix a bijection  $\rho : 2^{<\kappa} \rightarrow \kappa$ . For each branch  $b \in 2^\kappa$  put

$$A_b = \{\rho(b \upharpoonright \alpha) : \alpha < \kappa\}.$$

Each  $A_b$  has size  $\kappa$ . Distinct branches share only their prefixes through their first disagreement; their sets therefore intersect in fewer than  $\kappa$  points. The family  $(A_b)_{b \in 2^\kappa}$  is  $\kappa$ -almost disjoint, so Theorem 13.3 applies. Every resulting  $Y_{A_b}$  is supported in  $G_\kappa$ , hence belongs to  $K_\kappa$ . No order-preserving property of  $\rho$  is needed: any subset of the ordinal  $\kappa$  has a well-ordered increasing enumeration.

The group  $G_\kappa$  has cardinality  $\kappa$ , and  $|\mathbb{R}| \leq \kappa$  follows from  $2^{<\kappa} = \kappa$ . A Hahn series is specified by a coefficient function on that group, so

$$|K_\kappa| \leq |\mathbb{R}|^\kappa \leq \kappa^\kappa = 2^\kappa.$$

The constructed masks already give the reverse inequality. They provide the lower bound for the transcendence degree in (13.5); the set  $K_\kappa$ , which generates the extension, gives the upper bound.

Although the base field is a proper class, this last use of transcendence degree causes no size problem. Algebraic independence of a subset of the set  $K_\kappa$  is a definable set-theoretic condition: each polynomial involves only finitely many surreal coefficients. Zorn's lemma on subsets of  $K_\kappa$  gives a maximal independent

set over  $\mathcal{S}_{<\kappa}$ . Maximality makes the set-generated extension algebraic over that independent set. The usual exchange argument is finitary and applies unchanged. Thus the transcendence basis under discussion is a set.  $\square$

**Corollary 13.6** (Complex coefficients and measurable cardinals [09]). *The families in Theorems 13.3 and 13.5 are also algebraically independent over  $\mathcal{S}_{<\kappa}(i)$  inside **SC**. If  $\kappa$  is measurable, Theorem 13.5 applies, and its  $2^\kappa$  omnific integers all lie outside the equalizer of  $J$  and  $H_j$ .*

*Proof.* Split a putative polynomial relation with coefficients in  $\mathcal{S}_{<\kappa}(i)$  into real and imaginary parts. At least one part is a nonzero polynomial over  $\mathcal{S}_{<\kappa}$ , contradicting real algebraic independence. A measurable cardinal is strongly inaccessible, hence  $2^{<\kappa} = \kappa$  [18]. Every constructed mask has support of size  $\kappa$ , so Theorem 5.1 gives the disagreement assertion.  $\square$

The independence theorem itself is therefore not a large-cardinal theorem. For example, it applies at  $\omega_1$  under CH. The measurable-cardinal hypothesis adds the two embeddings whose disagreement it witnesses [09]. [merge] This is transcendence over the *domain* equalizer  $\mathcal{S}_{<\kappa}$ ; it does not decide 07's Question 21.6 on the joint algebraic dependence of the two *image* fields over  $C_j$ .

## 14 Defect algebra and a logarithmic homomorphism

### 14.1 What all old-image coefficients fail to see

**Proposition 14.1** (Coefficient invisibility). *For every  $a, x \in \mathbf{No}$ ,*

$$[\omega^{J(a)}] J(x) = [\omega^a] x = [\omega^{J(a)}] H_j(x). \quad (14.1)$$

*In particular,  $J$  and  $H_j$  preserve the coefficient at exponent zero. They preserve the finite-surreal valuation ring and its infinitesimal ideal, and induce the identity on its residue field  $\mathbb{R}$ .*

*Proof.* If  $a$  is in the support of  $x$ , its unique original index gives the coefficient in (5.2). A missing-image index cannot supply exponent  $J(a)$  by Theorem 5.1. If  $a$  is not in the support, the same range-reflection argument used in that theorem excludes  $J(a)$  from the support of  $J(x)$ . The identity for  $H_j$  is its definition. Setting  $a = 0$  gives the constant coefficient. The valuation identities in Corollary 5.2, together with  $J(0) = 0$  and order preservation, give preservation of the valuation ring and its maximal ideal. On a finite surreal, its residue is its coefficient at zero.  $\square$

The constant-coefficient case  $a = 0$  is also 09's (see the note after Theorem 11.1). Thus (10.4) detects genuinely new exponent positions. No coefficient extraction at an exponent in  $J(\mathbf{No})$  can see the defect. Both embeddings nevertheless have the same value-group action and the same residue action.

**Proposition 14.2** (Twisted product law [06, 07, 09]). *The additive map  $D_j$  satisfies*

$$D_j(xy) = J(x)D_j(y) + D_j(x)H_j(y). \quad (14.2)$$

*For  $c \in \mathcal{S}_{<\kappa}$  one has  $D_j(cx) = J(c)D_j(x)$ .*

*Proof.* Expand the right side as  $J(x)J(y) - J(x)H_j(y) + J(x)H_j(y) - H_j(x)H_j(y)$ . The middle terms cancel and multiplicativity gives the left side. The scalar identity follows because  $D_j(c) = 0$ .  $\square$

Equation (14.2) is a twisted derivation identity, not an ordinary Leibniz rule for a derivation on **No**. No claim is made that  $D_j$  is a Berarducci–Mantova derivation or a strongly additive derivation.

06 (Proposition “Algebra of the defect”) and 09 (Proposition “Defect identities”) state the same law in the equivalent symmetric form

$$D_j(xy) = H_j(x)D_j(y) + H_j(y)D_j(x) + D_j(x)D_j(y), \quad (14.3)$$

obtained by substituting  $J = H_j + D_j$  in  $J(xy) = J(x)J(y)$ . 09 adds that  $D_j$  is additive and  $\mathbb{R}$ -linear with kernel  $\mathcal{S}_{<\kappa}$ , and that for  $x \neq 0$

$$D_j(x^{-1}) = -\frac{D_j(x)}{J(x)H_j(x)},$$

the difference of  $J(x)^{-1}$  and  $H_j(x)^{-1}$ ; the quadratic last term in (14.3) cannot be discarded [09]. 06 shows that  $D_j$  is not a derivation relative to  $H_j$ : if the ordinary  $H_j$ -Leibniz identity held, then putting  $x = y = h_\kappa$  would force  $D_j(h_\kappa)^2 = 0$ , contradicting  $D_j(h_\kappa) \neq 0$  (Theorem 7.1) in a field [06].

## 14.2 Passing from a multiplicative discrepancy to an additive one

For  $x \neq 0$ , let

$$\eta_j(x) = \frac{J(x)}{H_j(x)}. \quad (14.4)$$

By Corollary 5.2, this ratio lies in  $1 + \mathfrak{m}$ . For  $u \in \mathfrak{m}$ , the formal series

$$\log(1 + u) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} u^n \quad (14.5)$$

is strongly summable by the Hahn–Neumann lemma. The usual formal power-series identities are legitimate after such infinitesimal substitution; in particular, this logarithm takes products in  $1 + \mathfrak{m}$  to sums [2, 3]. Only this near-one logarithm is needed, not a preservation theorem for the global surreal exponential.

**Theorem 14.3** (Logarithmic defect). *The map*

$$L_j : \mathbf{No}^\times \longrightarrow (\mathfrak{m}, +), \quad L_j(x) = \log \eta_j(x)$$

*is a group homomorphism with kernel  $\mathcal{S}_{<\kappa}^\times$ . If  $x$  has normal form (2.1) of length  $\theta \geq \kappa$ , then*

$$\text{LT}(L_j(x)) = \frac{r_\kappa^*}{r_0} \omega^{a_\kappa^* - J(a_0)}. \quad (14.6)$$

*Thus a nonzero additive infinitesimal invariant records every departure from the equalizer, after multiplicative normalization.*

*Proof.* Because both embeddings are multiplicative,  $\eta_j(xy) = \eta_j(x)\eta_j(y)$ . The logarithm identity gives  $L_j(xy) = L_j(x) + L_j(y)$ . If  $\eta_j(x) = 1 + u$  with  $u \neq 0$ , then the first term of (14.5) is  $u$ , while all higher powers have strictly larger valuation. Hence the logarithm is nonzero and has leading term  $\text{LT}(u)$ . It follows that  $L_j(x) = 0$  precisely when  $J(x) = H_j(x)$ , proving the kernel assertion.

Finally  $u = D_j(x)/H_j(x)$ . Its leading term is the quotient of the leading term in (5.5) by the common leading term from Corollary 5.2, namely (14.6). Its exponent is negative since  $a_\kappa^* < a_0^* = J(a_0)$ , so it is infinitesimal as required.  $\square$

For the basic witness,

$$\mathrm{LT}(L_j(h_\kappa)) = \omega^{-\kappa}.$$

The homomorphism formulation avoids introducing a quotient whose individual equivalence classes might be proper classes. We use its kernel as a class predicate; no extra axiom for forming hyperclasses or class-sized quotient objects is needed.

## 15 An exact exponential compatibility locus

This section adds the canonical Gonshor exponential. It is independent of the proof of the main defect and image-intersection theorems, but the same support calculation yields a second exact boundary.

Write every surreal uniquely as

$$x = p(x) + r(x) + \varepsilon(x),$$

where  $p(x)$  has only positive exponents,  $r(x) \in \mathbb{R}$ , and  $\varepsilon(x) \in \mathfrak{m}$ . Thus  $p(x)$  is its purely infinite part. We use the classical exponential facts that  $\exp(p)$  is a Conway monomial for purely infinite  $p$ , that the exponential on infinitesimals is its strong power series, and that the inverse logarithm of a monomial is purely infinite [2, Chapter 10][3, Section 5].

**Lemma 15.1** (Exponential transfer [06, 07]). *The elementary-induced embedding satisfies*

$$J(\exp x) = \exp J(x), \quad J(\log y) = \log J(y) \quad (y > 0).$$

*Proof.* Gonshor's exponential is defined by a uniform recursive cut on the canonical options of its argument, using exponentials of simpler options, finite Taylor polynomials, field operations, and their inequalities. The cut absoluteness argument from Lemma 2.6 applies to this recursion as well. Therefore the exponential computed in  $M$  on an old argument equals the one in  $V$ . Elementary transfer of the defining recursion gives the first identity. The second follows from the uniqueness of the inverse of the strictly increasing bijection  $\exp$ .  $\square$

06 includes the exponential in its absoluteness lemma, with the same recursion argument (the note after Lemma 2.6). 06 also proves the infinitesimal part of the next theorem as a separate proposition, for both maps and for power series in several variables.

**Proposition 15.2** (Formal evaluation [06]). *Let  $A \in \mathbb{R}[[T_1, \dots, T_n]]$  and let  $\varepsilon_1, \dots, \varepsilon_n$  be infinitesimal surreals. With strong formal evaluation,*

$$\begin{aligned} H_j(A(\varepsilon_1, \dots, \varepsilon_n)) &= A(H_j(\varepsilon_1), \dots, H_j(\varepsilon_n)), \\ J(A(\varepsilon_1, \dots, \varepsilon_n)) &= A(J(\varepsilon_1), \dots, J(\varepsilon_n)). \end{aligned}$$

*The same statements hold with complex coefficients and surcomplex infinitesimals after the extension of Section 12.*

*Proof.* The Neumann lemma makes the family of evaluated monomial terms strongly summable. It is countable, since there are finitely many variables. Both embeddings fix every coefficient and preserve finite products. For  $H_j$  use strong additivity; for  $J$  use Theorem 8.1, since  $\aleph_0 < \kappa$ . Ordered-field embeddings fixing the reals preserve the condition of being infinitesimal. The coefficientwise proof over  $\mathbb{C}$  is identical.  $\square$

This includes the Taylor exponential on  $\mathfrak{m}$  and the logarithm on  $1 + \mathfrak{m}$ . Arbitrarily large supports of the individual infinitesimals cause no problem: the *number of Taylor terms* is still countable. This is another distinction between support size and summation-family size [06].

**Theorem 15.3** (Exact exponential defect [06, 07]). *For every  $x \in \mathbf{No}$ ,*

$$\frac{H_j(\exp x)}{\exp(H_j(x))} = \exp(D_j(p(x))). \quad (15.1)$$

*In particular,*

$$H_j(\exp x) = \exp(H_j(x)) \iff |\text{supp}(p(x))| < \kappa. \quad (15.2)$$

*The companion commutes with the exponential on every finite surreal, including finite surreals with arbitrarily long infinitesimal support. For a positive  $y = r\omega^a(1 + u)$ , with  $r > 0$  real and  $u \in \mathfrak{m}$ ,*

$$\log(H_j(y)) - H_j(\log y) = D_j(\log(\omega^a)). \quad (15.3)$$

*Proof.* The companion preserves the purely infinite, real, and infinitesimal parts separately, because it preserves the sign of exponents. Since  $\exp(p(x))$  is a single monomial,

$$H_j(\exp(p(x))) = J(\exp(p(x))) = \exp(J(p(x))).$$

Strong linearity and infinitesimal power-series evaluation give  $H_j(\exp(\varepsilon(x))) = \exp(H_j(\varepsilon(x)))$ , while real exponential values are fixed. Applying  $H_j$  to  $\exp x = \exp(p(x))e^{r(x)}\exp(\varepsilon(x))$  and dividing by  $\exp(H_j(x))$  proves (15.1). Injectivity of the exponential and Theorem 5.1 give (15.2). For finite  $x$ , its purely infinite part is zero.

Finally use  $\log y = \log r + \log(\omega^a) + \log(1 + u)$ . The near-one logarithm commutes with  $H_j$  by its strong power-series formula. Also  $\log(H_j(\omega^a)) = \log(J(\omega^a)) = J(\log(\omega^a))$ . Subtracting the two decompositions gives (15.3).  $\square$

06 proves (15.1) and (15.2) with the same factorization, using Proposition 15.2 for the infinitesimal factor, and records the intermediate formula

$$H_j(\exp x) = \exp(J(p(x)) + r(x) + H_j(\varepsilon(x))). \quad (15.4)$$

It notes that a purely infinite series may have either sign: “purely infinite” describes its support, not its positivity; and it cites the classical facts that the exponential maps the group of purely infinite series onto the monomial group and agrees with the real exponential and the infinitesimal Taylor exponential on the other two parts [2, 13]. The identity (15.2) fails at every purely infinite input with at least  $\kappa$  nonzero terms [06]. The logarithmic identity (15.3) is 07’s.

**Corollary 15.4** (A scale-sized exponential discrepancy [06]). *The ratio in (15.1) is always a monomial. It is either exactly one, a positive infinitesimal, or larger than every positive real number. In particular*

$$H_j(\exp x) \sim \exp(H_j x) \iff H_j(\exp x) = \exp(H_j x),$$

*where  $y \sim w$  means  $y/w \in 1 + \mathfrak{m}$  for positive  $y, w$  (06 writes  $p \sim q$ ).*

*Proof.* Both  $J(p(x))$  and  $H_j(p(x))$  are purely infinite. Their difference is therefore purely infinite, possibly zero. Gonshor exponentiation maps this additive group to the monomial group. A nonidentity positive monomial is either infinitesimal or infinitely large relative to the reals. Apply (15.1).  $\square$

The two criteria of Corollary 5.6 and (15.2) are different. For example,  $h_\kappa$  has  $\kappa$  terms but is finite, so the Hahn lift fails to commute with  $\omega^x$  at  $h_\kappa$ , while commuting with  $\exp x$  there. For the purely infinite omnific integer

$$\zeta_\kappa = z_\kappa = \omega^\kappa h_\kappa = \sum_{\alpha < \kappa} \omega^{\kappa - \alpha},$$

both identities fail [06].

**Corollary 15.5** (Exponential closure without logarithmic closure). *The fields  $\mathcal{S}_{<\kappa}$  and  $C_j$  are closed under the Gonshor exponential but not under logarithms of positive elements. With this induced exponential, neither is an elementary substructure of  $(\mathbf{No}, \exp)$ , although both are elementary ordered subfields of  $\mathbf{No}$ .*

*Proof.* If  $x \in \mathcal{S}_{<\kappa}$ , then  $p(x) \in \mathcal{S}_{<\kappa}$ . By the preceding theorem and Lemma 15.1,

$$H_j(\exp x) = \exp(H_j(x)) = \exp(J(x)) = J(\exp x).$$

The equalizer theorem yields  $\exp x \in \mathcal{S}_{<\kappa}$ . The same closure for  $C_j = J(\mathcal{S}_{<\kappa})$  follows from exponential transfer.

The purely infinite element  $z_\kappa$  of Section 11 has  $\kappa$  outer terms, so it is not in  $\mathcal{S}_{<\kappa}$ . But  $\exp(z_\kappa)$  is a positive monomial and hence belongs to  $\mathcal{S}_{<\kappa}$ . Its unique logarithm in  $\mathbf{No}$  is  $z_\kappa$ , which is absent. Similarly  $J(\exp(z_\kappa))$  lies in  $C_j$ , whereas its logarithm  $J(z_\kappa)$  does not, by injectivity of  $J$  and the description of  $C_j$ . Thus the sentence asserting that every positive element has an exponential preimage holds in  $(\mathbf{No}, \exp)$  and fails in both smaller exponential structures. Their ordered-field elementarity follows from real closedness.  $\square$

**Corollary 15.6** (One image is exponentially closed and the other is not). *The image  $J(\mathbf{No})$  is closed under both exponential and logarithm. The image  $H_j(\mathbf{No})$  is not closed under exponential.*

*Proof.* The first assertion follows immediately from Lemma 15.1. For the second, use the classical order-preserving bijection  $g : \mathbf{No}^{>0} \rightarrow \mathbf{No}$  specified by

$$\exp(\omega^b) = \omega^{\omega^{g(b)}} \quad (b > 0).$$

The purely infinite exponential formula says that  $\exp(\sum r_\alpha \omega^{b_\alpha}) = \omega^{\sum r_\alpha \omega^{g(b_\alpha)}}$  when all  $b_\alpha > 0$ ; see [2, Theorem 10.13] and [3, Lemma 5.2, proof]. Define

$$p_\kappa = \sum_{\alpha < \kappa} \omega^{g^{-1}(-\alpha)}.$$

The exponents are strictly decreasing and positive, so this is a purely infinite normal form and  $\exp(p_\kappa) = \omega^{h_\kappa}$ . The definition of  $g$ , together with transfer of exponential and the omega-map, shows that  $J(g(b)) = g(J(b))$  and hence that  $J$  commutes with  $g^{-1}$ . Since  $-\alpha$  is fixed for  $\alpha < \kappa$ , all the displayed exponents are fixed. Therefore  $H_j(p_\kappa) = p_\kappa$ , so  $p_\kappa \in H_j(\mathbf{No})$ .

But  $h_\kappa \notin J(\mathbf{No}) = \Gamma_j$  by Theorem 7.1. The monomial  $\omega^{h_\kappa}$  consequently has its exponent outside  $\Gamma_j$  and is not in  $H_j(\mathbf{No})$  by (4.1). This proves the failure of exponential closure.  $\square$

**Theorem 15.7** (Two canonical closures of the common field). *Among class subfields of  $\mathbf{No}$  containing  $C_j$ , the least one closed under logarithms of positive elements is  $J(\mathbf{No})$ , and the least one closed under external set-indexed strong sums is  $H_j(\mathbf{No})$ . More explicitly, every element of  $J(\mathbf{No})$  has the form*

$$\frac{\log m_1}{m_2}, \quad m_1, m_2 \in \{\omega^{J(a)} : a \in \mathbf{No}\} \subseteq C_j, \quad (15.5)$$

*and every element of  $H_j(\mathbf{No})$  is a strong sum of elements of  $C_j$ . These two closures are incomparable and intersect in exactly  $C_j$ . In contrast, both the logarithmic field closure and the strong-sum closure of the domain equalizer  $\mathcal{S}_{<\kappa}$  are all of  $\mathbf{No}$ .*

*Proof.* The field  $J(\mathbf{No})$  contains  $C_j$  and is logarithmically closed by Lemma 15.1. Take  $J(x) \in J(\mathbf{No})$ . Choose  $d > 0$  so that  $z = \omega^d x$  is purely infinite, exactly as in Theorem 11.1. Then  $\exp z$  is a monomial. Both  $\exp z$  and  $\omega^d$  have one-term support and hence lie in  $\mathcal{S}_{<\kappa}$ . Set  $m_1 = J(\exp z)$  and  $m_2 = J(\omega^d)$ ; they are monomials in  $C_j$ . Exponential transfer gives

$$J(x) = \frac{J(z)}{J(\omega^d)} = \frac{\log m_1}{m_2}.$$

Any logarithmically closed class subfield containing  $C_j$  must therefore contain every  $J(x)$ , proving minimality and (15.5).

The field  $H_j(\mathbf{No})$  is strongly closed by Theorem 8.6. Every normal-form term  $r\omega^{J(a)}$  in one of its members already lies in  $C_j$ , so strong closure of any class containing  $C_j$  forces that member to be included. This proves the other minimality assertion. Incomparability and the intersection were proved earlier.

For  $\mathcal{S}_{<\kappa}$ , the identical shift argument without  $J$  expresses every  $x \in \mathbf{No}$  as a quotient of the logarithm of a monomial in  $\mathcal{S}_{<\kappa}$  by another monomial in  $\mathcal{S}_{<\kappa}$ . Also every surreal is a strong sum of real multiples of monomials, all of which lie in  $\mathcal{S}_{<\kappa}$ . This proves both final assertions. The minimality statements are explicit universal properties; they do not require forming an intersection of a collection of all classes.  $\square$

The two image fields are therefore canonically isomorphic as ordered valued fields over  $C_j$ , but their actual placement in  $\mathbf{No}$  has different compatibility with the canonical exponential. Neither their shared residue field and value group nor their common monomials detect this difference. The support criterion (15.2) is different from the whole-support equalizer criterion: the long-support finite number  $h_\kappa$  lies in the former locus and not the latter.

## 16 Supercompactness as a seed-coefficient system

The first support threshold depends only on the critical point. Thus it cannot by itself distinguish a measurable embedding from a stronger embedding with the same critical point. A richer family of coefficient probes does retain the classical seed information used for supercompactness.

### 16.1 Coding an arbitrary index set [09]

The measure coefficient used the special seed  $\kappa \in j(\kappa)$ . A seed can instead be an arbitrary element of an embedded set. Let  $I$  be a nonempty set, fix a bijection  $e : \theta \rightarrow I$  from an ordinal, and put

$$Y_{A,e} = \sum_{\substack{\alpha < \theta \\ e(\alpha) \in A}} \tau_\alpha \quad (A \subseteq I). \quad (16.1)$$

For  $s \in j(I)$  let  $\beta_s = (j(e))^{-1}(s) < j(\theta)$ .

**Theorem 16.1** (Seed coefficient formula [09]). *For every  $A \subseteq I$ ,*

$$[\omega^{q\beta_s}] J(Y_{A,e}) = 1 \iff s \in j(A). \quad (16.2)$$

*The coefficient is always zero or one. The resulting collection  $U_s = \{A \subseteq I : s \in j(A)\}$  is a  $\kappa$ -complete ultrafilter on  $I$ . It may be principal for some choices of  $s$ .*

*Proof.* Applying  $j$  to the defining mask gives

$$J(Y_{A,e}) = \sum_{\substack{\beta < j(\theta) \\ j(e)(\beta) \in j(A)}} \tau_\beta.$$

At the unique position  $\beta_s$  its coefficient is one precisely when  $s \in j(A)$ . Complements and finite intersections give an ultrafilter, and intersections indexed by a fixed ordinal below  $\kappa$  give  $\kappa$ -completeness, as in Theorem 10.1. If  $s = j(a)$  for some  $a \in I$ , then the resulting ultrafilter is principal at  $a$ , explaining the qualification.  $\square$

Changing the enumeration  $e$  merely changes which coefficient position carries the same seed. It does not change the derived ultrafilter. At the same time, a single derived ultrafilter need not determine the whole embedding  $j$  or its target  $M$ ; the theorem recovers exactly the ultrafilter data stated, not an arbitrary ambient embedding from one bit [09]. When  $s$  is not of the form  $j(a)$ , the index  $\beta_s$  is a missing-image index and the coefficient (16.2) is a coefficient of the defect  $D_j(Y_{A,e})$ , as in the next theorem.

## 16.2 The supercompact seed [07]

Assume  $\lambda \geq \kappa$ ,  $j(\kappa) > \lambda$ , and  $M$  is closed under  $\lambda$ -sequences from  $V$ . Let

$$I = P_\kappa(\lambda) = \{a \subseteq \lambda : |a| < \kappa\}, \quad s = j \text{ `` } \lambda.$$

Closure gives  $s \in M$ , and in  $M$  it has size at most  $\lambda < j(\kappa)$ , so  $s \in j(I)$ . These are the standard hypotheses and seed for a  $\lambda$ -supercompactness measure [6]. Choose a bijection  $e : \theta \rightarrow I$ . Let  $\beta_s < j(\theta)$  be the unique index satisfying

$$j(e)(\beta_s) = s. \tag{16.3}$$

For  $A \subseteq I$ , define the mask

$$m_A = \sum_{\substack{\alpha < \theta \\ e(\alpha) \in A}} \omega^{-\alpha}.$$

The construction depends on the enumeration  $e$ , which is explicitly part of the data.

**Theorem 16.2** (Seed extraction). *The index  $\beta_s$  is a missing-image index:  $\beta_s \notin j \text{ `` } \theta$ . Moreover*

$$[\omega^{-\beta_s}] D_j(m_A) = \mathbf{1}_{\{s \in j(A)\}} \quad (A \subseteq I). \tag{16.4}$$

*Consequently the seed coefficient recovers the fine normal  $\kappa$ -complete ultrafilter*

$$U_s = \{A \subseteq P_\kappa(\lambda) : s \in j(A)\}.$$

*Proof.* If  $\beta_s = j(\alpha)$ , then  $s = j(e(\alpha)) = j(a)$  for some  $a \in P_\kappa(\lambda)$ . Since  $|a| < \kappa$ , Lemma 2.5 gives  $j(a) = j \text{ `` } a$ . But  $s = j \text{ `` } \lambda$ , so injectivity of  $j$  implies  $a = \lambda$ , a contradiction. Thus  $\beta_s$  is missing from the pointwise index image.

The transformed mask has coefficient 1 at  $-\beta_s$  exactly when  $j(e)(\beta_s) = s$  belongs to  $j(A)$ . The companion has its exponents only at positions  $-j(\alpha)$ , so it contributes zero there. This gives (16.4).

Ultrafilter and  $\kappa$ -completeness follow by the same Boolean and short-index argument as in Theorem 10.1, with  $s$  replacing  $\kappa$ . For each  $\xi < \lambda$ , the seed contains  $j(\xi)$ , so

$$\{a \in P_\kappa(\lambda) : \xi \in a\} \in U_s;$$

this is fineness. If  $A \in U_s$  and  $f : A \rightarrow \lambda$  satisfies  $f(a) \in a$ , then  $j(f)(s) \in s = j \text{ `` } \lambda$ . Write  $j(f)(s) = j(\xi)$  for some  $\xi < \lambda$ . The fiber where  $f(a) = \xi$  then belongs to  $U_s$ , proving normality.  $\square$

Fine normal  $\kappa$ -complete ultrafilters on  $P_\kappa(\lambda)$  are the classical ultrafilter characterization of supercompactness, due to Solovay; see [6]. We have not discovered a new large-cardinal equivalence. The contribution is that the seed can be read as a coefficient of the same defect construction, and that its index is provably outside the termwise image. An intrinsic, enumeration-free coefficient geometry for these seed systems is a further research question, not a result claimed here.

### 16.3 Covering seeds and exact seeds [09]

For a cardinal  $\lambda \geq \kappa$ , an ultrafilter  $U$  on  $P_\kappa(\lambda)$  is *fine* if  $\{a : \xi \in a\} \in U$  for every  $\xi < \lambda$ . It is *normal* if, whenever  $A \in U$  and  $f : A \rightarrow \lambda$  satisfies  $f(a) \in a$ , there is a  $\xi < \lambda$  such that  $\{a \in A : f(a) = \xi\} \in U$ . These are the standard ultrafilter notions [18, 19].

**Proposition 16.3** (Covering seeds and exact seeds [09]). *Let  $I = P_\kappa(\lambda)$  and suppose  $s \in j(I)$ .*

- (i) *If  $j \restriction \lambda \subseteq s$ , then  $U_s$  is fine and  $\kappa$ -complete.*
- (ii) *If  $s = j \restriction \lambda$  belongs to  $M$  and is an element of  $j(I)$ , then  $U_s$  is normal and fine.*
- (iii) *For a fine ultrafilter on  $P_\kappa(\lambda)$ , the ultrapower seed represented by the identity map contains every  $j(\xi)$  for  $\xi < \lambda$ . If the ultrafilter is normal, that seed is exactly  $j \restriction \lambda$ .*

*Proof.* For each  $\xi < \lambda$ , let  $A_\xi = \{a \in I : \xi \in a\}$ . Its image is the set of  $b \in j(I)$  containing  $j(\xi)$ . A covering seed belongs to each  $j(A_\xi)$ , proving fineness; completeness is Theorem 16.1.

For normality in (ii), apply  $j$  to  $f(a) \in a$  on a  $U_s$ -large set. Then  $j(f)(s) \in s = j \restriction \lambda$ . Thus  $j(f)(s) = j(\xi)$  for some  $\xi < \lambda$ , and  $s$  lies in the image of the corresponding constant fiber. That fiber belongs to  $U_s$ .

For (iii), write the seed as  $[\text{id}]_U$  before transitive collapse. Fineness says  $[\text{const}_\xi]_U \in [\text{id}]_U$ , which gives one inclusion after collapse. Conversely, any element of this seed is represented by a function  $f$  for which  $f(a) \in a$  on a large set. Normality makes  $f$  constant there, so its collapsed class is some  $j(\xi)$ . This proves equality for a normal fine ultrafilter.  $\square$

Part (ii) is the ultrafilter half of Theorem 16.2, proved there by the same argument; 09 derives the membership  $s \in j(I)$  the same way: a supercompactness embedding with  $j(\kappa) > \lambda$  and  $M^\lambda \subseteq M$  has  $j \restriction \lambda \in M$ , and in  $M$  this set has order type and cardinality  $\lambda < j(\kappa)$ , because its external increasing enumeration has order type  $\lambda$  and the order type of a well-ordered set in a transitive model is absolute.

The usual ultrafilter characterization of strong compactness is the existence, for every  $\lambda \geq \kappa$ , of a fine  $\kappa$ -complete ultrafilter on  $P_\kappa(\lambda)$ . Supercompactness has the corresponding normal-fine characterization [18]. Through Theorem 16.1, these existing characterizations become coefficient-selection principles on the specified omnific masks. The difference between a seed *covering*  $j \restriction \lambda$  and an available seed *equal* to  $j \restriction \lambda$  is retained explicitly [09].

**Warning 16.4** (Translation is not a new consistency equivalence [09]). The preceding paragraph still quantifies over set embeddings, seeds and subset codes. It is a numerical presentation of established large-cardinal mechanisms, not a characterization in the bare ordered-field language of **No** or the bare ring language of **Oz**. Eliminating that extra structure is a substantive further question.

## 17 Strong compactness as a support cover; supercompactness as exact descent [06]

### 17.1 The set-theoretic principles and their witnesses

Fix an uncountable cardinal  $\kappa$  and a cardinal  $\lambda \geq \kappa$ , and use the fine and normal ultrafilters on  $P_\kappa(\lambda)$  of the previous section. To avoid ambiguity about local terminology, in this section the  $(\kappa, \lambda)$  *strong-compactness principle* means existence of a fine  $\kappa$ -complete ultrafilter on  $P_\kappa(\lambda)$ . The corresponding *supercompactness principle* requires the ultrafilter to be normal as well. Requiring the respective principles for every cardinal  $\lambda \geq \kappa$  gives the usual global large cardinal properties. The proofs below also identify the usual sequence-closed elementary embedding characterization of  $\lambda$ -supercompactness. The seed construction is standard [6, 12]; 06 gives the argument to make the exact translation to surreal supports explicit. The letter  $W$  denotes a fine ultrafilter in this section only.

**Lemma 17.1** (Cover and exact-seed characterizations [06]). *The  $(\kappa, \lambda)$  strong-compactness principle is equivalent to existence of a set-ultrapower embedding  $j : V \rightarrow M$  of critical point  $\kappa$  and a set  $s \in M$  such that*

$$j \text{ “ } \lambda \subseteq s \subseteq j(\lambda), \quad M \models |s| < j(\kappa). \quad (17.1)$$

*Such an embedding necessarily has  $j(\kappa) > \lambda$ . The supercompactness principle is equivalent to existence of such a set-ultrapower embedding with  $j(\kappa) > \lambda$  and  $j \text{ “ } \lambda \in M$ , or equivalently with  ${}^\lambda M \cap V \subseteq M$ .*

*Proof.* Start with a fine  $\kappa$ -complete ultrafilter  $W$  on  $P_\kappa(\lambda)$ . Its ultrapower is well-founded: a putative infinite descending membership chain would, by countable completeness, yield a pointwise descending membership chain on a nonempty common measure-one set, contradicting Foundation. Take its transitive collapse. The seed  $s = [\text{id}]_W$  is an element of  $j(P_\kappa(\lambda))$ , so it is a subset of  $j(\lambda)$  of internal size  $< j(\kappa)$ . Fineness says that it contains every  $j(\alpha)$  for  $\alpha < \lambda$ .

Completeness fixes all ordinals below  $\kappa$ : any function into such an ordinal is constant modulo  $W$ , since otherwise the intersection of the complements of all its fibers would be large and empty. Induction on the ordinal gives the assertion in the transitive collapse. On the other hand, if  $j(\kappa) = \kappa$ , then an internal enumeration of  $s$  of length  $< \kappa$  would be an external enumeration of a set containing  $\lambda \geq \kappa$  distinct elements, which is impossible. Hence the critical point is exactly  $\kappa$ . More generally, an internal cardinal bound  $\nu < j(\kappa)$  for  $s$  gives an external injection of  $\lambda$  into  $\nu$ . Since  $\lambda$  is an ambient cardinal,  $\lambda \leq \nu < j(\kappa)$ .

Conversely, suppose an embedding and  $s$  as in (17.1) are given, and let  $U_s = \{X \subseteq P_\kappa(\lambda) : s \in j(X)\}$  (06's  $W_s$ ). Because  $s \in j(P_\kappa(\lambda))$ , this is an ultrafilter. Preservation of intersections indexed below the critical point gives  $\kappa$ -completeness. Containment of  $j \text{ “ } \lambda$  in  $s$  gives fineness. This proves the cover characterization.

If the initial fine ultrafilter is normal, every member of  $[\text{id}]_W$  is represented by a selector  $f(a) \in a$  on a large set. Normality makes it equal to a constant  $j(\alpha)$ . Fineness gives the converse inclusion, so  $[\text{id}]_W = j \text{ “ } \lambda \in M$ . Conversely, if  $s = j \text{ “ } \lambda$  and  $j(\kappa) > \lambda$ , then its increasing enumeration in  $M$  has length  $\lambda < j(\kappa)$ , so it satisfies (17.1). For a selector  $f$ , the value  $j(f)(s) \in s$  equals  $j(\alpha)$  for some  $\alpha < \lambda$ . The corresponding fiber belongs to  $U_s$ , proving normality. Finally apply Lemma 9.3 to obtain the sequence-closure formulation. A sequence-closed embedding not initially presented as an ultrapower also yields this normal ultrafilter by the same exact-seed argument, and hence yields a set-ultrapower witness.  $\square$

### 17.2 Binary normal forms and coefficientwise covers

For  $t \subseteq \theta$  a set of ordinals, put

$$B(t) = \sum_{\beta \in t} \omega^{-\beta},$$

with its indices in increasing order. A *binary series below*  $\theta$  has this form. On this family define

$$B(t) \preccurlyeq_{\text{coeff}} B(t') \iff t \subseteq t'.$$

Equivalently every coefficient of the difference  $B(t') - B(t)$  is nonnegative. This coefficientwise order is not the ordinary numerical order on  $\mathbf{No}$ . In particular a small leading coefficient inequality cannot replace a support inclusion. For the canonical detector  $h_\lambda$  of (7.1),

$$J(h_\lambda) = h_{j(\lambda)}, \quad H_j(h_\lambda) = B(j \text{ `` } \lambda). \quad (17.2)$$

The second expression is always defined externally, even when it fails to belong to  $M$ .

**Theorem 17.2** (Support-cover characterization of strong compactness [06]). *The  $(\kappa, \lambda)$  strong-compactness principle holds if and only if there is a set-ultrapower embedding  $j : V \rightarrow M$  of critical point  $\kappa$  and a binary series  $\widehat{B} \in \mathbf{No}^M$  below  $j(\lambda)$  such that*

$$H_j(h_\lambda) \preccurlyeq_{\text{coeff}} \widehat{B} \preccurlyeq_{\text{coeff}} J(h_\lambda), \quad M \models |\text{supp}(\widehat{B})| < j(\kappa). \quad (17.3)$$

*Proof.* If  $s$  is the cover seed in Lemma 17.1, let  $\widehat{B} = B(s)$ , computed in  $M$ . Absoluteness identifies it with the ambient binary series, and the containments of sets in (17.1) become the coefficientwise inequalities in (17.3). The support map  $\beta \mapsto -\beta$  is a bijection in  $M$ , so the internal cardinal bounds are identical.

Conversely, take the normal form of  $\widehat{B}$  inside  $M$  and negate its exponents to obtain  $s \subseteq j(\lambda)$  in  $M$ . The inequalities in (17.3) say exactly that  $j \text{ `` } \lambda \subseteq s$ . The internal bound on the support gives the bound on  $s$ . Apply Lemma 17.1.  $\square$

**Theorem 17.3** (Single-number characterization of supercompactness [06]). *The  $(\kappa, \lambda)$  supercompactness principle holds if and only if there is a set-ultrapower embedding  $j : V \rightarrow M$  with  $\text{crit}(j) = \kappa$ ,  $j(\kappa) > \lambda$ , and*

$$H_j(h_\lambda) \in \mathbf{No}^M. \quad (17.4)$$

*For this equivalence  $h_\lambda$  may be replaced by any fixed surreal of support cardinal  $\lambda$ . It may in particular be replaced by the omnific integer*

$$\zeta_\lambda = \omega^\lambda h_\lambda = \sum_{\alpha < \lambda} \omega^{\lambda - \alpha}. \quad (17.5)$$

*Proof.* By Theorem 9.1, (17.4) is exactly the assertion  $j \text{ `` } \lambda \in M$ . Now use Lemma 17.1. The landing criterion depends only on support cardinality, proving the replacement claim. Every exponent in (17.5) is positive, so  $\zeta_\lambda$  is omnific. Alternatively, use the identity  $H_j(\zeta_\lambda) = \omega^{j(\lambda)} H_j(h_\lambda)$  and the fact that the nonzero multiplying monomial and its inverse both belong to  $\mathbf{No}^M$ .  $\square$

**Corollary 17.4** (Omnific support covers [06]). *The criterion (17.3) is equivalent to existence of a purely infinite binary-coefficient omnific integer  $\widehat{B}^+ \in \mathbf{Oz}^M$  whose support lies among the exponents  $\{j(\lambda) - \beta : \beta < j(\lambda)\}$  and such that*

$$H_j(\zeta_\lambda) \preccurlyeq_{\text{coeff}} \widehat{B}^+ \preccurlyeq_{\text{coeff}} J(\zeta_\lambda), \quad M \models |\text{supp}(\widehat{B}^+)| < j(\kappa).$$

*Proof.* Multiply all three series in (17.3) by  $\omega^{j(\lambda)}$ . This translates the support bijectively, preserves coefficientwise inclusion and internal cardinality, and makes every exponent positive. Dividing by that monomial reverses the construction.  $\square$

[merge] In this corollary  $\preccurlyeq_{\text{coeff}}$  is read on the shifted binary series, whose exponents are  $j(\lambda) - \beta$ ; 06 uses the same symbol.

### 17.3 What the distinction says, and what it does not say

Strong compactness provides an internally short support *containing* the transported old indices. Supercompactness makes the exact set of old indices an internal support. The cover may include extra monomials; exact descent permits none. This is the mathematical distinction behind the two criteria, rather than a change of terminology for numerical approximation.

At  $\lambda = \kappa$  a normal measure supplies both tests. At  $\lambda = \kappa^+$ , an ultrapower indexed just by  $\kappa$  cannot supply even the cover, by Lemma 9.5. Stronger witnesses, when they exist, must use sufficiently large index sets. One must not conclude that a measurable cardinal fails a stronger large-cardinal property merely because its particular normal-measure ultrapower has the short landing spectrum from Theorem 9.7.

These are equivalences with classical principles, not new consistency proofs. For example, the global existence of exact-descent witnesses for every  $\lambda$  is equivalent to supercompactness, and hence has exactly that hypothesis and relative consistency strength. No assertion that any of these large cardinals exists is made in ZFC alone [06]. For omnific integers the statements are not merely about an ambient field: the measure and compactness detectors can themselves be taken to be omnific integers [06]. [merge] 07's Theorem 16.2 reads the same normal seed as a defect coefficient; 06's theorems read it as membership of one value in  $M$ ; 09's Proposition 16.3 records the covering and exact seeds as ultrafilters. All three are translations of the classical fine and normal-fine characterizations.

## 18 Weak compactness as an omnific prefix principle [09]

There is a second large-cardinal mechanism that does not start with a measure: compactness of trees. 09 gives its explicit normal-form version, while retaining the exact topology and the known set-theoretic content.

### 18.1 Length-preserving binary codes

For  $\sigma \in 2^\alpha$  put

$$w_\sigma = \sum_{\xi < \alpha} (1 + \sigma(\xi)) \tau_\xi. \quad (18.1)$$

Every coefficient is one or two, so the length of the normal form is exactly  $\alpha$ . This convention is important. If one used coefficients zero and one without also recording the length, appending a zero bit would not change the number and the coding of tree nodes would not be injective. For  $\beta \leq \alpha$ , the normal-form truncation of  $w_\sigma$  to length  $\beta$  is  $w_{\sigma \upharpoonright \beta}$ . Thus the prefix order of binary strings is literally normal-form truncation on these omnific codes.

**Definition 18.1** (Omnific prefix compactness at  $\kappa$  [09]). Let  $\kappa$  be strongly inaccessible. The principle  $\text{OPC}(\kappa)$  says: whenever  $\mathcal{T}$  is a set of numbers of the form  $w_\sigma$  with  $\sigma \in 2^{<\kappa}$ , is closed under normal-form truncation, and has a node of every length  $\alpha < \kappa$ , there is a number

$$w_b = \sum_{\xi < \kappa} (1 + b(\xi)) \tau_\xi, \quad b \in 2^\kappa,$$

all of whose proper normal-form truncations belong to  $\mathcal{T}$ .

**Theorem 18.2** (A transported weak-compactness criterion [09]). *For a strongly inaccessible cardinal  $\kappa$ ,*

$$\text{OPC}(\kappa) \iff \kappa \text{ has the tree property} \iff \kappa \text{ is weakly compact.}$$

*Proof.* The last equivalence is the standard inaccessible-cardinal tree characterization of weak compactness [18]. For the first, transport  $\mathcal{T}$  through (18.1). At level  $\alpha < \kappa$  there are at most  $2^{|\alpha|} < \kappa$  nodes. Hence it is a  $\kappa$ -tree, and the tree property yields the desired branch.

Conversely, binary prefix trees suffice; this avoids an implicit assumption about unique limit nodes in an arbitrary tree. Let  $\mathcal{R}$  be a  $\kappa$ -tree, with each level  $\mathcal{R}_\alpha$  of size less than  $\kappa$ . For each  $\alpha$  choose a nonzero ordinal  $\delta_\alpha < \kappa$  and an injection  $\iota_\alpha : \mathcal{R}_\alpha \rightarrow 2^{\delta_\alpha}$ . Such a choice is possible, for example by taking  $\delta_\alpha$  large enough to code a well-ordering of that level. Define  $\Delta_\alpha = \sum_{\beta \leq \alpha} \delta_\beta$  by ordinal addition. Regularity gives  $\Delta_\alpha < \kappa$ , and these lengths are cofinal in  $\kappa$ .

For  $t \in \mathcal{R}_\alpha$ , concatenate the labels  $\iota_\beta(t \upharpoonright \beta)$  of its predecessors at all levels  $\beta \leq \alpha$ , including the label of  $t$  itself. Here  $t \upharpoonright \alpha = t$ , and each earlier restriction denotes the unique predecessor at that height. Take the binary tree of all prefixes of all such concatenated codes. Its height is  $\kappa$ , and each level has fewer than  $\kappa$  members by the strong-limit property. Applying OPC( $\kappa$ ) through (18.1) yields a binary branch.

At every completed block  $\Delta_\alpha$ , that branch determines a unique node of  $\mathcal{R}_\alpha$ , since  $\iota_\alpha$  is injective. Any sufficiently long code witnessing the branch prefix confirms that the nodes determined at two completed blocks are in the predecessor relation. They therefore form a cofinal branch of  $\mathcal{R}$ , as required.  $\square$

This is an exact coding of a classical tree principle. It is not a new proof of the relative consistency of weak compactness. Its value is to show which additional organization of omnific numbers carries the compactness information [09].

## 18.2 The branch is not an order-topological limit

The topology naturally associated with (18.1) is the prefix or truncation topology: a basic neighborhood fixes an initial normal-form segment. It is not the order topology inherited from **No**. Indeed, **Oz** is discrete in the inherited order topology because

$$(z - 1/2, z + 1/2) \cap \mathbf{Oz} = \{z\} \quad (z \in \mathbf{Oz}).$$

Thus the branch theorem cannot be rephrased as ordinary convergence of the omnific partial sums. Moreover every proper prefix has support smaller than  $\kappa$ , whereas the completed branch has support exactly  $\kappa$ . Demanding a branch *inside*  $\mathcal{S}_{<\kappa}$  would fail even for the full binary tree, regardless of weak compactness.

In generalized analysis, Carl–Galeotti–Löwe relate a specifically *weak*  $\kappa$ -Bolzano–Weierstrass principle to the tree property for suitable ordered fields, including their surreal-based field  $\mathbb{R}_\kappa$  [21]. Their hypotheses and notion of weak convergence must not be replaced by the usual real Bolzano–Weierstrass theorem. The prefix principle above makes a related distinction in a different, explicitly combinatorial setting [09].

## 19 What large cardinals change, and what they do not

### 19.1 No change to finite algebraic truth

Both  $J$  and  $H_j$  are elementary ordered-field embeddings. Their complex lifts are elementary embeddings of algebraically closed fields. Consequently they preserve every first-order formula in those respective field languages. The distinction made in this article is in additional structure: normal forms, external supports, indexed strong sums, and their coefficient probes. It would be wrong to describe (7.4) as a failure of an ordinary field axiom.

Likewise, a measurable cardinal does not alter the formula for adding two specified old surreal sign sequences. It supplies a nontrivial elementary embedding of the surrounding set universe, and that map moves certain *entire set-indexed presentations*. The finite identities still transfer; the pointwise index set does not exhaust the transferred one.

## 19.2 Why there is no Kunen contradiction

The set-theoretic map is  $j : V \rightarrow M$ , not an elementary self-embedding  $V \rightarrow V$ . After forming surreal numbers in  $M$ , we compose with the identity inclusion into the ambient  $\mathbf{No}^V$ . That inclusion is elementary in the *ordered-field reduct*, by real closedness. It is not being claimed elementary in a structure that recovers all of the surrounding set theory from birthdays or other codes.

This matters when comparing with the repository’s birthday-enriched structures [10]. An order-field embedding can preserve specific operations and a birthday identity on its image without making the full ambient birthday expansion an elementary extension. The extra quantifiers of that expansion can range over new sign codes. Accordingly, our construction is not an exception to Kunen-type obstructions for a set-theoretically faithful elementary self-embedding.

09 makes the same point [09]. By quantifier elimination for real closed fields, both  $J$  and  $H_j$  are elementary in the ordered-field language; similarly their complexifications are elementary in the language of algebraically closed fields of characteristic zero [20]. For these definable class structures, the assertion is interpreted formula by formula, or syntactically via quantifier elimination; it is not an assertion that a set-sized satisfaction relation for all classes has been constructed. The birthday function, the omega-map, normal-form truncation and an entire summable family are additional data, and pure field elementarity does not force their preservation. In particular, composing  $j : V \rightarrow M$  with the inclusion  $M \subseteq V$  does not yield an elementary set-theoretic self-embedding of  $V$ . The two field embeddings do not evade the usual restrictions on such self-embeddings. The enriched structures considered in the repository and the announced Chen–Hamkins–Yang work concern a stronger language [10, 22].

[merge] The birthday-cutoffs report proves the relevant obstruction for the full birthday-expanded class: every class elementary self-embedding of it is the identity, by transfer of Kunen’s inconsistency (`hset : thm : kunen` in `docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets`). Its correspondence between elementary embeddings of  $H_\kappa$  and of the birthday-bounded surreal structures (`hset : thm : embeddings`) acts on sign sequences exactly as  $J$  does, and the remark following it warns, as Lemma 2.5 does here, that  $j(A)$  must not be replaced by  $j \text{ “ } A$ : the transported sign sequence has domain  $j(b(x))$  and may have coordinates outside  $j \text{ “ } b(x)$ . Neither result is used above, and neither implies the defect formula.

## 19.3 A hierarchy of information

The first failure cardinal of positive strong summation is  $\kappa$ . The first missing coefficient on the masks  $h_A$  recovers the normal measure  $U_j$ . At a supercompact seed, the wider mask family  $m_A$  recovers the associated fine normal measure. These are three different levels of data:

$$\text{critical point} \quad \longleftarrow \quad \text{normal-measure coefficient functional} \quad \longleftarrow \quad \text{seed-coefficient systems.}$$

The arrows mean forgetting information, not an implication that every normal measure extends to every seed system. In particular, two distinct normal measures can have the same critical cardinal and the same equalizer field  $\mathcal{S}_{<\kappa}$ , while their defect functionals differ by Corollary 10.4.

## 19.4 Choice and the scope of set formation

The proof does use Choice, already present in ZFC, to enumerate an arbitrary set in Lemma 2.5 and to use the conventional measure characterizations. It uses Replacement with the embedding parameter to form pointwise images and inverse images of *sets*. It does not choose a basis for the entire proper class **No**, does not appeal to a class version of Zorn’s lemma, and does not form the collection of all class endomorphisms as a set or class of its own. A definable normal ultrapower provides all the specific maps needed. [merge] The merged material keeps this scope. 09’s Theorem 13.5 uses Zorn’s lemma only on subsets of the set  $K_\kappa$ ; 06’s Lemma 9.3 uses ordinary Choice for a set of representatives.

## 19.5 What is visible at each level of structure [09]

**Proposition 19.1** (Pure field sentences do not detect large cardinals [09]). *Every parameter-free sentence about **No** in the ordered-field language has the truth value prescribed by the complete theory of real closed fields. Every parameter-free field sentence about **SC** has the truth value prescribed by algebraically closed fields of characteristic zero. Accordingly, an additional large-cardinal axiom cannot change any of these truth values, provided the same ordinary axioms of surreal arithmetic are being used.*

*Proof.* Real closed fields and algebraically closed fields of fixed characteristic have complete theories, as follows from quantifier elimination [20]. The canonical surreal structures satisfy the respective theories. The same syntactic decision of each sentence therefore applies in every ambient model of the stated foundations.  $\square$

This standard observation rules out an overly simple research program: one cannot discover a parameter-free polynomial-field sentence whose truth in the surreals switches exactly when a measurable cardinal exists. The positive theorems instead concern class embeddings, normal-form coefficients, families of prescribed cardinality and tree branches [09].

| Data retained                                  | Information obtained                                  | Important qualification                                 |
|------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------|
| Pure ordered field                             | Real closed field theory                              | No large-cardinal detection by parameter-free sentences |
| Normal forms and a support cutoff              | $\mathcal{S}_{<\kappa}$ and long-support independence | Available for uncountable $\kappa$ in ZFC               |
| A measurable-cardinal embedding                | Exact equalizer and additional-term formula           | Conditional on the embedding’s set-theoretic origin     |
| Omnific masks and a coefficient seed           | A normal or derived ultrafilter                       | Uses all subset codes and coefficient operations        |
| Covering or exact seeds on $P_\kappa(\lambda)$ | Fine or normal-fine measures                          | Standard strong/supercompactness mechanisms             |
| Prefix-coded omnific trees                     | Weak compactness at an inaccessible                   | Truncation topology, not the inherited order topology   |

(09’s table.) A birthday predicate is another way to recover substantial set-theoretic information, as the repository and the Chen–Hamkins–Yang announcement emphasize [10, 22]. 09 does not need to resolve the exact expressive strength of that predicate. Its main comparison uses a more concrete invariant: the set of additional indices in one transformed normal form.

**Why this is not simply a change of notation [09].** Several individual ingredients are standard. Nevertheless the exact equalizer puts them together in a restrictive way. The two maps have identical actions on every monomial, identical leading coefficients, the same induced action on values, and infinitesimally close relative outputs. Both are elementary as pure ordered-field embeddings and both preserve the omnific ring. Yet their agreement locus is a sharply defined, proper-class real closed subfield, and they disagree on an explicitly maximal-cardinality independent family inside one fixed set-sized Hahn field. The discrepancy is not an arbitrary error term. Its support is the exact complement  $j(\theta) \setminus j''\theta$  in an internally transformed ordinal index set. Thus the theorem converts a specific set-theoretic failure of pointwise surjectivity into a full normal-form identity, which contains more information than preservation or failure of one infinite sum alone.

## 19.6 What has actually been established [06]

The central conclusions are exact rather than asymptotic. The first comparison threshold is strict support size  $< \kappa$ ; the normal-measure landing threshold is nonstrict support size  $\leq \kappa$ . The defect is the actual series on the omitted indices, not an error estimate. Every nonzero coefficient survives. The landing criterion holds for each individual input and is insensitive to coefficients. These features distinguish the results from a generic construction of an embedding or a general assertion that large cardinals produce nonstandard algebraic structures.

The exponential theorem identifies a different threshold, on only the positive-exponent part. This supplies a concrete separation between preserving all strong sums and preserving global exponential structure: the Hahn lift has the first property and fails the second on exactly the specified inputs. The set-transport map has the opposite global pattern: it preserves Gonshor exponentiation, but its strong summation law stops at the critical cardinal.

## 19.7 Hypotheses which cannot be silently removed [06]

The target must be transitive and the canonical constructions must be identified through the absoluteness interface. The seed-closure converse uses a *set-ultrapower* presentation; it was not proved for an arbitrary elementary embedding into an arbitrary inner model. The no-landing result above  $\kappa$  uses an index set of cardinality  $\kappa$ , not merely that the critical point is  $\kappa$ .

The support sizes in the comparison theorems are external cardinalities. The bound in the compactness cover theorem is an internal cardinal bound in  $M$ . The parameter  $\lambda$  in the compactness theorems is an ambient cardinal. These distinctions prevent several tempting but incorrect shortcuts in the proofs.

No automorphism classification, Kunen inconsistency result, new large-cardinal consistency theorem, or first-order characterization in the pure surreal field is established. In particular, one should not infer a large cardinal from the existence of an arbitrary non-strong field map; such behavior already has constructions unrelated to the universe embeddings used here.

# 20 Position relative to the literature and the supplied repository

## 20.1 Classical ingredients and proposed contributions

The following separation is part of the claim, not merely an editorial qualification. Conway normal forms, the real closedness of  $\mathbf{No}$ , the algebraic closedness of  $\mathbf{No}(i)$ , Hahn reindexing, and Gonshor's exponential are established background. Normal measures derived from embeddings, the ultrapower construction, and the

fine-normal-measure characterization of supercompactness are also established background. The article uses these results to obtain an explicit comparison inside one ambient surreal field.

| Item                                     | Status in this article                                                                                                        | Relevant precedent                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Conway normal forms and strong summation | Classical input; conventions and absoluteness requirements made explicit.                                                     | Conway, Gonshor, van den Dries–Ehrlich, Ehrlich [1, 2, 3, 4].                              |
| Monomial reindexing                      | Classical Hahn construction; elementary proof supplied.                                                                       | General Hahn-field theory and the strong maps studied in [5].                              |
| Critical-index defect                    | Candidate original exact formula, including its first term and exclusion of image exponents.                                  | Standard distinction $j(I) \neq j \text{ “ } I$ supplies the set-theoretic mechanism.      |
| Equalizer and intersection               | Candidate original package: exact short-support equalizer, common-image formula, incomparability, and coefficient projection. | Real-closed-field facts are classical; the present pair is constructed from the embedding. |
| Strong-sum escape                        | Candidate original intrinsic closure threshold for $J(\mathbf{No})$ , not only a non-commutation example.                     | Strong summability itself is classical.                                                    |
| Measure coefficient                      | Candidate original arithmetic realization. The derived ultrafilter and its properties are not new.                            | Normal ultrapowers and seed measures [6].                                                  |
| Mask criterion                           | Classical ultrafilter criterion transported to an explicitly specified Hadamard algebra.                                      | Direct-product homomorphisms and ultrafilters [7].                                         |
| Exponential boundary                     | Candidate original compatibility locus for this companion and its two-closure description.                                    | Gonshor’s exponential normal-form formula [2, 3].                                          |
| Omnific and surcomplex transfer          | Consequences proved here from the real construction; no separate large-cardinal assumption.                                   | Canonical integer parts and quadratic extensions are classical.                            |

The phrase “candidate original” means that this formulation and proof package was not located in the targeted material inspected for this article. It does not assert that no specialist has previously observed some or all of the consequences. In particular, elementary embeddings and Hahn series are mature subjects; the mechanism is simple once the correct two images are compared.

06 draws its own line between established ingredients and proposed contributions [06]:

| Ingredient or statement                                               | Status in manuscript 06                                        |
|-----------------------------------------------------------------------|----------------------------------------------------------------|
| Conway normal forms; Hahn arithmetic; infinitesimal formal evaluation | Classical prerequisites, principally [1, 2].                   |
| Termwise exponent lift; strong versus ordinary additivity             | Established machinery; compare [5].                            |
| Ultrafilter seeds; fine and normal ultrafilters; seed-closure lemma   | Classical set theory, with proofs included where used [6, 12]. |
| Exact equalizer and missing-index defect; precise exponential ratio   | Proposed comparison results proved there.                      |

| Ingredient or statement                                                    | Status in manuscript 06                                                 |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Universal individual-input landing criterion and normal-measure trichotomy | Principal proposed contributions, derived from the preceding machinery. |
| Coefficient recovery of the normal measure; omnific witness versions       | Proposed explicit constructions, not a new definition of measurability. |
| Support covers and exact descent for compactness principles                | Exact reformulations; no extra consistency-strength claim.              |

Its literature check was targeted, not exhaustive. It included the primary sources in the bibliography and the repository’s relevant indices and scope statements. A proof being short once two theories are placed together does not establish priority. Conversely, a result can be a useful new formulation without settling a named conjecture [06]. 06 adds that Kaplan–Krapp–Serra already construct a non-strongly additive automorphism, so mere existence of such a map is not a contribution; ultrapower seeds, normal measures and sequence-closure criteria are established set theory [6, 12]; the novelty 06 proposes lies in the exact *interaction* of these constructions, especially the universal landing criterion for each individual normal form.

09’s comparison [09]. Hahn lifting and strong additivity are established subjects. Kuhlmann and Serra study coefficient and exponent lifts and strongly additive Hahn-field automorphisms [16]. Kaplan, Krapp and Serra apply related ideas to the surreals and distinguish strong additivity from preservation of additional exponential structure [5]. The maps here are generally *embeddings*, not automorphisms; 09’s proposed addition is the exact critical-point equalizer and its term-by-term discrepancy formula. The field  $\mathcal{S}_{<\kappa}$  belongs to the cardinal-bounded Hahn-field tradition [15]; 09 does not claim the short-support field construction as new, and its proof includes singular uncountable cardinals and separates this cardinal condition from birthday cutoffs, which are studied classically in [3]. Connections between large cardinals and surreal-based generalized analysis already occur in Carl–Galeotti–Löwe [21]; their weak Bolzano–Weierstrass principle is important precedent for Section 18, and no claim to have initiated that connection is made. 09’s dependency ledger marks the exact discrepancy and the almost-disjoint independence theorem as its main proposed contributions; the absoluteness lemma, the short-support field and the strong lift as standard constructions with proofs included; the normal and seed coefficient formulas as explicit coding results with the seed theory credited; the numerical measurability criterion as an infinitary criterion, not a bare-field equivalence; and the strong compactness, supercompactness and weak compactness statements as established set theory transported to the codes.

## 20.2 The repository snapshot actually inspected

The repository was read at commit

0865f043aec113c14c69ef45006bbc7546a4e75a.

The targeted inspection comprised the root `README.md`, the `docs/README.md` report index, the `READMEs` for *Surreal Fields Across Set-Theoretic Universes* and *Birthday Cutoffs and Hereditary Sets*, and the former article’s LaTeX preamble [8, 9, 10]. The style here follows that report’s restrained theorem-centered layout. This was not a line-by-line audit of the whole repository or its formalization ledger.

The cross-universe report describes omitted cuts, fresh sign sequences, external saturation, first new birthdays, and Prikry examples. The birthday report describes hereditary-set interpretations and the recovery of set-theoretic information from birthday-enriched structures. Its `README` explicitly attributes the announced

full-class bi-interpretation to Chen–Hamkins–Yang. Those earlier themes are relevant context but are not used as unproved premises of our theorems.

The present article takes a different route. It does not add birthday coding to recover the entire set universe, and it does not change the universe by forcing. It compares the pointwise elementary action and its strong monomial extension inside a fixed universe. The output is an arithmetic defect whose support and coefficients can be calculated exactly. The constant-term discussion of omnific integers is also kept separate from the repository’s proposed set-sized quotient theorems [11].

The root README reports existing formal Hahn infrastructure, including summable families and exponent-embedding maps. That makes a substantial part of the algebraic proof a plausible formalization target. It does *not* mean the large-cardinal construction, the absoluteness lemma, or any theorem specific to this article is already checked in Lean. The repository itself warns that research reports, finite tests, and formal proof coverage are different kinds of evidence.

06 and 09 inspected the same snapshot [06, 09]. 06 read the root README, the documentation index, directory trees, and the READMEs of the birthday-cutoff and universes reports, and records the identifier as a “tree object”, not asserted to be a commit (it is in fact the commit 0865f04, “Merge reviewed documentation after angular multiplicity validation”; Appendix D). 09 read the main documentation and the introductions and scope statements of the universes, birthday-cutoffs and bounded-support reports. Both describe those reports as 06 and 07 do: birthday interpretations, elementary-embedding transfer and forcing comparisons belong to the project and are not advertised as new; none of the new proofs requires an unreviewed repository theorem. 09 adds that the repository’s *Bounded-Support Hahn Arithmetic* concerns support bounded in the *order* of a specified exponent group [23], while its independence theorem concerns support bounded in *cardinality*, with a coefficient field containing short-support surreals with arbitrary exponents outside the selected set-sized Hahn field; these are genuinely different base fields. 09 also retains the repository’s separation of written arguments, finite tests and checked Lean declarations. 06’s Remark after Theorem 11.4 keeps its results apart from the repository’s work on omnific automorphisms and set-sized quotients.

### 20.3 The collection at the placement commit [merge]

The sources’ repository statements were checked at the placement commit 21375f8 and at the time of writing. The reports they cite are unchanged since the snapshot, except the set-sized-quotients report, which has since been extended (07 cites only its catalogue entry). The statements stand: the universes report proves absoluteness of the arithmetic and of cuts (`univ:prop:absolute`), not of normal forms; the bounded-support report (`bst:`) bounds supports in order, not in cardinality; the root README describes Lean infrastructure in which strictly increasing additive exponent embeddings preserve Hahn sums, and no statement of this report is formalized. The relations to the neighbouring reports are:

- `docs/foundations-and-computation/definable-surreals-and-omnific-integers (dsn:)`. Lemma 2.6 supplies the omega-map and normal-form absoluteness that its `dsn:q:support` leaves unproved, for same-ordinal inner models with the same reals (Remark 2.9); that report now carries a reciprocal remark. The question itself (classify the intermediate support fields) stays open.
- `docs/surreal/omnific-preserving-automorphisms (opa:)`.  $J$  gives a negative answer, under a measurable cardinal, to the first two clauses of `opa:as:q:embeddingstrong` (Remark 11.6); that report records this, writing  $J$  as  $\hat{J}$ .
- `docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (hset:)`. Its embedding correspondence `hset:thm:embeddings` and Kunen transfer `hset:thm:kunen` are

the birthday-enriched side of Section 19; its remark on  $j(A)$  versus  $j \text{ “ } A$  is the same distinction as Lemma 2.5.

- `docs/foundations-and-computation/surreal-fields-across-universes (univ:)`. Its `univ:prop:absolute` is the arithmetic part of Lemma 2.6. Its forcing and saturation themes are the subject of Questions 21.11 and 21.20; none of its named questions is answered here.
- `docs/surreal/transcendence-over-bounded-support (bst:)`. Maximal transcendence over order-bounded supports there; Theorem 13.5 is over cardinality-bounded supports. Neither implies the other.
- `docs/surreal/set-sized-quotients-of-omnific-integers (osq:)`. The constant-term discussion of Section 11 is kept apart from its quotient theorems, as all three sources say.
- `docs/surreal/independent-surreal-copies (isc:)`, written concurrently as part of batch 31 (781b19e) and now cited by label. It identifies  $H_j$  as its strong exponent lift for the exponent map  $J$  (`isc:thm:lift`), and relates its linear-disjointness theorem for full Hahn fields (`isc:thm:sliceddisjoint`) to Question 21.6 without answering it (see the note after that question):  $J(\mathbf{No})$  is not a full Hahn field (Theorem 8.6), and  $C_j$  is a proper class. The question stays open here.

## 20.4 Relation to current automorphism work

Kaplan–Krapp–Serra [5] provide a contemporary framework for strongly additive maps, monomial actions, valuation structure, and additional surreal operations. We use their definitions and general Hahn context, not a particular non-strong automorphism construction from that paper. Our maps are nonsurjective embeddings, and their valuations are transported by  $J$  rather than fixed pointwise. Results that assume bijectivity or pointwise preservation of the value group must not be applied to this pair without checking those hypotheses.

No claim is made to settle one of that paper’s numbered questions. Rather, our construction answers a new precisely stated family of structural questions under an explicit large-cardinal hypothesis. A specialist comparison with the unpublished details of related projects would be appropriate before submitting a priority claim.

## 21 Further research questions and concrete next steps

The following questions are proposed by this article. They are not presented as an established list of famous open problems. Each isolates something not proved above and indicates what a meaningful answer would require.

**Question 21.1** (Optimal consistency strength). Suppose two ordered-field embeddings  $P, Q : \mathbf{No} \rightarrow \mathbf{No}$  fix  $\mathbb{R}$ , agree on all ordinals and monomials,  $Q$  is strongly linear, and their equalizer and image intersection satisfy the formulas of Theorem 6.1 for an uncountable  $\kappa$ . Suppose also that  $P$  fixes every ordinal below  $\kappa$ , moves  $\kappa$ , and commutes with the omega-map. Must  $\kappa$  be measurable, or can such a pair exist without an ambient elementary embedding of set theory?

The obstruction to a converse is precise: the first defect coefficient is not automatically a character for the Hadamard product merely because  $P$  is multiplicative for ordinary surreal multiplication. Theorem 10.9 solves the converse only when the coefficient-algebra structure and its small-union law are available.

**Question 21.2** (Classification of strong companions). For an arbitrary real-fixing omega-map-compatible ordered-field embedding  $P$ , classify the equalizer of  $P$  with the Hahn reindexing map  $T_P$ . Which equalizers are cardinal support bounds, and which depend on the shape, cofinality, or iterated structure of the exponent support?

The construction  $T_P$  is always available by Lemma 4.1. What is special to the present large-cardinal setting is the exact missing-index formula and the absence of cancellation in it.

**Question 21.3** (Recovering an embedding from fewer coefficient probes). What is the least family of masks and coefficient positions that determines  $j$  on a specified hereditary-size universe  $H_\lambda$ ? Can normal-form probes of bounded complexity recover more than the derived normal measure at the critical point?

Using all subsets of all ordinals with all coefficients would retain far more information than a single normal measure. The meaningful task is to specify a restricted family and prove a sharp reconstruction or nonreconstruction theorem.

**Question 21.4** (Measure order from defect data). Can the Mitchell ordering, or another comparison relation between normal measures on  $\kappa$ , be recovered from a natural algebraic relation between their coefficient functionals  $\chi_j$ , without naming the ambient ultrapower models as extra predicates?

Corollary 10.4 gives injective detection of individual measures. It gives no criterion for their relative position in an ultrapower hierarchy.

**Question 21.5** (Intrinsic supercompact seed systems). Can the seed construction of Theorem 16.2 be formulated without choosing an enumeration of  $P_\kappa(\lambda)$ ? Is there a natural coordinate-free compatibility law for the resulting systems as  $\lambda$  varies that detects the appropriate closure of  $M$ ?

Any proposal must preserve fineness and normality and must distinguish an actual seed in  $j(P_\kappa(\lambda))$  from an arbitrary missing-image index. Merely having many extra coefficients does not imply supercompactness.

**Question 21.6** (Joint algebraic dependence of the images). Are  $J(\mathbf{No})$  and  $H_j(\mathbf{No})$  linearly disjoint over  $C_j$ ? More modestly, for a finite tuple  $x$ , can one decide the transcendence degree of  $H_j(\mathbf{No})(J(x))$  over  $H_j(\mathbf{No})$  from the missing-image supports?

Every single element outside the intersection is transcendental over the opposite real-closed image field. That does not decide joint algebraic independence of a tuple, and the exact intersection theorem is not a substitute for a disjointness proof. *Related result* (batch 31). The independent-copies report docs/surreal/independent-surreal-copies proves that full Hahn fields on two subgroups  $G_1, G_2$  of one ordered group, set or class, meet in the full Hahn field on  $G_1 \cap G_2$  and are linearly disjoint over it (isc:thm:sliceddisjoint). This does not answer the question. The field  $H_j(\mathbf{No}) = \mathcal{H}(\Gamma_j)$  is the full Hahn field on the subgroup  $\Gamma_j$  by (4.1), but  $J(\mathbf{No})$  is not a full Hahn field on any subgroup: it is not closed under positive strong sums of  $\kappa$  terms by (8.9), whereas a full Hahn field on a subgroup is closed under every set-indexed strong sum of its members. Nor is  $C_j$  such a field: it contains the monomials  $\omega^{-\alpha}$ ,  $\alpha < \kappa$ , but not their strong sum, which has  $\kappa$  terms (Theorem 6.1). The question stays open.

**Question 21.7** (Alternating the two closures). Starting with  $C_j$ , what field is obtained by alternating logarithmic field closure and external strong-sum closure? Do finitely many alternations stabilize, and can the result be characterized directly from the exponent image  $J(\mathbf{No})$ ?

Theorem 15.7 determines each first closure exactly. Their incomparability makes the mixed closure a genuinely different question. One should distinguish closure under sums from closure under all normal-form subseries or the omega-map.

**Question 21.8** (Differential compatibility). For a specified canonical surreal derivation, what is the exact locus on which  $J$  or  $H_j$  commutes with it? Can a differential analogue of (15.2) be expressed using support or the iterated logarithmic structure of the exponents?

The exponential locus is settled here. A derivation may depend on further normal-form and logarithmic data, so its preservation is not a consequence of being an ordered-field embedding. No differential claim should be imported without verifying the actual chosen derivation.

**Question 21.9** (Definability in weaker signatures). Which of  $\mathcal{S}_{<\kappa}$ ,  $C_j$ , and the coefficient projection  $\Pi_j$  are definable in natural reducts of the structure with the maps  $J, H_j$ ? Can any of the large-cardinal information survive after the embedding symbols and the Hadamard product are forgotten?

With both maps named,  $\mathcal{S}_{<\kappa}$  is the immediate equalizer formula. The question concerns losing symbols, not that trivial definition. Pure field elementarity alone cannot recover the external strong-sum threshold.

**Question 21.10** (Extending the coefficient-projection isomorphism). Under what explicit additional hypotheses does the isomorphism  $\Pi_j : J(\mathbf{No}) \rightarrow H_j(\mathbf{No})$  extend to an automorphism of a larger specified surreal subfield, or of the full ordered field? Which extension requirements on monomials, simplicity, or exponentiation obstruct it?

The given isomorphism fixes the common field and all common monomials. It is not globally multiplicative on  $\mathbf{No}$ , so its existing definition cannot simply be reused as the desired extension.

**Question 21.11** (Forcing and lifted embeddings). If an embedding  $j$  lifts through a forcing extension, which of its old coefficient defects are preserved and which new supports appear? Can the lifting criterion be expressed through a commuting square of strong companions and coefficient projections?

This would connect the present arithmetic construction to the repository's cross-universe saturation work. One must first specify the ground and extension, verify the lift, and distinguish new index sets from changed cardinalities; no automatic preservation is asserted.

**Question 21.12** (A reusable formal theorem package). Can the exact defect, image intersection, and closure theorems be formalized in a way that separates abstract ordered Hahn algebra from the set-theoretic embedding interface? What is the smallest verified interface sufficient for the elementary-embedding application?

The proposed architecture in Appendix A treats the critical-point and absoluteness obligations as explicit theorems to be proved, not as silently trusted axioms. This would make the genuinely new algebraic consequences independently reusable and auditable.

[merge] The twelve questions above are 07's. The nine questions of 06 and the thirteen of 09 follow, each set with its source's own framing; all 34 are kept, and where two sources ask related questions the overlap is noted rather than merged, since their formulations differ. One question is partly answered inside this report: the exponential half of Question 21.30 (09) is settled by 06's and 07's Theorem 15.3. All others remain open here.

## 21.1 Questions of manuscript 06 [06]

06's questions are proposed directions, not assertions that each is a previously published open problem. They are designed to distinguish new mathematical content from immediate reformulations of the theorems already proved.

**Question 21.13** (An intrinsic converse without a universe embedding). Can a suitably enriched surreal structure characterize which pairs of maps arise as  $(J, H_j)$  from an elementary universe embedding?

The coefficient formula recovers a candidate measure from a map on the codes  $\tilde{h}_A$ , but field multiplication alone does not give coefficientwise intersection of binary supports. Determine a minimal additional operation or relation—for example controlled coefficientwise multiplication on a designated coding family—which makes the ultrafilter laws follow intrinsically. A genuinely new converse would have to construct the ultrafilter or embedding, not assume it in the language from the outset. (Compare Question 21.1 and Question 21.22; the criteria Theorem 10.9 and Theorem 10.10 add exactly such a coefficientwise operation.)

**Question 21.14** (Which landing spectra occur?). For a fixed critical point  $\kappa$ , classify the downward-closed classes  $\mathcal{L}(j)$  realized by set-ultrapower embeddings. How does this spectrum constrain the least possible cardinality of an index set?

The normal-measure answer is exactly the cardinals at most  $\kappa$ . The index-size obstruction supplies a necessary bound whenever an internally short cover exists. Beyond these statements, investigate which successor, limit, or singular endpoints can occur and whether two embeddings with the same landing spectrum can have substantially different support-cover spectra.

**Question 21.15** (Extender targets and failure of the seed–closure converse). For elementary embeddings not generated by one set-ultrapower seed, what additional hypothesis makes  $j \restriction \lambda \in M$  imply closure under all external  $\lambda$ -sequences of elements of  $M$ ?

The surreal landing theorem still reduces to seed membership, but the proof of sequence closure uses a single evaluation parameter  $[\text{id}_I]$ . An extender presentation may require a family of parameters. A useful theorem would identify the necessary seed support and express it as a strengthened descent condition for a family of normal forms.

**Question 21.16** (Higher coefficients and derived measures). What structural information about an embedding is recoverable from the whole coefficient profile of  $J(\tilde{h}_A)$  as  $A$  varies?

The explicit normal form already gives  $[\omega^{-\beta}] J(\tilde{h}_A) = 1 + \mathbf{1}_{j(A)}(\beta)$  for every  $\beta < j(\kappa)$ . Thus it records the ultrafilters derived from all such seeds. The nontrivial next problem is to recover relations among these ultrafilters—factor maps, equivalences, or iteration data—from relations among coefficient observables, rather than merely restating the membership formula. (Related: Questions 21.3 and 21.25.)

**Question 21.17** (Iteration and multilevel transport defects). Develop a coherent comparison theory for iterated elementary embeddings and successive Hahn lifts, keeping track of which intermediate values actually belong to the next domain.

A formal composition of termwise lifts is straightforward in a common ambient universe, but iteration through inner models introduces descent conditions. Determine whether a hierarchy of first missing indices gives an invariant of the iteration independent of the chosen presentations. Any proposed cocycle identity must specify these domains before performing algebra with the defects. (Related: Question 21.24.)

**Question 21.18** (Canonical derivations and transport). How do  $J$  and  $H_j$  interact with the Berarducci–Mantova derivation or other specified strongly additive surreal derivations?

First prove the necessary absoluteness or naturality theorem for the chosen derivation; it is not a consequence of field elementarity alone. Then compare differentiation of the missing-index formula with transport of derivatives. The goal is an exact commutation locus comparable to (15.2), not a generic statement about differentiating formal sums. (Related: Questions 21.8 and 21.30.)

**Question 21.19** (Birthday cost of a critical-support detector). For a fixed  $\kappa$ , what is the least birthday of a surreal with  $\kappa$  nonzero normal-form terms? What is the least birthday of an omnific witness to failure of exponential compatibility?

06's theorems use the explicit  $h_\kappa$  and  $\zeta_\kappa$ , but do not optimize their birthdays. Sharp answers would connect this support-based theory to the repository's birthday investigations without confusing the two invariants. Normal-form length, support cardinal, and sign length must be tracked separately throughout.

**Question 21.20** (Forcing and lifted coefficient codes). When an elementary embedding lifts through a forcing extension, how do the code families  $\tilde{h}_A$  and their defect coefficients change for new subsets  $A \subseteq \kappa$ ?

For old subsets the original measure code is fixed data, while new subsets require an extension of the measure. Formulate a preservation theorem distinguishing persistence of old coefficients, extension of the coding domain, and changes to the landing spectrum. This could connect the present theory to forcing across surreal universes without assuming that ground-model and extension normal-form families coincide. (Related: Question 21.11.)

**Question 21.21** (Local restrictions and omega-preserving automorphisms). Which set-sized restrictions of  $J$  extend to automorphisms of a surreal structure containing the monomial map, or both the monomial map and Gonshor exponentiation?

The full  $J$  cannot be such an automorphism by Proposition 7.4. A local extension theorem, or a concrete local obstruction, would nevertheless bear on the richer automorphism questions in [5]. Homogeneity of the pure ordered field does not by itself settle the question for the monomial-enriched structure.

## 21.2 Questions of manuscript 09 [09]

09's questions are proposed by that manuscript. Except where a published source is explicitly named, no claim is made that each is a previously formulated open problem. Their purpose is to specify what a next theorem should say and which extra hypotheses would have to be controlled.

**Question 21.22** (Removing coefficientwise multiplication). How far can the extra coefficient algebra be removed from a numerical criterion for measurability? In the compatible-field-embedding criterion of Theorem 10.10, is preservation of Hadamard multiplication on all masks implied by weaker assumptions about order, the omega-map, truncations, and  $< \kappa$ -strong additivity?

A useful first target is preservation only of disjointness of mask supports, together with binary-valued coefficients at the designated probe. Any proposed converse should explicitly explain how it yields a *set* of subsets of  $\kappa$  and why that set is closed under fewer than  $\kappa$  intersections. Bare field elementarity cannot supply this by itself.

**Question 21.23** (The bare exact-equalizer principle). For which uncountable cardinals  $\kappa$  can there exist ordered field embeddings  $E, H : \mathbf{No} \rightarrow \mathbf{No}$  agreeing on every monomial, with  $H$  fully strong and  $\text{Eq}(E, H) = \mathcal{S}_{<\kappa}$ , without assuming that  $E$  comes from a set embedding? What is the consistency strength of adding omega compatibility for  $E$ ?

The current theorem proves one implication from measurability. It does not prove that the equalizer statement alone implies measurability, nor that it is impossible at an ordinary uncountable cardinal. Separating the algebraic condition from an encoded ultrafilter would clarify the genuine large-cardinal content of the main construction. (Related: Question 21.1, which adds more conditions on the pair.)

**Question 21.24** (Iterated discrepancies). Given an iteration of normal ultrapowers, can its successive missing-index sets be described by a normal-form calculus that is functorial under composition? Which terms distinguish the pointwise image of a pointwise image from the full iterated image of an ordinal index set?

A concrete starting problem is a two-step iteration with critical points  $\kappa$  and  $j(\kappa)$  inside the appropriate models. Each step has its own formula (5.3); the difficulty is comparing the strong lift of the composite with the composite of the two strong lifts when their domains and internal series differ. Statements about all maps must specify the universe in which each summability assertion is evaluated.

**Question 21.25** (What one coefficient forgets). Which additional families of seed coefficients determine the seed hull or an extender associated with  $j$ , rather than only one normal measure? Can the passage from one seed to several seeds be expressed by explicitly compatible operations on the corresponding omnific masks?

A single normal measure does not in general specify every possible embedding that derives it. A successful reconstruction theorem should state exactly which factor of the embedding is recovered. Compatibility under projections between finite tuples of seeds is a natural first milestone.

**Question 21.26** (Mitchell-order information). Can comparison or iteration of measures be recognized in a system of coefficient probes that is stable under the induced surreal embeddings? In particular, can membership of one measure in another ultrapower be encoded without merely reinstating the full membership relation as an extra predicate?

This is deliberately stronger than coding each ultrafilter separately. The problem is to represent the relation between the codes in a useful surreal language, not simply to assign another number to another set. (Related: Question 21.4.)

**Question 21.27** (Beyond  $2^{<\kappa} = \kappa$ ). When  $2^{<\kappa} > \kappa$ , what is the largest family of explicitly constructed positive omnific masks algebraically independent over  $\mathcal{S}_{<\kappa}$ ? Can one still attain  $2^\kappa$  in  $K_\kappa$  by replacing the almost-disjoint branch construction?

Theorem 13.3 applies to every available  $\kappa$ -almost-disjoint family, so one route is combinatorial. Another would replace private coordinates by a stronger support-separation argument. A complete answer should distinguish limitations of the particular masks from limitations on the transcendence degree itself.

**Question 21.28** (Singular support cutoffs). For singular uncountable  $\kappa$ , which analogues of the equalizer and summability-threshold results can be realized by abstract Hahn embeddings? Can cofinality and support cardinality produce different equality loci?

The field  $\mathcal{S}_{<\kappa}$  is already real closed for singular  $\kappa$  (Theorem 3.1). An ordinary nontrivial elementary embedding of the universe cannot have such a critical point under the present hypotheses. Thus an affirmative algebraic construction at singular  $\kappa$  would necessarily use a mechanism different from the normal-ultrapower one.

**Question 21.29** (Other coefficient fields and ordered groups). How much of Theorem 13.3 survives for Hahn fields with coefficients of positive characteristic, nondiscrete valuation on the coefficients, or exponent groups without the displayed free rational subspace?

The selected coefficient in the proof has multiplicity one, so it avoids one familiar source of cancellation in positive characteristic. The main remaining issue is to formulate the correct small-support coefficient base and to supply enough independent exponent coordinates. This offers a plausible extension beyond surreal numbers without changing the core separation argument.

**Question 21.30** (Exponential and differential refinements). How do the pair  $(J, H_j)$  and the discrepancy interact with the Gonshor exponential and with a specified surreal derivation? Can one calculate an exact equality locus for either operation analogous to Corollary 5.6?

09 proves no compatibility with these operations, and adds that an approach through their recursive definitions must verify absoluteness, and an approach through transseries must verify all relevant summability conditions. The questions about enriched automorphisms in [5] provide context, but those automorphisms and the generally nonsurjective embeddings here should not be identified. [merge] *Status*. For the exponential this is answered by Theorem 15.3 (06, 07): the exact locus of  $H_j \circ \exp = \exp \circ H_j$  is  $|\text{supp}(p(x))| < \kappa$ , and  $J$  commutes with  $\exp$  everywhere (Lemma 15.1), with the absoluteness of the recursion verified there. The derivation half is open, as in Questions 21.8 and 21.18.

**Question 21.31** (Set-sized restrictions). For which pairs of birthday cutoffs can the main pair of embeddings be restricted to set-sized real closed surreal fields, retaining the same support equalizer on the common domain? What are the optimal birthday bounds on the images and the discrepancy?

The identity  $b(J(x)) = j(b(x))$  is a starting bound for one map. The strong lift may need a different support-to-birthday estimate. A complete theorem should name its cutoffs and closure properties rather than appeal to the proper-class theorem informally. (Related: Question 21.19.)

**Question 21.32** (Omnific arithmetic without masks). Can any part of the measure-recovery mechanism be expressed using only ordinary omnific addition, multiplication, order and a small number of named maps or parameters? In particular, how much of normal-form coefficient extraction is definable in such a structure?

The lack of nontrivial idempotents in an integral domain makes a naive identification with a Boolean algebra impossible. One could instead seek interpreted quotients or definable families. This direction should be compared with the repository's omnific definability and quotient projects, not assumed to follow from them.

**Question 21.33** (Higher compactness in truncation spaces). Can stronger branch principles on two-parameter or seed-indexed families of omnific normal forms distinguish weak compactness, strong compactness and supercompactness using one coherent truncation-space framework?

The distinction between a covering seed and an exact seed suggests a specific invariant to retain. Any topological formulation must specify the topology and the allowed cardinalities of covers or branches; the ordinary inherited order topology on  $\mathbf{Oz}$  is too discrete for the prefix argument. (06's support covers and exact descent, Section 17, are one such pair of invariants, stated with embeddings rather than branches.)

**Question 21.34** (A checked exact-equalizer theorem). Can the main theorem be formalized with an abstract elementary-embedding interface whose only large-cardinal inputs are the critical-point and normal-form transport properties, while separately proving the bridge from a genuine ultrapower?

A practical first deliverable would formalize Hahn subseries partition, strong exponent lifts, and the private-coordinate independence argument in a set-sized Hahn field. The second deliverable would formalize Lemma 2.5 and the exact comparison theorem for an ordinal-indexed normal-form transport. Only the third would connect that interface to actual surreal sign sequences and ultrapowers. This ordering would keep the algebraic verification independent of the largest foundational infrastructure task. (Related: Question 21.12.)

## 22 Conclusion

The difference between  $j(I)$  and  $j \text{ “ } I$  becomes a literal normal-form difference in surreal arithmetic. Comparing the elementary-induced map with the strongly linear map having the same monomial values exposes that difference without sacrificing finite algebra.

The outcome is stronger than a single counterexample to summation preservation. There is an exact equalizer, an exact intersection, a canonical isomorphism of two incomparable immediate image fields, and an intrinsic positive-family escape threshold. Their common field has two different canonical enlargements: logarithmic closure produces the elementary image, while strong-sum closure produces its Hahn companion. One coefficient of the difference recovers the derived normal measure; shifting the code makes it purely infinite and omnific. The same mechanism works over complex coefficients and at supercompact seeds.

These statements answer the structural questions posed in this article under a measurable-cardinal hypothesis. They are offered for independent verification and comparison with the literature, not as a claim that the large-cardinal hierarchy or a named longstanding conjecture has been settled. The strongest next step is to test the converse: how much of the set-theoretic embedding can be forced by the arithmetic signature alone?

[06] The more global issue is not whether the termwise series exists—it always exists in the ambient surreal field—but whether it exists in the target model. For every individual normal form that question is exactly seed membership. In a set-ultrapower it is exactly sequence closure. Normal measures give the strict/non-strict two-threshold picture, while fine and normal fine measures give support covers and exact descent at larger cardinals. The constructions remain meaningful in the surcomplex field and in the omnific-integer rings, where explicit coefficient codes recover normal measures. Thus large-cardinal hypotheses have a precise manifestation in arithmetic of canonical infinite supports, without invoking a new finite algebraic law, a blanket automorphism theorem, or an unproved consistency claim.

[09] The short-support field is not merely an incidental agreement set. It is real closed, has a canonical omnific integer part, and admits an explicit large algebraically independent family of long-support omnific extensions. At a measurable cardinal these are simultaneously independent witnesses to the failure of full strong additivity, and the measure itself appears as a coefficient of the same sort of number. The remaining challenge is to reduce the extra language needed to observe these effects. Normal-form support, coefficientwise multiplication, truncation trees and seed data make the set-theoretic information visible. The pure field language does not. The exact boundary between those two levels is a concrete research program rather than a reason to conflate ordinary surreal arithmetic with the entire large-cardinal hierarchy.

## A Proof dependencies and a formalization plan

### A.1 Dependency order

The proof does not use the repository’s unreviewed set-theoretic bi-interpretation or quotient theorems as assumptions. Its main dependency chain is

- canonical surreal/Hahn theory + elementary embedding with critical point
- ⇒ absolute normal forms and the missing-index lemma
- ⇒  $J$ ,  $H_j$ , exact defect and exponent-range exclusion
- ⇒ equalizer, intersection, coefficient projection, strong escape
- ⇒ omnific, surcomplex, and mask-coefficient consequences.

The logarithmic-defect homomorphism adds only the standard near-one Hahn logarithm. The exact exponential locus and two-closure theorem add the canonical Gonshor exponential and its purely infinite normal-form formula. The supercompact seed theorem adds the separate  $\lambda$ -sequence-closure hypothesis on  $M$ .

## A.2 A proposed Lean interface, not a formalization claim

A realistic implementation should begin with a universe-bounded Hahn workspace, then state separately how that workspace represents canonical surreal normal forms. The following components are logically distinct.

| Component             | Obligation                                                                                                                             | Verification status here                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Exponent reindexing   | An additive order embedding preserves reverse well-ordered support, finite convolution, and summable families.                         | Written proof; repository describes related infrastructure.                  |
| Canonical bridge      | Sign sequences, Conway arithmetic, the omega-map, and normal-form reconstruction agree with the Hahn representation.                   | Classical input; not rebuilt or kernel-checked here.                         |
| Universe absoluteness | Old normal forms and their strong sums are unchanged on passage from $M$ to $V$ .                                                      | Written reduction to the constructive sign-expansion theorem.                |
| Embedding interface   | Images of index functions; fixed coefficients; $j(\theta) \setminus j \text{ “ } \theta$ ; critical-index and range-reflection lemmas. | Written proofs; no Lean set-theory implementation supplied.                  |
| Defect theorem        | Split a transformed summable family into image and missing-image indices; prove first-term and support exclusion.                      | Written proof; finite deletion identity checked separately.                  |
| Image geometry        | Exact intersection, projection isomorphism, real closedness, immediacy, and strong-sum escape.                                         | Written proofs only.                                                         |
| Measure interface     | Indicator masks, Hadamard characters, completeness, normality, and supercompact seeds.                                                 | Written proofs; finite Boolean characters checked only as a finite analogue. |
| Exponential interface | Absolute recursive exponential, monomial values on purely infinite inputs, and infinitesimal power-series evaluation.                  | Classical input and written consequences; not formalized here.               |

The measurable-cardinal assumption must not be inserted as an unnamed “surreal summation axiom.” A useful interface can abstract the exact properties of  $j$  for the algebraic proofs, but the application to a normal ultrapower still needs an explicit model of the set-theoretic construction. That boundary is essential for avoiding a formal proof whose interesting mathematics has merely been assumed.

### A.3 06's dependency map and formalization route [06]

06's main proof chain is

canonical normal forms and absoluteness  $\implies (J, H_j)$  and their array formulas  
 $\implies$  equalizer and missing-index defect,  
 array formulas and internal support extraction  $\implies$  universal landing criterion,  
 landing criterion and ultrapower seed lemma  $\implies$  sequence-closure detector,  
 landing criterion and index-size obstruction  $\implies$  normal-measure trichotomy.

The coefficient-recovery theorem uses the array formula directly. The strong-compactness criterion uses support extraction and the classical cover seed; the supercompactness criterion adds exact seed membership. The exponential formula uses the equalizer theorem, infinitesimal formal evaluation, and the classical decomposition of Gonshor exponentiation. The surcomplex and omnific statements are restrictions or coefficient-field extensions of these arguments.

No Lean proof or automated verification of the new large-cardinal claims is included. The repository's existing Hahn layer is relevant infrastructure, but its presence is not formal verification. A useful formalization should isolate four interfaces. First, prove the algebraic reindexing theorem for a Hahn field over a set-sized coefficient field and an ordered-group embedding, together with the finite-fiber proof of strong additivity. Second, formalize the array statement: retaining an injected subset of a nonzero ordered coefficient array gives equality precisely when no indices were lost, and the least omitted index determines the leading term of the defect. These are independent of large-cardinal foundations. Third, give an explicit normal-form/sign-sequence bridge with the absoluteness theorem for transitive universes. This is the most foundationally sensitive interface and should not be replaced by an unproved coercion between internal and external Hahn fields. Fourth, formalize ultrapower seeds, the representative-function proof of sequence closure, and the index-size union obstruction. The final theorems then become short compositions of these interfaces. Finite truncation experiments can check coefficient masking and convolution identities, but they cannot validate the critical-point or nonprincipal-measure assertions: a finite set does not carry a nonprincipal ultrafilter. For that reason 06 includes no finite simulation advertised as evidence for the large-cardinal steps.

### A.4 09's dependency ledger and formalization boundary [09]

09's main argument is organized so that neither a conjectural exponential identity nor a repository bi-interpretation theorem is needed.

| Result                                             | Required ingredients                                                            | Status in 09                                      |
|----------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------|
| Absolute normal forms,<br>Lemma 2.6 (second route) | Absolute sign-prefix recursion; Conway<br>normal forms; truncations are simpler | Written proof relative to<br>standard foundations |
| Short-support field, Theorem 3.1                   | Real closed divisible-group Hahn fields; finite<br>coefficient support bounds   | Standard construction, proof<br>included          |
| Strong lift, Theorem 4.3                           | Increasing additive exponent embedding;<br>finite Hahn convolution              | Standard lifting mechanism,<br>proof included     |
| Exact discrepancy, Theorem 5.1                     | Normal-form transport; subseries partition;<br>first missing index              | Main proposed contribution                        |
| Summation threshold and omega<br>obstruction       | Exact discrepancy; explicit ordinal families                                    | Proposed consequences with<br>proofs              |

| Result                                           | Required ingredients                                                                          | Status in 09                                           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Independence, Theorem 13.3                       | Small rational coefficient span; private coordinates; maximum total degree                    | Main proposed contribution; no large cardinal required |
| $2^\kappa$ family, Theorem 13.5                  | Independence theorem; binary-branch almost-disjoint family                                    | Proposed application with explicit cardinal hypothesis |
| Normal and seed coefficient formulas             | Absolute mask construction; standard seed ultrafilters                                        | Explicit coding results; seed theory credited          |
| Numerical measurability criterion, Theorem 10.10 | Strong additivity on disjoint masks; Hadamard compatibility; measurable-cardinal construction | Infinitary criterion, not a bare-field equivalence     |
| Strong/supercompactness interpretation           | Fine and normal-fine ultrafilter characterizations                                            | Established set theory transported to the codes        |
| Omnific prefix principle                         | Binary coding of a $\kappa$ -tree; inaccessible tree characterization                         | Established weak compactness transported to the codes  |

09’s mathematical proofs are written proofs, not Lean proofs. No finite program can test the existence of a measurable cardinal by enumerating examples, and no such test is offered as evidence. The repository contains checked Hahn and algebraic infrastructure, but 09’s claims have not been imported into that development or built against its Lean version. A future formalization should state the elementary-embedding interface as assumptions and separately verify which assumptions arise from the chosen set-theoretic construction.

## B Verification, limitations, and an audit checklist

### B.1 Finite regression checks

The accompanying program code/`finite_regression.py` uses only the Python standard library and exact rational arithmetic. It tests finite Laurent-polynomial reindexing, the deletion-of-indices coefficient identity, the twisted product identity, a finite exponent shift, and the exhaustive Boolean-algebra characters of a four-point set. Results are saved in `data/finite_regression_results.json`. [merge] These are the delivery names. In this directory the program is `code/07-critical-point-defects-finite_regression.py` and its recorded outputs are `data/07-critical-point-defects-finite_regression_results.json` and `data/07-critical-point-defects-finite_regression_console.txt`. Run without arguments, the program writes to the delivery name `data/finite_regression_results.json`; rerun it on a copy with an explicit `-output` path. 06 and 09 ship no programs.

These tests can detect an incorrect sign, a wrong coefficient position, or confusion between pointwise and convolution products in a finite identity. They do not model a nontrivial elementary embedding of the universe. On a finite Boolean algebra every ultrafilter is principal; the exhaustive check confirms that fact and cannot certify the existence of a nonprincipal complete ultrafilter. There is no numerical experiment for measurability in this package.

The PDF is compiled from the standalone source and visually inspected. No Lean kernel, Wolfram kernel, or other proof assistant was used to verify this article’s new theorems. The source and finite program are provided so that those limited forms of verification can be reproduced.

## B.2 Points requiring particular attention in an independent review

The most important review target is the passage from the internal normal form of  $j(x)$  to its external canonical normal form. That passage uses Lemma 2.6 and the constructive, not arbitrary, Hahn identification. Next, verify that the missing-image exponents cannot lie in  $J(\mathbf{No})$ ; this range-reflection argument is the source of both the intersection theorem and the intrinsic escape theorem.

Further checks concern the separation of outer support from birthday; the restriction to positive nonzero terms in the exact family-cardinality criterion; the distinction between a zero coefficient probe and a zero whole defect; and the use of Hadamard multiplication in the character theorem. In the exponential section, check the uniform recursive absoluteness and the monomial value of a purely infinite exponential. The supercompact theorem requires its actual seed to belong to  $M$ , not just an external set of the right apparent cardinality.

Every assertion about classes in this article reduces to a specified class map and set-indexed operations, or to a universal property of such maps. No proof uses a proper-class normal-form support. No claim is made that  $H_j(\mathbf{No}) \subseteq \mathbf{No}^M$ , or that  $J(\mathbf{No}) = \mathbf{No}^M$ ; neither identification is part of the construction. All image and intersection statements occur inside the fixed ambient  $\mathbf{No}^V$ . [merge] Both identifications in fact fail for a normal-measure ultrapower:  $H_j(x) \notin \mathbf{No}^M$  once  $x$  has more than  $\kappa$  terms (Theorem 9.7), and the ordinal  $\kappa \in \mathbf{No}^M$  is not in  $J(\mathbf{No})$  (Lemma 2.5).

## B.3 06's audit checklist [06]

| Potential mistake                                               | Correction used in the proofs                                                                                           |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Replacing $j(\lambda)$ by $j \text{ `` } \lambda$               | They coincide exactly below the critical point. The first missing index for larger domains is $\kappa$ .                |
| Assuming termwise images belong to $M$                          | This is the conclusion of the landing theorem, not part of the definition of $H_j$ .                                    |
| Zero coefficients in a purported normal form                    | All input coefficients are nonzero. This makes every omitted term detectable and is essential to the equalizer theorem. |
| Identifying support size with birthday                          | A monomial has support size one regardless of the complexity of its exponent.                                           |
| Using arbitrary coefficientwise convergence as strong summation | Strong summability requires finite incidence at every exponent and reverse well-ordered union of supports.              |
| Inferring a normal measure from any non-strong field embedding  | The measure recovery theorem assumes an elementary universe embedding and its canonical transport.                      |
| Using external size in the cover condition                      | The bound $ s  < j(\kappa)$ is evaluated in $M$ . The union obstruction is then proved with external representatives.   |
| Claiming $j(\kappa)$ is inaccessible in $V$                     | Only its inaccessibility in $M$ is used; the power-set agreement yields the external lower bound needed in the proof.   |
| Applying the closure converse to arbitrary embeddings           | A set-ultrapower representation is explicitly required in Lemma 9.3.                                                    |
| Reading coefficientwise inclusion as numerical order            | Binary supports, not their leading values, define $\preccurlyeq_{\text{coeff}}$ .                                       |
| Calling either lift an automorphism                             | Explicit omitted range elements prove that both are proper embeddings.                                                  |
| Treating a classical reformulation as a new consistency theorem | Cover and exact-descent criteria are proved equivalent to the corresponding fine-ultrafilter principles.                |

06's delivered PDF was compiled, checked for unresolved references and overfull boxes, and visually inspected; that document verification is not a machine verification of the mathematics.

#### B.4 09's specific error checks [09]

The crucial checks are mathematical, not numerical sampling. All normal-form terms in (5.1) are nonzero; otherwise an additional index might disappear and the equalizer conclusion would fail. The transformed exponent sequence is strictly decreasing, so the complement in (5.3) cannot cancel. The index  $\kappa$  really is the first missing one, even when the length is much larger than  $\kappa$ . The support bound for the coefficient span in Theorem 13.3 uses only finitely many coefficient supports, so it remains valid at singular uncountable  $\kappa$ . The branch code uses coefficients one and two, not zero and one, to retain the node's length. The ultrafilter code instead uses zero and one, where zero deletion is intended. Coefficientwise multiplication is always explicitly separated from ordinary surreal multiplication. The equalizer is explicitly separated from a fixed field. Finally, all constructions use set-sized supports and set-sized families, although their ambient fields are proper classes. 09's delivered PDF was compiled from its source; 09 reports no undefined references, missing citations, overfull boxes or LaTeX warnings, and visual review, and states that these checks concern document production, not the proofs.

#### B.5 Claim summary

Under (1.1), the exact defect, support equalizer, image intersection, projection isomorphism, positive escape, measure coefficients, omnific and surcomplex transfers, exponential locus, and two closures are proved in the text, relative to the stated classical inputs. The supercompact construction is proved with its additional closure hypothesis. The mask criterion is a proved reformulation of the classical measure characterization. Publication priority, consistency-strength converses, joint algebraic independence, differential compatibility, and all twelve research questions remain outside the claims of this article.

[merge] For the merged report: 06's landing criterion, landing spectrum, one-number test and normal-measure trichotomy are proved under their stated hypotheses (a set-ultrapower presentation for sequence closure, an index set of size  $\kappa$  for the trichotomy); its support-cover and exact-descent characterizations and 09's seed, covering-seed and prefix-tree statements are proved translations of classical principles; 09's short-support field, almost-disjoint independence and  $2^\kappa$  family are proved in ZFC (the last under  $2^{<\kappa} = \kappa$ ); 09's compatible-field-embedding criterion is a proved infinitary equivalence. The 34 research questions of the three sources remain outside the claims, except the exponential half of Question 21.30. Appendix C lists every non-claim.

### C Non-claims, by source [merge]

Every limitation, disclaimer and priority caveat stated by any source is kept, with its place in this report; the delivered proof-status files are shipped unchanged. The lists are grouped by source; items are paraphrased only where the source's own words appear elsewhere in the report.

#### Manuscript 07 (base).

- 07.1 Not refereed, not Lean-verified; publication priority not certified; no solution of a named published conjecture and no new consistency proof for a large-cardinal axiom (title page, Section 1.4).

- 07.2 No claim that arbitrary non-strong surreal embeddings require a measurable cardinal; the two embeddings are not identified as automorphisms; equality on monomials does not imply equality on a completed field (Section 1.4).
- 07.3 The proofs are not theorem-prover output; the finite checks test finite identities only and cannot test an elementary embedding or a nonprincipal complete ultrafilter (Section 1.4, Appendix B).
- 07.4 The literature and repository comparison is targeted, not an exhaustive priority search; “candidate original” means not located in the material inspected, not that no specialist has observed it; a specialist comparison is appropriate before any priority claim (Section 20).
- 07.5 Strong sums are not order-topological limits (Remark 2.2).
- 07.6 The sign-expansion theorem is a classical input, not reproved; an arbitrary Hahn-field identification would not suffice (Remark 2.8).
- 07.7 Hahn reindexing is standard; the derived measure, its properties, ultrapowers and the fine-normal characterization of supercompactness are classical; the mask criterion’s ultrafilter characterization is not new (Sections 4, 10, 16, 20).
- 07.8 The equalizer  $\mathcal{S}_{<\kappa}$  and the common image  $C_j$  are not identified (Remark 6.2); exact intersection gives transcendence, not linear disjointness or joint algebraic independence of the image fields.
- 07.9 No unconditional cardinality criterion for signed families (remark after Theorem 8.1); a zero probe does not mean a zero defect; the character theorem concerns the Hadamard product only; the mask criterion is not a characterization in the pure ordered-field language.
- 07.10 No real-valued probability measure on all surreal sets is introduced (after Theorem 10.1); no new factorization theorem for set-sized quotient rings (Section 11).
- 07.11 The surcomplex maps fix a chosen orientation  $i$  and say nothing about arbitrary automorphisms of the pure surcomplex field (Section 12).
- 07.12  $D_j$  satisfies a twisted identity; it is not claimed to be a Berarducci–Mantova or strongly additive derivation; only the near-one logarithm is used for  $L_j$ , and no hyperclass or class-sized quotient is formed (Section 14).
- 07.13 The supercompact construction depends on the enumeration; no new large-cardinal equivalence; an enumeration-free seed geometry is a question (Section 16).
- 07.14 No exception to Kunen-type obstructions; the inclusion is elementary only in the ordered-field reduct; the information hierarchy’s arrows mean forgetting, not extension (Section 19).
- 07.15 Choice is used only for sets: no basis of  $\mathbf{No}$ , no class Zorn lemma, no collection of all class endomorphisms (Section 19).
- 07.16 No claim to settle a numbered question of Kaplan–Krapp–Serra; results assuming bijectivity or pointwise preservation of the value group must not be applied to this pair unchecked (Section 20).
- 07.17 The repository’s Lean Hahn infrastructure does not mean any theorem here is checked; the proposed Lean interface is not a formalization claim; the measurable-cardinal assumption must not become an unnamed axiom (Section 20, Appendix A).
- 07.18 No claim that  $H_j(\mathbf{No}) \subseteq \mathbf{No}^M$  or  $J(\mathbf{No}) = \mathbf{No}^M$  (Appendix B).
- 07.19 The twelve questions are proposals, not famous open problems; consistency-strength converses, joint algebraic independence, differential compatibility, recovery of the ambient embedding, classification of

image pairs and an enumeration-free supercompact invariant are not established (Section 21, delivered proof status).

- 07.20 The repository inspection covered selected root documentation, the catalogue, two report READMEs and a source preamble; it was not a full audit or exhaustive novelty search; no repository theorem is an unproved premise; the repository was not modified, no Lean files were written and the repository build was not rerun (Section 20, delivered README and proof status).
- 07.21 The finite models are deliberately not finite elementary embeddings; there is no finite analogue of the large-cardinal assumption; the program does not verify elementarity, measurability, class recursion, normal-form absoluteness, transfinite summation, supercompactness, or the article (delivered proof status and recorded data).

### Manuscript 06.

- 06.1 An AI-assisted draft with written proofs, not refereed or machine-checked; the proposed contributions are the comparison, defect and descent statements, not the Hahn-lifting or ultrapower-seed machinery; not a priority certification; no named published conjecture resolved (Section 1.4).
- 06.2 All large-cardinal implications retain their hypotheses; no consistency strength beyond the classical ultrafilter principle; no large-cardinal axiom is claimed proved in ZFC (Section 1.4, Section 17).
- 06.3 The repository's birthday, hereditary-set, embedding and forcing work is not repackaged as new; no new proof needs an unreviewed repository theorem; mere existence of a non-strong map is not a contribution (Section 20).
- 06.4 The absoluteness lemma is an interface, not a claim that the Hahn fields of  $M$  and  $V$  coincide (Remark 2.7).
- 06.5 The maps are proper embeddings, not automorphisms; the Kaplan–Krapp–Serra automorphism classification (their Question 5.6) is not answered; no conflict with Kunen (after Proposition 7.4).
- 06.6 The “one number” test is relative to  $j$  and  $M$ ; no pure-field formula defines supercompactness (after Corollary 9.4).
- 06.7 The seed–closure converse is proved only for set-ultrapowers; the no-descent result above  $\kappa$  uses an index set of size  $\kappa$  and is not a theorem about every embedding with critical point  $\kappa$ ;  $j(\kappa)$  is inaccessible in  $M$ , not asserted in  $V$  (Sections 9, 19).
- 06.8 Coefficient extraction and the monomial scale are specified extra structure, not claimed definable in the pure field language (after Corollary 10.5).
- 06.9 The compactness criteria are equivalences with classical principles; a measurable cardinal is not shown to fail a stronger property because of its normal-measure ultrapower's landing spectrum (Section 17).
- 06.10 No global surcomplex exponential or logarithm is assumed (after Theorem 12.3).
- 06.11 Preserving  $\mathbf{Oz}$  is not preserving it pointwise; not a theorem that all endomorphisms of  $\mathbf{Oz}$  come from large cardinals (Section 11).
- 06.12 No automorphism classification, Kunen inconsistency result, new consistency theorem or first-order characterization in the pure surreal field; no large cardinal may be inferred from an arbitrary non-strong field map (Section 19).

- 06.13 No Lean proof; the repository's Hahn layer is not verification of 06; finite experiments cannot validate the critical-point or measure assertions, so none is supplied; the archive contains no purported formal proof (Appendix A).
- 06.14 The literature check was targeted; the Conway and Gonshor books were not retrieved in full; a Wikipedia page served as orientation only; not performed: peer review, Lean or other proof-assistant verification, computation of a nonprincipal ultrafilter (delivered proof status).
- 06.15 The questions are proposed directions, not published open problems (Section 21); the repository inspection was a scope and overlap check, not a full audit (Section 20).

### **Manuscript 09.**

- 09.1 An unrefereed AI-assisted manuscript; its contributions are proposed, not certified priority claims; not formally verified; no established named open problem is claimed settled (Section 1.4, delivered README).
- 09.2 Merely constructing a Hahn lift, observing that embeddings fail to preserve long sequences, or coding an ultrafilter numerically would not by itself be the result (Section 1.3).
- 09.3 No existence or consistency of any large cardinal; no nontrivial  $(V, \in) \rightarrow (V, \in)$ ; the structure questions for all surreal exponential automorphisms are not settled (Section 1.4).
- 09.4 The maps are embeddings, not automorphisms; the short-support field construction is not claimed new; no claim to have initiated the large-cardinal and generalized-analysis connection; repository bi-interpretation results are not used as assumptions (Section 20).
- 09.5 Definability alone does not give absoluteness; the reals remark does not put every set of reals or of old surreals into  $M$  (Section 2).
- 09.6 Support cardinality, birthday and order-bounded support are different bounds; individual transcendence is much weaker than independence; the equalizer is not a fixed field (Sections 3, 5).
- 09.7  $D_j$  is not an ordinary derivation; continuity is an interval statement about a class order; not every family of size  $\kappa$  fails; the omega incompatibility concerns a specified monomial action only (Sections 5, 8, 14).
- 09.8 No conflict with repository claims about maps to set-sized rings: the embeddings have proper-class codomains (Section 11).
- 09.9 The independence theorem is not a large-cardinal theorem (Section 13).
- 09.10 Measure recovery is not a proof that an arbitrary field embedding yields a measure; a zero coefficient is not a zero defect; masks are not ordinary idempotents; the numerical criterion is infinitary and not first-order in the bare field or omnific-ring language (Section 10).
- 09.11 A derived ultrafilter need not determine  $j$  or  $M$ ; the seed and compactness statements are numerical presentations of classical mechanisms, not new consistency equivalences (Section 16).
- 09.12 The prefix principle is an exact coding of a classical tree principle, not a new relative consistency proof; the branch is not an order-topological limit; the Carl–Galeotti–Löwe hypotheses must not be replaced (Section 18).
- 09.13 Parameter-free field sentences cannot detect large cardinals; the exact expressive strength of the birthday predicate is not resolved (Section 19).

- 09.14 Every sum is set-indexed; there are no proper-class sums; the omega-map is not the Gonsior exponential; pure field elementarity does not give elementarity in an enriched birthday language or a self-embedding of  $V$  (delivered proof status, items 1–10).
- 09.15 The proofs are written, not Lean proofs; no finite program tests a measurable cardinal; the claims were not imported into the repository’s Lean development; the literature check does not certify that no equivalent result exists elsewhere; the repository comparison was targeted and no repository file was modified (Appendix A, delivered README).
- 09.16 The questions include genuine gaps and are not an exhaustive list of published problems (Section 21).

### Added by the merge.

- M.1 No statement of this report has a Lean implementation mapping; independent proof review and formalization are pending.
- M.2 Remark 2.9 covers same-ordinal inner models with the same reals only; the classification asked in `dsn:q:support` stays open.
- M.3 Remark 11.6 is conditional on a measurable cardinal (or a class embedding with critical point); whether an **Oz**-reflecting non-strong embedding exists without large cardinals, and the clauses on dense images, are not addressed.
- M.4 Of the 34 questions only the exponential half of Question 21.30 is answered here; Theorem 13.3 does not decide Question 21.6.
- M.5 The merge found no contradiction between the three sources; its corrections (Appendix D) are bibliographic and concern the repository identifier, not mathematics.
- M.6 A clean build and the rerun finite regression certify no transfinite statement.

## D Provenance and merge decisions [merge]

**Sources.** The report is built from three manuscripts of batch 30, committed as archives in 190d301 and placed in 21375f8. Each archive was matched to its manuscript number by the hashes of its proof-status file.

| No. | Archive                                                     | Pin     | Shipped here                                                                                                                                 |
|-----|-------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 06  | Surreal_Large_Cardinals_Critical_Supports.zip (27-page PDF) | 0865f04 | proof status, build helper (prefixed 06-critical-supports-)                                                                                  |
| 07  | Surreal_Large_Cardinals_Research.zip (32-page PDF)          | 0865f04 | this article’s base text; proof status, build helper, finite regression and its three recorded outputs (prefixed 07-critical-point-defects-) |
| 09  | surreal_large_cardinals_article.zip (30-page PDF)           | 0865f04 | proof status, build helper (prefixed 09-support-geometry-)                                                                                   |

All three pin the same commit 0865f043aec113c14c69ef45006bbc7546a4e75a (“Merge reviewed documentation after angular multiplicity validation”), 55 commits before the placement. The manuscripts’ .tex,

PDFs and READMEs of 06 and 09, and 07's PDF, are not shipped; 07's delivered README was replaced by the collection README.

### Where the merge had to choose.

- *Base.* 07, as decided at placement: on the shared theorems 06 and 07 have the weakest hypotheses and 07 is broader. 09's headline assumes a normal-measure ultrapower, though its arguments need no more than 07's.
- *Printed once.* The first missing index, the absoluteness lemma, the induced embedding, the strong companion, the defect formula and equalizer, the leading-term corollary, the omega-map locus, the small-family threshold and first failure cardinal, the derived normal measure, the distinguishing corollary, the zero-probe warning, the character theorem, the omnific equalizer and constant terms, the surcomplex transfer, the twisted product law, exponential transfer and the exponential locus. Each carries its sources.
- *Second routes.* 09's proof of the absoluteness lemma (limit cylinders and truncation simplicity); 09's direct real-closedness of  $\mathcal{S}_{<\kappa}$  beside 07's proof through the embeddings; 06's direct support counting for the short fields; 09's fiber argument in the weighted evaluation; 06's formal-evaluation proposition for the infinitesimal factor.
- *Kept separately, as different statements.* The three measure codes  $h_A$ ,  $\tilde{h}_A$ ,  $Y_A$  and their omnific shifts; the mask criterion (07) and the compatible-field-embedding criterion (09); 07's supercompact seed coefficient, 06's exact-descent and cover characterizations and 09's covering and exact seeds; the three sources' questions, with overlaps noted.
- *Printed from a source where 07 took the fact as background.* 06's proof of the critical-point facts (Lemma 2.3).
- *Headline theorems.* 06's and 09's summary theorems are printed in Section 1.3 with proof pointers, not as separate proofs.

### Discrepancies found and corrected.

- 06 records the snapshot 0865f043... as a tree object, "not asserted to be a commit". It is a commit; its message is quoted above. The delivered 06 proof status is unchanged.
- 07 gave Gonshor's book as Lecture Note Series 124; 06 and 09 give 110, which is correct. The bibliography now says 110. (The definable-surreals report records the same correction for another manuscript.)
- 06's bibliography links the universes and birthday-cutoff reports on the main branch; 07's and 09's link them at the snapshot. The merged bibliography keeps 07's entries.
- 06 states the counting proof of the short fields with "regularity and uncountability"; only uncountability is used (note after Corollary 3.2). This is a remark, not a correction of a false statement.
- 07's text names its program and outputs by their delivery paths; the shipped paths are given in Appendix B.

No false theorem and no contradiction between the sources was found.

## References

- [1] John H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976. Classical source for surreal arithmetic, normal forms, and omnific integers.
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- [8] Vladimir Reshetnikov and contributors, *Surreal*, repository snapshot 0865f043aec113c14c69ef45006bbc7546a4e75a, root README.md and docs/README.md, consulted 23 September 2026. <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a>. Repository statements and formalization coverage are distinguished from independently checked proofs in the present article.
- [9] *Surreal Fields Across Set-Theoretic Universes*, repository research report, README and source preamble inspected at the snapshot in [8]. <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/foundations-and-computation/surreal-fields-across-universes>. Research context only; its new theorems are not premises here.
- [10] *Birthday Cutoffs and Hereditary Sets*, repository research report, README inspected at the snapshot in [8]. <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets>. The README credits the full-class bi-interpretation announcement to Chen–Hamkins–Yang. That announcement is not claimed as new here.
- [11] *Set-Sized Quotients of Omnific Integers*, catalogue entry in docs/README.md at the snapshot in [8]. Relevant report directory: <https://github.com/VladimirReshetnikov/Surreal/tree/0865f043aec113c14c69ef45006bbc7546a4e75a/docs/surreal/set-sized-quotients-of-omnific-integers>. Cited only to identify the repository’s existing constant-term theme.  
[merge] Entries added from manuscripts 06 and 09 (as delivered; 06’s *conway*, *gonshor*, *kks*, *neeman*, *repo*, *birthday*, *universes* and 09’s *Conway*, *Gonshor*, *DE*, *KKS*, *Repo*, *RepoUniverses*, *RepoBirthdays* are the entries above).
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- especially Definition 2.10 and the surreal specialization. <https://arxiv.org/html/1810.03029v2>. [06]
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